MBIS624-26S1-LecA-Te-s1-low

Meeting started: Mar 23, 2026, 10:10:00 PM
Meeting duration: 114 minutes
Meeting participants: Speaker 1, Speaker 2, Speaker

View original transcript at Tactiq.

Transcript

00:00 Speaker 1.: Ready to start.
00:01 Speaker 2.: Perfect timing. Okay, so I assume everybody here is in the MBIS program. So at least we have a kind of homogenous course here. Everybody is from the same program. And are you all in your first semester? Great. So this is a very introductory foundational course. So in a way.
00:33 Speaker 1.: a lot easier than some of the others.
00:36 Speaker 2.: So, great way to ease into the program, but at the same time, I do appreciate that as you are just starting, you might have some questions that... the more advanced students wouldn't have. So please feel free to email or come to me if you've got questions that are not strictly course related,
00:59 Speaker 1.: because
01:00 Speaker 2.: this is where we start. So I do appreciate that you would have
01:07 Speaker 1.: different
01:09 Speaker 2.: needs. And we do trade courses differently. So the course is currently still in the system, that's why I put BI, business intelligence and analytics, but really that is a little bit
01:25 Speaker 1.: redundant
01:28 Speaker 2.: because business intelligence is actually the use of analytics in business. So the newer term is business analytics. So very often when we will talk about business intelligence or business analytics, we really mean the same. In some
01:48 Speaker 1.: of the
01:49 Speaker 2.: books, which to be honest doesn't really matter, but in some

of the books they say, okay, business intelligence is only about descriptive analytics where we look at
02:00 Speaker 1.: the
02:02 Speaker 2.: present and past information to gain understanding of what has been happening. So if you encounter articles like that, I just wanted you to be aware. But in this course, I'll be using business analytics and business intelligence pretty much interchangeably. So today, first class, first semester, we'll take it. sort of easy, a bit of a course introduction, and then what actually business analytics or intelligence is, and what is the approach that we will be taking, because the goal of this course is, yes, to learn some analytics techniques
02:53 Speaker 1.: to
02:55 Speaker 2.: prepare you for more advanced courses, but also, more importantly, our industry partners have asked that we learn how to approach business problems and to translate those business problems into data questions because what happens in life is they're not going to come and tell you, can you please
03:26 Speaker 2.: do a time series for something. They're just going to come and say, while I'm having this problem, what do you think we can do about it? So we'll be practicing translating problems into specific questions starting today. But as I said, Very gentle start. In this course, we actually are very fortunate to have in the second term a visiting professor who is experienced and
03:56 Speaker 1.: specializes
03:58 Speaker 2.: in analytics. So he,
04:01 Speaker 1.: Casey
04:02 Speaker 2.: Chen, he comes just after the break. So he'll be taking the second term and will be taking
04:13 Speaker 1.: Well,
04:14 Speaker 2.: some of the fun bits with AI and predictive analytics. And I will be doing the first term, so we will be focusing more on descriptive and diagnostic analytics. what perhaps allow books called business intelligence. And Emily is our tutor. She will be taking the tutorials. Tutorials start from next week. No tutorials this week. And I have just... opened the tutorial allocation system.
04:50 Speaker 1.: I

04:50 Speaker 2.: purposefully had not opened it up so that people don't turn up for tutorials this week. So you can allocate yourself to one of the tutorial sessions. Lots of spaces there. Feel free to come here in front. Nothing scary.
05:15 Speaker 2.: So unlike some of the other more advanced courses, this course has a bit more of hands-on assignments. So the lab assignments, there are two of them. One in term one will actually end over the break. And one in term two.
05:37 Speaker 1.: The
05:38 Speaker 2.: assignments will
05:39 Speaker 1.: be
05:41 Speaker 2.: testing your knowledge in what you've learned. learned in tutorials, and a little bit in lectures as well. So,
05:49 Speaker 1.: 15%.
05:51 Speaker 2.: You're also given, because I'm a big believer in giving credit for any work, you'll be given...
06:02 Speaker 1.: credit for
06:04 Speaker 2.: submitting tutorial solutions. So you go to a tutorial or you do it by yourself. Some people prefer to do that. And you will have a submission link to submit your file with your solution. The purpose is to acknowledge that you are doing the work regularly instead of waiting until the assignment is released. So how do we mark the tutorial submissions? There are quite a lot of you and only two of us, so what we do is we just look to see whether it's more or less right. And if it's more than 50% right, it's a pass. and you get 100%. So it really is a
06:55 Speaker 1.: way of giving
06:57 Speaker 2.: you some credit for doing the good work. Nice and easy, and of course the scary stuff with the term test and the final exam. It really is not hard because in this course, the way we've set it up this year because of the arrangement with Casey, The term test will be only on the first five weeks on my material. It will be sometime in week six, still waiting for the schedule. It will be based on very short sort of description of a business, and then you'll have some questions. So very similar to what we do in class. As to the final exam, it will be on the second part that Casey is taking, and he will tell you what that is going to be about.

07:55 Speaker 2.: Unlike all of my other courses, this is not going to be open book because Really, it's a bit too foundational to make it an open book, but you will be allowed to bring in
08:10 Speaker 1.: what
08:12 Speaker 2.: we call a cheat sheet, which is one sheet. Okay, just to preempt all the questions I usually get, whatever font, whatever. filled up the sheet, don't care. One A4 side, which if
08:28 Speaker 1.: you want a
08:29 Speaker 2.: size 6 font and you want to read with a magnifying glass, that is your choice. I don't really care. So, because every time I say that, people are like, oh, how big is the font? How big are the margins? Okay, have a go.
08:45 Speaker 1.: Fill it up.
08:46 Speaker 2.: So that's all about the arrangements and the assessments. And I understand that some of you were anxious because the course outline was not available. We had to. kind of, I had to step in at very short notice. I do teach other courses in this area. So it was kind of only last week that
09:11 Speaker 1.: I realized I'll
09:14 Speaker 2.: be taking the first half. So, but it's all under control.
09:18 Speaker 1.: We're all good.
09:19 Speaker 2.: So the course outline is now on and we're in business. Any questions?
09:28 Speaker 1.: All good? Okay, so, actually,
09:36 Speaker 2.: because most of you are not in any of my other courses, I do want to say that I expect you to participate in class. This is an expectation. You don't get marks for it. It's just for fun. So, what that means is that you don't get evaluated. There is nothing wrong that you can say. So,
09:59 Speaker 1.: let's start
10:02 Speaker 2.: with some opening conversation about business analytics and understanding how dealing with data fits in with business goals. What happens is every business collects now a lot of data. So they have access to,
10:27 Speaker 1.: we
10:28 Speaker 2.: generate data all the time. So, for example, today already you would have generated a lot of data. Every time you log in, for example, the information about what you do on computer networks is recorded. Every time

you go into a room after hours and swipe your card,
10:50 Speaker 1.: it's
10:52 Speaker 2.: recorded. They know who it is, what time, which door, right? So we have a lot of data. And actually in this room, we're also generating data, right? Because we are having a video recording. And that is also a type of data. And any text that you put, any post on social media, that is also data. So businesses have...
11:26 Speaker 2.: access to huge amounts of data, but
11:30 Speaker 1.: they
11:31 Speaker 2.: have to think about how they can use this data to actually gain some insights that are relevant to their business. So the question here is, what are those business insights for? You don't have to answer yet, we'll have a couple of examples and then we'll figure it out. What are those business insights for and how do we start actually, you see how this one looks? quite linear. It looks like, oh, here we have data, we do something with it, and then we get a business insight. And this is very often how we do things, especially in undergraduate courses. Because we say, okay, here's your lovely, perfect data set, and please perform this technique, and you follow the steps, and then you get the result, and you say, here's my result, figure it out.
12:24 Speaker 2.: But what happens in business is a bit of two ways, right? Because they might say, well, we have a problem with this. You have to figure out what data you need for that. Then you get some insight. So the business insight, and this is, you know, if you want to remember one thing from today, I really want you to think about the fact that any data results need to be linked to a specific decision.
12:52 Speaker 2.: the business has to make. So it's not enough to say, oh, this is what the average
12:58 Speaker 1.: number
13:01 Speaker 2.: of people attending classes is, right? Because then I have to think, okay,
13:07 Speaker 1.: and?
13:08 Speaker 2.: Why do we care? What decisions can we make about that? So business insights are always related to actions and decisions that businesses have to make. So to give you some examples, I do like looking, we work with

the supermarkets. Doesn't have to be countdown. We do have examples from foodstuffs as well. But this one is first with countdown. Again, not picking on
13:39 Speaker 1.: them.
13:42 Speaker 2.: We'll have... Foodstuffs have prepared a lovely presentation for us too. This one is just on...
14:10 Speaker 1.: Unfortunately, that wasn't the experience for one Auckland shopper on Sunday. Henry thought he was getting his willies worth with his promotion of half-price skincare products at West City. But at the checkout, he was charged full price. I felt very unhappy and angry, especially when the store manager told me that today's half-price sign is for the next week. It's not a surprise to advocacy group ConsumerNZ. For a customer to experience a pricing error is really, really common. So common, in fact, that in August, the Grocery Commissioner estimated Kiwis were losing tens of millions of dollars a year from supermarket pricing errors. I'm going to bring up the mandatory requirement for supermarkets.
15:00 Speaker 2.: So I just wondered a bit about the problem. We're all probably, we have experienced it. If you've been here more than a week, you have experienced it already in the supermarkets. So now what I would like you to think about,
15:16 Speaker 1.: having
15:17 Speaker 2.: heard the problem that Henry had, So to us, it's very clear what the problem is for customers, right? They're paying tens of millions of dollars more than they should. Now, if you were in the business, from the business point of view, what is the problem for the business? So this is my first question. What is the problem for the business? And my second question is, What would be some questions that you'll be asking that can be answered with data? So what are some things that you, if you were in that situation in countdown, you would be asking questions about this to... to figure out what to do, or just what other information you would have liked to have. So remember, we are just starting, so you could generate any questions. So what was the, the first one was, why is this a problem for, is it a problem for countdown? And if it is, why is it a problem? What is the problem? Do they actually have interest in fixing it? And the second question is, if you were to help them, what is some further information that you would like to get for this problem where customers

are overcharged? So how long do you need for that? Everybody awake?
16:50 Speaker 1.: So... Let's say three
16:52 Speaker 2.: to four minutes. Talk to other people, because that is the way. If you haven't met each other, say hi, give your name. This is a nice starting position. But I want to hear talking.
19:22 Speaker 2.: Okay, my analytical skills tell me that there's been enough noise around here to generate the answers. Okay, so this is actually a bit of a tricky one. Why is this a problem for countdown? Because in some ways, you might think, okay, well, there are actually customers are buying at a higher price than they should. They're making more money than they should.
19:52 Speaker 2.: Yeah, why is this a problem? Yep. One, two, three, right. Right, so that was a similar conversation because I decided to be in a team. So we had a similar conversation that obviously this affects the reputation of the business as a whole and as you said, a specific store. Let's see if you're following up this with a data question. But one of the issues in New Zealand that you may or may not be aware of is that we actually have only two supermarket businesses, and they both have very, very similar issues. There was a huge Commerce Commission report.
20:56 Speaker 2.: last year or the year before that we were looking at with food staff. So in a way, yes, normally this would be a problem if it was a competitive environment, and in general, obviously, every business would like to have
21:11 Speaker 1.: loyal,
21:12 Speaker 2.: trusting customers. Is there another problem? You had the same, and here you had, oh. There was somebody here? Right. Can you also say your name? Because I am determined to remember the names by the end I leave this class.
21:33 Speaker 1.: Nicola.
21:40 Speaker 2.: Right. So actually, compliance risk is the main issue for the supermarkets, although, yes, from
21:48 Speaker 1.: a good business perspective,
21:51 Speaker 2.: they do care about the reputation and loyalty of customers. But in this case, this is why I kind of left the Commerce Commissioner talking a bit about how common this is, is that there are regulations about it, and there is financial... So there is a bottom line problem as well. It's not all intangible.

There is actually a repercussion if this keeps happening. And in fact, for the same supermarket in Australia, because I was checking it out with some colleagues, they have to, if they ever have a situation like that, they have to give the product for free.
22:41 Speaker 2.: well, here it's not as common, like I have tried it. You actually have to push it. Usually they just try
22:48 Speaker 1.: to fix it for you. So here it's a little bit less, there it's, I've
22:54 Speaker 2.: tried it there. There's no conversation. They say, oh, we've made a mistake. It's free. So in addition to the penalties, obviously they're also going to lose money if they're giving free products. So this is quite important because sometimes when you first hear about the problem, your job as an analyst is also to understand the magnitude of the problem. and how bad it is, how important it is to fix it, because we sometimes have to prioritize the problems, right? So in this case, obviously now, how are we going to figure out the magnitude? Did anyone have questions about that?
23:42 Speaker 2.: There are data questions. Okay, what questions did you have?
23:46 Speaker 1.: Let's start with that.
23:47 Speaker 2.: What questions did you generate that you're going to ask to understand more about this situation? about the promotions?
24:10 Speaker 1.: Okay,
24:11 Speaker 2.: so when they get notified, or when does it start, or both? So you could check that, yes, you could check how much notice
24:22 Speaker 1.: do the stores get about
24:24 Speaker 2.: the promotions. How would that help? Now nobody's gonna say what their questions are.
24:38 Speaker : Yeah. Yes. Yeah.
24:43 Speaker 1.: Yeah. Yeah, yeah. Right,
24:54 Speaker 2.: exactly. So that could help you to discover this is really good direction because the first thing is for us to figure out, well, why is this happening?
25:06 Speaker 1.: What's potentially
25:08 Speaker 2.: the root cause that we want to check? So if you look and see, okay, they were notified too early and in this case, they put up the marketing material before the promotion was in place, then this can be isolated to specific

store management where they're not paying attention to the date, right? They've put it early. So then we can follow up.
25:37 Speaker 1.: with
25:38 Speaker 2.: more data questions from your question to see, because you see in that video, they showed, this is why I actually liked it, because there are lots of videos complaining about stuff, but this one, it, there you go,
25:59 Speaker 1.: it had
26:01 Speaker 2.: the receipt, right? So in the receipt, you could actually see which store and exactly what time, actually further down, so you could see everything. So then, what we can do, if we're in the central, right, we could actually, this is called post data point of sale. It is available to... Woolworths, and you can see how many stores are having this kind of discrepancies. Because now, the problem in this case is they were actually charging the right price. for that time because the promotion was not in place yet. So you would have to match the complaints against what happened and then see how many stores. So you see how it's never really that simple.
26:56 Speaker 2.: Another reason, what other questions did you have? Anyone here from the front? Raising your hand, but now not sure. Yeah,
27:07 Speaker 1.: now not sure. What's your name? My name is Julia. Julia, right. Sorry, I didn't. So we have discussed, like, we thought the problem would be coming from probably the data. Yeah. Data received, so because now that we get the data, it's fed into the system, so maybe the problem...
27:29 Speaker 1.: Yeah, and
27:30 Speaker 2.: that is another option, right? When you don't have the single source of truth, maybe in the promotions it says one thing and it's sent to the store, but actually in the system, and this is very, very common with New World actually, in the system the data is different. So we have mismatch in the data. So we would then want to see for all the promotions in that system, for that product at that date, and in the system, the actual prices, do they match or not? So this is discoverable by data. So why were you hesitating? Great question. Okay, what else can we ask about this?
28:12 Speaker 2.: So if it was actually a problem with your problem, with the data, we then want to see, because you see how they were saying tens of millions of dollars, right? But remember we said the impact on us is potentially

fine, or if they actually give it for free. So that would be recorded, so you could actually see, okay, as a result of these discrepancies, how many times have we given it for free, and How much did that cost us? That is discoverable in the data. How would we discover it? Well, every time they have to do that kind of transaction, it has a separate code. We could check for the codes, and we could check whether it happens in some stores more often than in others. So we can see, is this a location-type problem, that these guys are always doing something, or is it a system-kind problem?
29:12 Speaker 2.: And then you can actually also see whether it happens across the board with all the promotions or some products seem to always be missed. You know, is there a problem with a price off? Ice cream, for example. Sorry, that's my favorite.
29:32 Speaker 2.: So there's actually quite a lot about this problem that we can assume. And this is your job to challenge assumptions. And so you can start with an assumption. Okay, you started with might be in the system. You think, okay, it's something with the store, right? So you start with that assumption, but then you check. You know, this is my assumption or idea, hypothesis. I need to go and check what is happening. You know, is it store related? How much are we paying? How much does it cost us? All of these questions can be generated, and then you actually get some business insights about a decision. What decision can they make once they discover that information?
30:23 Speaker 2.: So if they discover that it's,
30:26 Speaker 1.: let's
30:28 Speaker 2.: start with your situation
30:30 Speaker 1.: because that's kind of what was happening, that
30:33 Speaker 2.: in some stores they're putting the promotion labels before the promotion is actually in place.
30:42 Speaker 1.: Okay,
30:44 Speaker 2.: what decision can they make and who's going to make it? So we've discovered the problem. Oh yeah, yeah, definitely. That's already a rule. But in the long term, how are we going to fix this problem? Because they're losing money, because they're giving them half price.
31:21 Speaker 2.: Because this is a human problem we have, right, if they're putting. So in a way, as IT people, we find it very difficult when

31:32 Speaker 1.: we
31:33 Speaker 2.: have to fix the human problem, right? And so we keep trying to think, well, what is a way of fixing this? And in this case, at the end of the day, it's the store manager who... takes that cost, but we could maybe give them information about how much money they have lost because of that, and that if there is a fine, they will be paying the whatever portion of that, so there is some incentive for them to actually do it right.
32:09 Speaker 1.: Sorry,
32:09 Speaker 2.: you had
32:10 Speaker 1.: ideas? Yeah.
32:21 Speaker 2.: Ah, okay, so yeah, that's a really, so if they discover that sometimes people are putting it early, then you could actually tell them, okay, the signs have to be put on Monday or Sunday night, 12 o'clock, midnight if it's open, right? So we could actually make that a rule, and so they can follow that rule without... too much explanation, so that's another way. So that's for the human behavior, making a decision how we are going to implement this
32:52 Speaker 1.: rule
32:54 Speaker 2.: so that it's followed by everybody consistently. In the non-distance future, probably, we have some robots to put the promotion material, and they're always going to do it right. Okay, so what about the other situation with the system, which actually is more difficult, right? Because we have a mismatch between what's been advertised in the promotion and the prices that are in the system when you scan. This is actually where people lose more money.
33:30 Speaker 2.: What can we do about that?
33:56 Speaker 2.: Yeah, yeah, or figure out how I would be asking, well, how does the information, you know, why does it have to be in two different places? I mean, we can, you know, is it integrated? How does it follow? You know, when people make decisions about promotions, it's actually quite easy to automate it to go straight into the system. So clearly we need to, it's a system design decision, which, believe it or not, is an easier one for us, because there are a lot of things we can do about it, as you're probably already starting to work on in 601. So, sorry, you had something. What was your name?
34:41 Speaker 1.: Vicky. Vicky, right.

35:09 Speaker 2.: Yeah, yeah, you're right. So this is your situation. So yours was more proactive, saying, let's make it so that it doesn't happen again. But you are probably a little bit more pragmatic and say, well... It will probably happen again, so can we have some kind of a reactive flag that when it happens the first time and this customer service code is put in to return money or change, it triggers it in the system and... changes it immediately, so that at least we limit the impact. So you can see how, based on the data, we can understand,
35:49 Speaker 1.: so
35:51 Speaker 2.: especially with business intelligence or descriptive analytics, which we will be doing in the first time, a lot of it is actually about better understanding of how are we doing, what are the problems, how bad are they, you know, kind of more understanding of the business, but you do have to link it to decisions that can be made.
36:14 Speaker 2.: Now, because I'm just a little bit conscious of time, in this one, in this link, if you want to read it, this was a little bit more positive, because this was a complaint. In this other one, what Countdown had done after the COVID lockdown, they created this certainty model, which is analytic, so it's great. fun, where it's based on your loyalty, on your customer number, and your loyalty cards. So how many people have loyalty cards already? I know some of you have arrived recently. Right, so most of the time in supermarkets, so why do you have a loyalty card?
37:04 Speaker 2.: Points, so you get some rewards, and? Yeah, get discounts, right? So, basically, we think that we're getting something, and we are. Sometimes we're getting a discount, and I don't know where these points go, but, you know, supposedly points. And what the business gets is actually our information. our data,
37:34 Speaker 1.: because it
37:36 Speaker 2.: gets, when you go to the supermarket, what you buy, how often, if something is on special, do you buy more of that, or do you buy something only when it is on special, which means that it's not something you would have bought otherwise, so they get all of that information about us, and,
38:02 Speaker 2.: If you have completed it correctly, they also get your demographic. So they know how old you are and your address. not in my loyalty cards. But that gives them a lot of information to profile, right? So what

they did with this information, they had this predictive model, which is something you'll be doing in the second term, where you try to, based on
38:31 Speaker 1.: what's been happening, predict what will
38:33 Speaker 2.: happen in the future. So they were predicting whether a particular customer is going to be shopping online or in store, what they and
38:46 Speaker 1.: what
38:46 Speaker 2.: they will be buying. So, of course, if it's just for one person,
38:52 Speaker 1.: well...
38:53 Speaker 2.: Yeah, okay, that's a bit useful, but when you aggregate it, to say for every single person we can predict that, what that means is then for a particular store, depending on now how you store the data, for a particular store you could actually predict how many customers are going to come at any given time.
39:17 Speaker 2.: And you can predict how many customers are going to buy Hokey Pokey ice cream. So how does it help them to
39:27 Speaker 1.: make
39:28 Speaker 2.: a decision? If you know how many customers are going to come at any given time into a
39:36 Speaker 1.: supermarket, how
39:38 Speaker 2.: does that help? So the inventory levels, especially if you know what they're going to buy,
39:47 Speaker 1.: and
39:48 Speaker 2.: promotion if it matters, yes, and staffing, right? Because sometimes you would have noticed they have a lot of people in the checkout and sometimes not. So for supermarkets, what do you think is the most valuable resource in a supermarket?
40:10 Speaker 2.: Something that they really try to optimize. Yes, cost,
40:20 Speaker 1.: but
40:21 Speaker 2.: okay, let me put it a different way so that we think about it. Why would they care how many customers would like to buy Hokey Pokey ice cream? So they order, okay, so what if they have ordered more ice cream than they need? Well, economy of scale, but that's already there with the supermarkets, the economy of scale, but for the particular store. Okay, that's true. That's true. That is helpful because they might say, okay, these people are

coming and they don't buy that stuff, so we'll put it on sale because they'll be coming for this other one. But let's go with a specific product. If we know specifically about this particular product that we have predicted that 100 boxes of ice cream will be sold,
41:24 Speaker 1.: how
41:26 Speaker 2.: is that helpful? Yes, but this is related to cost. Yes. Okay, that's a great point. So if you say have ordered 500 boxes, but actually only 100 of them are going to be purchased on that day, that means that you have paid for, because the supermarkets pay for the supplier, right? 400 more.
42:10 Speaker 2.: As you said, the money is already there. You're not using it for something else. And what else? Where are these 400 boxes going to be? The storage space is the most valuable
42:26 Speaker 1.: resource for supermarkets.
42:28 Speaker 2.: Think about the freezer section. Tiny, tiny bits, right? And if you know, you're going to have... people buying more ice cream than, or more hokey pokey than chocolate, or if you know, actually if we don't have hokey pokey, they're still going to buy ice cream, so I'm not going to care about that. But you see, it helps them to make decisions, because if you have an empty space, so it's just as bad if you haven't ordered, like if you're going to have 100, but 50 have been ordered. What that means is that 50 customers will buy it, the other 50 would be annoyed but they also might not buy it right so we've lost revenue that we could have had and we have an empty space in the freezer sitting there empty not being used for something that people can actually buy so every time you see an empty space This is how I'm going to convert you to being a little bit crazy. Going into supermarkets and I always
43:32 Speaker 1.: look, where is the empty space? They got their
43:35 Speaker 2.: data wrong. That is actually wasting the space. Yes, yes, but if you at the same time have a space that's filled and nobody's using it, well, they're paying for that storage. For no reason. So this is why actually the certainty prediction models are very, very important. So you kind of have to think, okay, why is this an insight? How is it useful? So always try to think, what's the impact of that decision?
44:10 Speaker 2.: Okay, so here we come to, because we like to have models and frameworks, but this one actually, this is not in your book. You have slightly

different one in your book, but I thought this was a little bit easier to understand if you're beginning. So every time you have a data...
44:35 Speaker 1.: problem,
44:36 Speaker 2.: one way to break it down is to think about the steps that you go through where you
44:43 Speaker 1.: add value to data.
44:45 Speaker 2.: So the first thing that actually to add value to data is to provide some context. So for example, at the supermarket, if you have the post data for all the stores, all the products, all the time, it's just huge amounts of data, right? So when you provide a context and say, okay, I want to provide a context in terms of the type of ice cream and the shop, so that we say, okay, I want to see how much ice cream is being sold in this shop. over different days, then we're actually providing context in terms of the specific shop, the specific product, and then when we get that data within the context, it becomes more meaningful for that store, right? They can see what's happening. If they just see lots and lots of pause data, huge amounts, well, it's very difficult to make sense of it. So even by... structuring it, sorting it, extracting only bits of it, it becomes more meaningful. Now, that is still sorted information about
45:57 Speaker 2.: sales of ice cream in the store over a different time. But when it becomes knowledge is when you actually start to gain some understanding of what that means. You start to see some patterns. For example, you start seeing that the greatest amount is always on a Friday or it's always three o'clock. So for example, when we had one of the talks by foodstuffs, they said that actually
46:25 Speaker 2.: they have, I'll just go with one, they have one
46:28 Speaker 1.: product that
46:30 Speaker 2.: they always make sure to deliver, they time the delivery to
46:35 Speaker 1.: 10
46:37 Speaker 2.: minutes before 3 o'clock, and it's a huge amount, so you don't want that product to stay for a long time in the shop, and they say, okay, it goes before 4. What happens at 3 o'clock? On a work day. This might need you to sort of think a bit more because some of you haven't
47:01 Speaker 1.: been here. Schools.
47:03 Speaker 2.: Schools are let out. What do you think is the product? Yeah, it's not ice cream. If it was me, probably ice cream. But no, no. These kids are

just really not.
47:18 Speaker 1.: Drinks. Yeah, Coke.
47:21 Speaker 2.: Yeah, that's it. So they said the 1.5 liter bottles of Coke,
47:27 Speaker 1.: they just go,
47:29 Speaker 2.: they come and go. So hardly any storage required. So you can see how when you actually see in your information all the data for different products and you see, oh, okay, all of the sales were between.
47:49 Speaker 2.: 10 to 3 and 4 o'clock, we don't need to have it, then it becomes understanding, right? So it's the understanding that then leads to the decision, let's time the delivery to have it when it's needed. So you would need to understand that value chain
48:11 Speaker 1.: and
48:14 Speaker 2.: to be able to apply it in examples like the one we had with Woolworths.
48:20 Speaker 1.: So
48:20 Speaker 2.: that's the sort of exam question you could get. So you see, we've already done one exam question. And the reason why there are no model answers is because you could see already you had lots of different answers and they were all good. So it's not just one. Okay, so that's my promise that all the exam questions type things we are going to do in class. Now, obviously, I'm not going to give you the same problem. But let's see now how it works if you've been paying attention. So, actually, no. First, I wanted to give you an example with a different kind of data. Because with the supermarket, we were dealing with numbers,
49:13 Speaker 2.: right? Or if it's a text, it's structured, it's always a supermarket name and ID and so on. So it's all structured in a table. We can query it easily and all
49:27 Speaker 1.: good.
49:28 Speaker 2.: But we also have a different type of data for some reason that's in the next chapter. So we'll talk more about that next week. But I already wanted to to bring your attention to it, that a different kind of data for businesses is actually information they get from social media. So in this case, in this example, what happened was this company, it's an American company, and it's similar to what we get here, Finnish, the dishwasher tablets, and so they

produce this Dishwasher tablets, they call Tide Pots.
50:13 Speaker 2.: And they had
50:15 Speaker 1.: a bulk of
50:19 Speaker 2.: Instagram posts. So, these are all the Instagram posts for a year. From people who were, lots of space here, and one here, right? Opposite me in my line of sight. So from everybody who has put that they reside in the US, right? So you can imagine, this is like millions if not billions of posts because people post on Instagram for all sorts of things, right? They're not just looking for the detergent. So of course because
51:02 Speaker 1.: then
51:02 Speaker 2.: they put the context that they wanted
51:05 Speaker 1.: to only look at the
51:07 Speaker 2.: posts that had
51:09 Speaker 1.: mentioned their
51:11 Speaker 2.: Tide Pods. That makes sense, because that's what their company is. So, of course, now immediately we are... probably hugely reducing the number of posts, but there's still a lot, people might be mentioning it for all sorts of reasons. And so then they wanted to understand, so then the other context was that they wanted to understand how people felt about it. You know, was it largely positive, negative, neutral, why were they mentioning the pod? So the context was
51:50 Speaker 2.: that it
51:51 Speaker 1.: was
51:52 Speaker 2.: for the Tide Pods only, and then they wanted to understand the sentiment of customers, so that's how then it becomes an information, and then once they understand the sentiment, they want to use that for a marketing campaign.
52:18 Speaker 2.: Okay, so that was actually the knowledge they needed how can we use this information for marketing and Then of course the decision is what to put in the marketing campaign So this is not something for us to work together because we don't have time for you to look at the posts and they're actually a little bit old so it took me quite a bit of time to find them. But what was really interesting when they did the second part where they looked, what are the common themes, right? What do people talk about? So they first went

and looked, okay, we're going to look at negative and positive, right? If it's positive, you might say, okay, when people are positive, this is a text mining, you look whether they are happy or this was good.
53:17 Speaker 2.: washed well and so on, but when they were negative, they found one very common theme that they were not aware of. And what was happening was, and you probably can't see that very well, and I told you that this is a dishwasher tablet, but apparently a lot of kids, let
53:43 Speaker 2.: me just put that a
53:45 Speaker 1.: bit bigger.
53:48 Speaker 2.: Yes. Well done. If I had a chocolate, I would have given it to you. A virtual chocolate comes to you. So a lot of kids thought that this was a sweet candy. Yeah. So what happened is in all of the posts, they found that when people are talking about the Thai pots, they're also talking about candy because that, of course, is the American word. And they couldn't figure out.
54:17 Speaker 2.: how was that relevant? So they started looking and most of them were also negative because the kids were trying to eat the dishwasher tablets. So you can say, okay, that is real understanding of the sentiment, but also the problem. And so while you might say, okay, now the decision regarding the marketing campaign is to try and correct that. impression, right, about the candy. But you also might say, oh, actually, this could lead to a design decision to make the appearance of these parts different so that people don't
54:59 Speaker 1.: see it as a
55:01 Speaker 2.: candy. So even though their original idea was to think about how can they do a marketing campaign about that, well, they actually did a marketing campaign. They said, oh, well, you know. It's so good that it looks like candy, but it's not, or something like that. But the point is that it gives them a good idea about what people really relate to with their product. And it also made them realize that they had a problem that they didn't know they had. So... We already have gone through the information chain with the supermarket example. But what I wanted to show you with this one is that data is not always about numbers. Sometimes it's about feedback. It could be audio, visual, all kinds of things where we can get insights.
55:58 Speaker 1.: So.
55:59 Speaker 2.: This is now the bit where I check what you've been paying

attention to. So if you have Instagram or TikTok or whatever you fancy posts about the quality of a new
56:15 Speaker 1.: Toyota SUV, I
56:20 Speaker 2.: want you to go with what's the data, what's
56:24 Speaker 1.: the context.
56:26 Speaker 2.: What would be information, what would be knowledge, and what decision that could be related to? My PowerPoint slides are trying to rush me, but there you go. Also in this class, are we going to have a break? Or do you prefer to finish early? All right, every time I ask this question, there's a deep thought. Deep thought. All right, how many people would like to have a break now and for the rest of the term? Okay. How many people would like to finish early? Oh, this is a very close call. All right. Break again. And there are lots of undecided. So this is the time to join our side. Break. I'm actually counting it. Okay. And finishing early.
57:32 Speaker 1.: Wow.
57:36 Speaker 2.: All right. I'll try to do both, but let's have a break. We'll have a shorter break. We'll have a shorter break, and then we finish a little bit early. How about that? Five minutes, and think about your problem. I'd like to have breaks before I give you a task.
59:44 Speaker : Thank you. Thank you. So I would say it's probably better to make a point to this. Thank you.
01:00:39 Speaker 1.: Okay, so in this question, are the posts about the quality of the Toyota? No, no, any post. So they basically get active. So they could be good or bad? It could be anything.
01:00:54 Speaker 2.: It could be about Toyota or not. So any Instagram post for a period of time.
01:01:01 Speaker 1.: So then you obviously would be reducing
01:01:05 Speaker 2.: it to . And then you might say, okay, and the information is where we try to figure out the quality. So it's not the sentiment or the people's specific quality.
01:01:19 Speaker 1.: Sorry, what was your name? Right. So in
01:01:24 Speaker 2.: a similar way, I probably have shortened the question,
01:01:27 Speaker 1.: but in a similar way to the tight ports, you know,
01:01:32 Speaker 2.: where they just had the ground ports, right?

01:01:34 Speaker 1.: So they start off with a big bulk of those, and they take
01:01:40 Speaker 2.: out only the ones that have the
01:01:42 Speaker 1.: mention of the new model. And from there on, that's the context. Great to see people are using the
01:01:59 Speaker 2.: brake to actually do work.
01:02:00 Speaker 1.: I like that. Thank you.
01:02:41 Speaker : Thank you. Thank you.
01:03:35 Speaker : Thank you. All right.
01:03:55 Speaker 2.: Thanks to everybody who came to talk to me and didn't leave me bored during the break. I always appreciate it. And I really like, especially that you are talking about the pods and the supermarkets, because that's the whole point. That's how we... associate and remember things. So now let's talk about Toyota, yeah? So this would be a quick one because it's the same as what we've done already. So we, I have slightly shortened,
01:04:31 Speaker 1.: remember?
01:04:33 Speaker 2.: Shorter break. So I've slightly shortened the process, but actually providing a little bit of the context. So the data will be all the Instagram posts, right? For a period of time. But we already know this is about the specific model of Toyota SUV. So that is our context. the specific model. Now, how are we going to go about the information? What information are we going to get? So we've got all of the
01:05:11 Speaker 1.: posts
01:05:15 Speaker 2.: that
01:05:16 Speaker 1.: mention this model.
01:05:18 Speaker 2.: What information can we get that will give us insights, knowledge related to quality? So we might go with things people like or not like, which may be about the quality, might be about other things. So the first thing you would have to do, if you are the analyst in that situation, is to go to the business and say, what do you mean by quality? Because they might actually think only about... quality in terms of technical parameters, but they actually might mean about experiential quality, you know, what people really like.
01:06:05 Speaker 2.: Yes, yes, might be about comfort, experience, so this is important to set up front, because otherwise you might be doing some great work, and they're like, oh, actually we... Didn't mean that, we meant something

else. And I've seen quite a bit of that going on with our internship sometimes, and that's why I try to talk about it. So always make sure that you have these conversations, but really good. So the quality might be things that people really like or not like, because we'll say, okay, quality is about...
01:06:42 Speaker 2.: customer satisfaction and we are going to look at it or ask and make sure that you have the context right about what quality
01:06:55 Speaker 1.: they're looking for,
01:06:56 Speaker 2.: because sometimes they have narrower views like design or something like that. But we are gonna go with this one now, because you were first. So it's basically more or less about satisfaction. So we will then have, in terms of information, so you see how now we have put more context into it. The quality is the satisfaction, liking, not liking, and we're going to be able to
01:07:25 Speaker 2.: organize, as you said, strengths and weaknesses, or it might be, we might think it's a strength, but they don't like it. So actually, likes and dislikes, right? So it's perception.
01:07:45 Speaker 2.: Of the post, right? So we might actually look, yes, at the date of the post and see whether we have some
01:07:52 Speaker 1.: trends.
01:07:54 Speaker 2.: You know, did it start when, you know, do we have like phases? Is it related to a different event, for example? Might be related to the season, for example. If they really liked that it can drive in very bad weather, maybe that's specific to that season, and there is also better understanding. Right.
01:08:26 Speaker 2.: Don't fight. Everybody will have a chance. Yeah. Okay. Oh, well, that's another perfect descriptive thing that can be done. We can say, okay, now that we see
01:08:44 Speaker 1.: we
01:08:45 Speaker 2.: have these likes and dislikes, is there also mention of a competitor? And for both situations, because if they're liking something, they might say, oh, and this is so much better than Honda or Mazda or whatever else, or they might like it but say, well, but Honda is still better. So then you might say, okay, that would help us to identify for this particular model,
01:09:19 Speaker 2.: who are the competitors that are doing it better than us, or if they're saying, well, we are better than them, then that tells us which

competitors we're. So you see, then that's a different context, different decision, right? So what else could we get from dislikes and dislikes as information?
01:09:53 Speaker 2.: Okay, so this is a really good point. You see how before with the pods, we were told that it's about a marketing campaign. So we would be looking more for things that the
01:10:06 Speaker 1.: marketing guys would want to highlight.
01:10:08 Speaker 2.: But sorry, I keep forgetting to ask your name. So Hassan was saying that this could be about the engineering design. So then we are obviously, yes, we are going to look at the competitors because we know what their design is like, but we are also going to look for quality. liking or disliking of specific design issues, you know, how the dashboard presents or things specific to the car design that people like rather than general views.
01:10:48 Speaker 2.: Yeah, okay, so that's another point, that when you make a decision, you know, when you go to the design guys, you can say, okay, this model is actually already out. I mean, we understand that, but for the next model, or if they have something they really like, they might say, oh, we could introduce this in other vehicles, not just into the SUV. So you can see how there are lots of options where you can answer this in many different ways depending on the context. You know, is it the engineering design? You see how I left it here? I didn't say. So that you had. All the options. Anybody had other ideas not related to design?
01:11:36 Speaker 1.: Yeah, yeah.
01:11:48 Speaker 2.: Absolutely, and actually one of the things is, because we'll be doing this in this course, we'll be applying to different disciplines, marketing and so on, and I think sometimes there's misconception that marketing is all about advertising, but actually understanding the needs for design, getting into the designer teams, is also a function, so yeah, that's a really good point. Somebody else was like?
01:12:31 Speaker 2.: Okay, so I totally agree with you with some qualifications. So remember that, so the point that was made is we could look at the demographics and see whether we have a certain demographic that really likes it. So this could be our target customer, right?
01:12:52 Speaker 2.: And I totally agree with you, but I would probably, as a data analyst, would kind of verify this by also looking at the sales data, because

it could also be that certain demographics might be posting more to Instagram, because remember, and so that would give us a hypothesis or idea, and then we think, okay, we need to check this also with... some other data. So it's a great approach because then you know who your target audience is and no point targeting everybody because they're the ones who actually buy it and like it and post about it, right? So we have a good understanding of our customers. So that's a different type of understanding that would then
01:13:50 Speaker 2.: possibly say, what difference does it make who your target audience is? Well, first, you'll be targeting promotion messages to them. So for marketing, you'll be also designing to... because they're the ones who buy and post and so on, so very important part as well of the business. So you can see that depending on which decisions you link it to or which needs of the business you have, lots and lots of ideas. So if I've left it like that in the exam, then you can go with whatever ideas
01:14:29 Speaker 1.: you have.
01:14:31 Speaker 2.: So don't feel our people had lots of different ideas. Well, that's great. All of them were great.
01:14:51 Speaker 2.: Okay, so first of all, the exam question would never be worded like this because
01:14:57 Speaker 1.: it's just too open.
01:15:00 Speaker 2.: People could write a whole page and give lots of, right? So if this was an exam question, a good thing it's recorded because I would say something like,
01:15:11 Speaker 1.: considering
01:15:16 Speaker 2.: the information value chain, Describe each component, each element of the value chain, and for each element, provide an example in the context of Instagram post for
01:15:32 Speaker 1.: Toyota
01:15:34 Speaker 2.: SUV. So to now answer your question,
01:15:38 Speaker 1.: if
01:15:40 Speaker 2.: I say an example, then it's one example. That's all I ask. So every time somebody here gave an example, that was one answer. Please don't write more than you are asked to do. I don't like long answers. And I haven't done this in this course, but in open

01:16:05 Speaker 1.: book
01:16:06 Speaker 2.: tests, I actually limit the number of words. Because sometimes people just... You see how you have lots of ideas and you want to share all of them. And in class, that is great. In the exam, it's an exam. You have a limited time. Do what you need to do to get your marks and move on. Did that answer your question? Yeah, so it's never going to be, and actually it's a really good point, because I do say we are going to do exam type questions, but I'm a bit lazy at actually structuring, wording it exactly like an exam question.
01:16:48 Speaker 2.: So it's a little bit more open to. generates conversation, the exam question would be easier than this, because it would actually specify everything you have to do. And usually I have A,
01:17:02 Speaker 1.: B, C with marks,
01:17:05 Speaker 2.: so that you know how many marks you get. That also is usually quite a good sign about how much you are expected to write. If you've got five marks, it's not half an hour, is it? But we are also going to have a special session in the week of the test with a practice test, which
01:17:29 Speaker 1.: I
01:17:31 Speaker 2.: will release in advance. You all can write it, and I will tell you how well you're doing. So as I said, this is an introductory course. We are giving lots and lots of help and guidance. to set you up on the right path. So if, I do appreciate that some of you here thinking of careers in analytics and others are just taking this as an extra and might go more into technology management, but whichever person you're going to be, it's not possible, especially I think within the... year or two, AI is really coming, that really needs data, so analytics is going to be even bigger, so even if you are a decision maker, so for example, yesterday we were talking about something related to AI, and they needed to make a decision, and they said, well, we need the data for that,
01:18:41 Speaker 1.: and
01:18:44 Speaker 2.: They don't know what data they need, so they are talking to the analyst and say, to make this decision, what kind of data can we... get so you still have to understand enough that evidence is what supports decisions and then to kind of have some understanding of how you look at the evidence

and challenge it because another thing is when people are very passionate about data sometimes they present
01:19:15 Speaker 2.: their view of the information so for example you might present only what people liked about Toyota
01:19:22 Speaker 1.: but
01:19:23 Speaker 2.: maybe there was some really really strong dislikes and if we don't provide a balanced view so if if you as a decision maker see only the likes and say okay they like this so that's all good you don't think to ask oh but What are the things they didn't like and how many of them? Another thing I get all the time is somebody says, oh,
01:19:49 Speaker 1.: this
01:19:50 Speaker 2.: group of students are
01:19:53 Speaker 1.: really
01:19:54 Speaker 2.: interested in
01:19:55 Speaker 1.: X.
01:19:58 Speaker 2.: And I'm like, okay, how do you know? Oh, we talked to two of them and they were very interested. I said, oh, so two people from this group.
01:20:08 Speaker 1.: are very interested.
01:20:10 Speaker 2.: Does that tell us whether all the people in this huge group that has thousands of people is interested in X? Because if we do X, we can't do Y. So we have to make decisions. And they're like, oh, we haven't thought about it. I said, well,
01:20:25 Speaker 1.: please, please check with at least.
01:20:28 Speaker 2.: a representative sample, right? So sometimes decision makers take these anecdotal evidence, and sometimes some of us are louder about what we like, right? You might have... 100 posts and 20 of them are the same person, for example. So it's still just one person, right? They feel strongly about it. So we really need to pay attention as decision makers to what does the data actually tell us. And data scientists,
01:21:03 Speaker 2.: okay, I don't know whether any of you were in yesterday's class, but data scientists, sometimes they talk their own language. to talk about, if you're in that position, you like to talk about predictive diagnostic, this test, that test, that measure, that validity, and the decision makers, one, cannot understand it, you know, the R square, yesterday I thought was very funny, but

two, they think, okay, this is some kind of technical thing, so I don't have to worry about it, I'll leave it. them but actually what I'm trying to show you here is that there is technical part to the analysis
01:21:48 Speaker 1.: but actually it's a business it's a business
01:21:50 Speaker 2.: decision so very often the analysts are the ones they have to kind of
01:21:57 Speaker 1.: bridge
01:21:57 Speaker 2.: that conversation. And we had a very good example, which I'm not
01:22:03 Speaker 1.: sure if I put here, but a
01:22:05 Speaker 2.: couple of years ago, I was talking, because I do a lot of research in health, and I was talking to the people in the hospital, and they have really fancy analytics there. I mean, it's great fun to go and see. And I said, oh, you're so fortunate to have a whole team.
01:22:24 Speaker 1.: of
01:22:25 Speaker 2.: analytics guys and these dashboards that are live, dynamic. I was just in heaven looking at it. And they said, oh, yeah, but sometimes, you know, we just want to know what to do to get more beds. And we just don't have enough beds in admission. And I thought, oh, okay. This is a typical problem where, you know, the organization or business has a problem and it hasn't been translated into, well, this is the information that helps you with your problem. So this kind of translation, you might have great data, great analysis of it, but if you don't report it and explain, we have a problem. So here we come to
01:23:14 Speaker 2.: Again, a definition, but in the definition, what I was trying
01:23:18 Speaker 1.: to say here, because we already talked about decisions and
01:23:21 Speaker 2.: conclusions to address business questions, different disciplines, different fields have different needs. So what, for example, the accounting guys would need would be different from the marketing, and the analysts actually do have to have an understanding of these needs. So, for example,
01:23:43 Speaker 2.: For example, we had a talk by Carrie Jones, she was the analytics person in Woolworths, countdown at the time, and she said it took her six months to learn what the grocery business was about. You know, all these

things about what's important, what's the most valuable resource, like shelf space, how do they make this decision,
01:24:07 Speaker 1.: how do they deal with suppliers. Every
01:24:10 Speaker 2.: business is different, and that is why in the course we'll be having a lot of different applications so that you get
01:24:19 Speaker 1.: exposure to that too.
01:24:24 Speaker 2.: We come to, so we have the information value chain, and we come to the second sort of framework for today, and that's the SOAR model. I don't expect you to remember the exact words in the SOAR model, although you have cheat sheets so you can write it, but I do expect you to understand the idea behind it, because what I was talking about earlier is
01:24:52 Speaker 1.: about
01:24:53 Speaker 2.: that it's not enough to just analyze the data, is actually visualized and represented in this model. So let's have a look.
01:25:02 Speaker 1.: Oh,
01:25:03 Speaker 2.: the analytics mindset. This is what we've been told.
01:25:07 Speaker 1.: that if
01:25:08 Speaker 2.: you have the right mindset for analytical thinking, you have to be able to ask the right questions, we already talked about that, you have to actually be able to deal with data, so MB623
01:25:20 Speaker 1.: does
01:25:22 Speaker 2.: most of the data management, although we'll have a little bit in the second tutorial, and apply analytic techniques, which we'll be doing in this course, but interpreting share, present the results to the decision makers so they understand what they mean. So that is actually what the SOAR model is as well. The first one is to specify the question as we've been doing and the question needs to be
01:25:53 Speaker 1.: able to be addressed by data.
01:25:56 Speaker 2.: So for example, if you have a problem and say something like,
01:26:02 Speaker 1.: well,
01:26:04 Speaker 2.: should they be paying their staff more? Well, okay, good question, but now you have to tell me how we're going to know from the data, right? So what are the things that we are going to be looking at the data

01:26:19 Speaker 1.: to...
01:26:21 Speaker 2.: to address that business question. So we'll be doing a lot of this translation of business problems and questions into what data. So once you have that, you start thinking, okay, where is this data? For example, we had earlier
01:26:38 Speaker 1.: the
01:26:41 Speaker 2.: problem with countdown, right? And we said, okay, we want to know the the promotions list, the promotion prices, the system prices, when are they loaded, when
01:26:53 Speaker 1.: are they loaded. Where is this data, right?
01:26:56 Speaker 2.: Is it in the system? Is it in the promotion? Is it somewhere else? Do we have a business rule about it? So we need to... Yeah, this is what happens. I've done it to kind of speed me up, but it doesn't seem to work.
01:27:09 Speaker 1.: I'm still talking a lot.
01:27:11 Speaker 2.: So obtaining the data is about making sure you have it, but also to think where we can find this data. Because sometimes people just, oh, let's go into the system. But sometimes maybe, let's go to the students and ask them. They're also a source of data.
01:27:32 Speaker 2.: analyze the data which is what we've been doing in this course and report the results again we'll be doing that visualization so for today all we have focused on is specifying the question you'll be doing this in example in of that in the first tutorial and a couple of tasks in power bi to start you off very very simple
01:28:02 Speaker 2.: Please note, because somebody was asking me in the break,
01:28:05 Speaker 1.: you
01:28:07 Speaker 2.: can do the tutorials by yourself. You don't have to turn to the tutorials if you are quite
01:28:13 Speaker 1.: confident,
01:28:14 Speaker 2.: but Power BI really doesn't work on Macs. We've tried all sorts of things, so you either have to go to a lab, have a... Microsoft kind of machine or use the remote access. There is a remote access to university
01:28:32 Speaker 1.: computers.
01:28:35 Speaker 2.: So we've worked with this translating, but let's do one of them. How is the customer demographic profile, age, gender, income, changing

for typical customer shopping at McDonald's? So is
01:28:52 Speaker 1.: this
01:28:52 Speaker 2.: answerable by data? How are we going to find it out? Okay, so then you'll be getting the customers who use the app. we will be getting some information about the customer profile, but only if they use the app. So I don't know whether there are other ways of doing it. So typical customer shopping, so what I'll be asking then is, do we get data
01:29:31 Speaker 1.: about
01:29:33 Speaker 2.: the customer demographic in any other ways than from the app? And if it's only from the app, you have to acknowledge that we already have a certain demographic profile that uses the app. There are some people who don't use it. Now, if that is our target profile, what else do we have? In McDonald's, we also have the self-service stuff, right? So sometimes they could actually introduce a customer profile to encourage people to log in with it, maybe give them a discount if this is what they really wanted. But if they currently don't have it,
01:30:11 Speaker 2.: From the data in the app, we can say, we can check for each year what
01:30:19 Speaker 1.: are the groups in
01:30:20 Speaker 2.: terms of income. So you could have groups, you could have averages. Next week when we start talking about the specific techniques, you will see how
01:30:31 Speaker 1.: different analytic
01:30:33 Speaker 2.: results would give you a different result. For example, if I go with the
01:30:38 Speaker 1.: average age over the years.
01:30:41 Speaker 2.: I'll see, for example, in 2000, the average age was, say, 21,
01:30:50 Speaker 1.: something.
01:30:51 Speaker 2.: And now the average age is 18. So you might get a conclusion that your customers are getting younger. But if I was doing other techniques where I was doing quartiles or percentiles and say, actually the average is that, but the top one is actually seven. So even though the average, because there might be 80-year-old people coming to to shop or have a coffee,
01:31:24 Speaker 1.: actually

01:31:25 Speaker 2.: the majority of them are younger, and it's just the average overall is 21. So
01:31:32 Speaker 1.: different
01:31:33 Speaker 2.: analytic techniques, even descriptives, give you different insights. So it depends what you are after, and this is how you translate into, oh okay, this is in the data, but these are different ways I can get that data. That's where we get to
01:31:53 Speaker 1.: the
01:31:56 Speaker 2.: next one, which is obtaining, so from the app, because there was no other option, analyzing different analytics would give me different insight, and then you report the result and say, okay, on the whole, this is what the average is, but we can't make conclusions from the average, and here, look at these other ways to group our customers.
01:32:20 Speaker 2.: So one,
01:32:22 Speaker 1.: oh, okay.
01:32:24 Speaker 2.: I have actually 15 minutes. That's pretty good. Let's work on a scenario. So this is very common. I encounter this almost in every project that I work, even when it's not related to capacity. But people always want to hire more employees. That's what it is. We always need more people. People are always burned out. They're always stressed. Too much work. Okay, and managers always say, Yes, but the budget is constrained. It's a tough time in the business. We don't have money. And then all the different department reps have to push for getting more employees, right? So this is quite a typical problem. So in your department, the manager comes and asks you how you could make a business case. So how can you justify? How can you support your request for more people? So that's what the business case is, right? You want to justify the need for it. You need to provide evidence.
01:33:42 Speaker 1.: So the
01:33:45 Speaker 2.: business problem, insufficient resource to handle current workload. This is how it's presented. And so I want you to translate this problem into several,
01:33:59 Speaker 1.: at least two,
01:34:01 Speaker 2.: there's an exam type limit, so I would probably say at
01:34:06 Speaker 1.: least two, which means two in

01:34:08 Speaker 2.: the exam. But here, it means as many as you like, but minimum two. Questions that you
01:34:15 Speaker 1.: will ask from the
01:34:18 Speaker 2.: data to support this case. And then, say why the decision maker would care.
01:34:31 Speaker 1.: Because remember,
01:34:32 Speaker 2.: they need to also,
01:34:35 Speaker 1.: they
01:34:37 Speaker 2.: need to approve, so it has to be based on their interest. Is it clear what you have to do? Questions that you ask the data analyst need to look for in the data.
01:34:48 Speaker 1.: Say,
01:34:49 Speaker 2.: how are you going to find this data? And then, Why would the manager care about it? How long do you need for that?
01:35:01 Speaker 1.: Two, three,
01:35:06 Speaker 2.: well, five minutes, because that's what we have. Five minutes, and I'll be walking around
01:35:12 Speaker 1.: as a floating resource. Noise
01:35:22 Speaker 2.: level needs to be higher. I know you're tired.
01:37:07 Speaker : Thank you. Thank you. Thank you.
01:38:32 Speaker : Thank you. Thank you.
01:39:40 Speaker 2.: Thank you. All right. Next week, I'll definitely find a way to get to the back.
01:40:05 Speaker 1.: Because always
01:40:07 Speaker 2.: talking to the people in front of me, and they might stop sitting here. All right, so I don't want to influence in any way what you're going to say, although I heard some really interesting stuff. What do you think? How would you approach this?
01:40:29 Speaker 1.: This is problem. Yep.
01:40:39 Speaker 2.: Okay, so that is one way to go about it. So you can say, okay, how do we know what is a reasonable workload per person, right? Because we are told that the workload is too high. So what is the measure of a reasonable workload so that I can then say this workload when it is distributed to this number of people.

01:41:12 Speaker 2.: requires two more people, right? So that is one way of doing it. So depending on what it is, in some situations, it will be easier, for example, if this is a call center, you could have number of calls, different kind of calls, because some of them are more complicated,
01:41:31 Speaker 1.: how
01:41:33 Speaker 2.: long people are taking to resolve it, if it's the same category,
01:41:37 Speaker 1.: so lots
01:41:39 Speaker 2.: of information. But if it is, say, in the hospital,
01:41:47 Speaker 1.: you'll
01:41:49 Speaker 2.: be looking at, well, how many appointments, what were people presenting with, because currently we have these problems with emergency departments, right? So it depends. You might be coming with... blood all over you so that we have one problem, broken arm, it could be a different problem. So we need to categorize it and say, oh, okay, actually, it seems that for
01:42:23 Speaker 1.: this
01:42:25 Speaker 2.: specialty, say cardiology, we are getting 10 people per hour and They take 15 minutes per person minimum, so four people per hour is the
01:42:44 Speaker 1.: workload.
01:42:46 Speaker 2.: Okay, but then we had another question, which is do we decide on the reasonable workload based on how long people take on average? Or do we actually check when we have different people doing the same tasks, how long they take? Because then we might have a different problem. Some people who are taking longer... might need some training, or they may be more junior, so we can say, actually, for a junior person, this is
01:43:27 Speaker 1.: what the workload,
01:43:29 Speaker 2.: reasonable workload looks like, because they're still learning, and for this experience, guys, this is what the workload looks like, so when we are planning for it, so you can see how many questions we can have that are, They are there because in every job there are measurable tasks, and we can actually see. And one of our interns, was it two years ago, or I think first semester last year, had to, it was supposed to be an analytics, it was an analytics internship, but it was about the city council

01:44:09 Speaker 1.: service desks.
01:44:13 Speaker 2.: And they have two or three lines, and they were trying to decide whether they should have a dedicated line for one type of problem. And so they were trying to discover which type of problem takes
01:44:29 Speaker 1.: longer, so
01:44:31 Speaker 2.: they get a dedicated line, so they only deal with them. And so what... Well, they had this data was the person serving the customer. So, for example, what kind of problems you could have. You might be paying your rates, you might be paying for the dog licensing, you might have some complaint. So they're dealing with different categories of problems. And they have data because every time a person comes, the person at the service desk has to log in the start,
01:45:07 Speaker 1.: and
01:45:09 Speaker 2.: then when they finish, they log the end. But what they found was, and we'll talk about that next week, missing data. Because they were logging how long it takes them, but they didn't always type what the query was about. So some of the category information was missing because they only would log it if they had to do something. So for example, if the people are coming for advice or information, there was nothing for them to record, so
01:45:42 Speaker 1.: they...
01:45:42 Speaker 2.: so they didn't record it. So what they asked our students to do as part of the data collection, they had to go a few days to different locations and they had these clickers. and they had to actually time the tasks and write the categories, and then they extrapolated that to the bigger data to have the right way of replacing the missing data. So I thought that was a little bit of an interesting way of doing it, but it was more impartial because when you have to do it, you might not be quite... Right, so you see, that is, going back to your suggestion, that is one way where we can determine the workload, for example. What else are we going to ask in this situation?
01:46:42 Speaker 2.: So this is our department. Oh, so you mean, is it a group in the department or the whole department?
01:46:49 Speaker 1.: Yeah, maybe that's more related to the manufacturer department. Okay.
01:46:54 Speaker 2.: Okay, so yes, we could do that. Or even within the

department, as I was saying, emergency department, different specialties might actually... So again, if you're just looking at the average, it would be wrong. So we need to look at the different groups who are doing different tasks. So good point. What else might we be? So we have two questions already. The first question was how do we determine the reasonable workload? And by data, we said we can look at how long people take for different tasks on average or for each person. So this is And then yours was, yes, we can look how long they take, but for different types of roles. So that's the second question. Next, yes.
01:47:54 Speaker 2.: Oh, okay, so we could get data. So remember, we are data people, so we can say sick leave is recorded, so we can check whether, you know. how many people in terms of the full number of people every day, you know, do we have the full, so it might be sick leave, it might be annual leave, it might be anything, so to say, well, how many people we actually have on average, so yeah, good, third question.
01:48:28 Speaker 1.: So we can also ask, is it because of the infrastructure problem or the resource problem?
01:48:34 Speaker 2.: How are we going to find that in the data? If
01:48:36 Speaker 1.: the workload is not being managed, we can check on about if there is something missing in the infrastructure that we
01:48:42 Speaker 2.: can provide more. How are we going to check that in the data? Because remember, some of this is not data. It's a good one, but not. So think about that.
01:48:54 Speaker 1.: I think we can go to the finance also. Yes.
01:49:21 Speaker 1.: So if that is lower than the finance standard, then probably yes, they need to have more.
01:49:28 Speaker 2.: Right, so we could have, I'll just kind of summarize. So we could look at the financial data, and we could look in the financial data for many different things. For example, we could look at the revenue that departments generate versus the cost. sort of look at the ratio and say, okay,
01:49:53 Speaker 2.: you seem, in addition to the workload, you seem to be more deserving, for example. Or we could look similarly to the example with Woolworths. We can say, okay, when we are,
01:50:07 Speaker 1.: we
01:50:09 Speaker 2.: have too high a workload,

01:50:11 Speaker 1.: what's the impact?
01:50:13 Speaker 2.: What's the impact of that? For example,
01:50:16 Speaker 1.: you
01:50:17 Speaker 2.: know how you were asking about the sick leave and how many people are actually there to determine the workload? Well, we could also look at that to see whether we actually have increased sick leave because people are stressed, so that just compounds the problem. So that is also a cost. And we can also see, does this influence the performance? So that was kind of what you were saying, right? Because you can say, if we have that many people, we take this long to serve a customer, and this then costs us revenue.
01:50:54 Speaker 2.: So... financials are very important when you want to make a business case. You want to say, well, this is the cost of us not having an extra person, because very often we treat extra resource as a cost, but we don't think if we have that, if we put that extra cost, how much extra revenue we will have, which we may have or not. we might say, oh, okay, if we have two more people still the same revenue, then it's not a very strong case.
01:51:30 Speaker 2.: Another question I had here is if you look to see different people in the same role taking different average time to do a task, then we should be checking what can we do. Then we can say, okay, actually if everybody was taking that long, then we don't need extra people. So what we do need actually is to bring the skill up of everybody to be able to do it in. in that shorter time.
01:52:01 Speaker 2.: What would be the question that you would ask then, though? Because what does it assume? That everybody does an equally good job, right? So somebody might be very quick, but they might be very bad at it. And so maybe people keep coming back. So you need to challenge your assumption that the quality of work is the same no matter how long it take. So that is also a question to see and say, what's the performance of that person who takes 15 minutes and what is the performance of that person who takes half an hour?
01:52:37 Speaker 2.: Do you see how you can't help but make assumptions? Sometimes based on experience, sometimes prejudice. So as an analyst, you cannot afford to make assumptions. Well, actually, no, that's not true. We all make assumptions, but you have to be aware of them, and you have to

challenge them.
01:53:04 Speaker 2.: If such information is available, benchmarking is also very, very useful to see, okay, for similar, say if it's an emergency department, for other places in New Zealand, you know, how are they doing, what do they do, that's comparable, yeah. The turnover, right, that's another measure we can see.
01:53:43 Speaker 2.: Yes, yes, yes. That's what I said, the revenue. Oh, we're being kicked out.
01:53:47 Speaker 1.: Okay, so
01:53:48 Speaker 2.: I'll see you next week, but before I see you next week, you have the tutorial. So you'll see Emily on Tuesday, and I'll see you next week on Thursday. Thank you, guys.
01:54:17 Speaker 2.: You should just butt in any time. I am unstoppable.
01:54:24 Speaker 1.: And yesterday, I actually...
01:54:32 Speaker 2.: I'm very bad at this.

View original transcript at Tactiq.

MBIS624-26S1-LecA-Te-s1-low-2

Meeting started: Mar 23, 2026, 10:10:19 PM
Meeting duration: 111 minutes
Meeting participants: Speaker 1, Speaker 2, Speaker

View original transcript at Tactiq.

Transcript

00:00 Speaker 1.: All right, let's hope this works. This is quite noted.
00:21 Speaker 1.: Okay, guys, I'm always reluctant to interrupt, you know, happy noise. But let's start. So, okay, before we start, because I do appreciate that usually towards the end we are kind of rushing. So this is almost the end of week two. And hopefully you guys are all... finalized and settled, but are there any questions before we start? Anything that worries you, concerns you, not sure about?
01:03 Speaker 1.: Oh, very good question, because I actually was going to talk about that. So the situation with the textbook is that this is a textbook that was predetermined. And as much as possible, I'm trying to go with. the textbook.
01:26 Speaker 1.: But in the second half, as I had already mentioned, we have a visitor. And in my communications last week, it looks like not a lot of the textbook will be used. And so I would say...
01:43 Speaker 1.: Hold on to buying the book because it might not be as useful in the second half. And in the first half, I'll make sure, and in fact, I'm doing this today. I'm trying to put more information on the slides, which is not usually my style, so that you actually have the information that you absolutely need. So we do have also... So the book in the library available and for one of the chapters I will provide a handout of our photocopy because the copyright allows that. So that, you know, in case you don't buy the book, you have options. But really good question. Normally in courses we do say, well, you know, we're sticking with the book and the book is the reference.
02:42 Speaker 1.: Because of the uncertainty about what exactly is going to be

used in the second half, I would say it might not be as useful. So really good question. Anything else? You've had your first tutorial. Easy enough? How long did it take you to do it? Be honest. one hour okay that's that's good so because when so every tutorial gets tested with about four people beforehand and some of them said that wasn't enough you know, that we should put more. And I was like, well, you know, it's first week, people are, so, okay, so this is good, this is good. It's validating my decision. All right, so a little bit about today's class. So first of all, last week.
03:48 Speaker 1.: So as usual, the first two weeks, a lot of movement, people adding courses, dropping courses. So I try to keep it more or less. at a level where people can catch up, but today we are going to have our first, what I would call, well, actually last week we had one too. So anything that we apply to a particular real situation or a case is kind of a simulation of your exam questions. So, you know, participate as much as you can in those exercises because that's our practice and preparation for the exam. So last week we have covered two important elements.
04:44 Speaker 1.: that you need to focus on. The first one is we practiced a lot how you translate business problems to questions that can be addressed by data.
04:59 Speaker 1.: So taking in a general problem, I think we had the one with people being burned out. their workload too high and they needed more people. We also had the one with countdown. And so you have some business problems and then you have to think, well, okay, what are the questions? that I can address with data, that one, would help me to understand the problem better, and two, to see how it can be addressed based on the answers. So, and you have practiced a bit of that in the tutorials as well. So the second part is... And today, as I said, I am a strong believer in repetition, so we'll do a little bit of that too. The second part is we looked at the information value chain.
05:58 Speaker 1.: And it's not, I don't want you to remember it as a theoretical model, but more as a way that you move from having data, which is not processed, just lots and lots of transactions or facts, to something that actually has a meaning to an action, decision, or insights. So what, do I have a question? Brilliant. What are the elements of the information value chain?
06:35 Speaker 1.: Right data that was the easy one everybody who talks first has

it easy because they have a lot more choice so we start with data which is the facts we're collecting the data in a particular context so we had with countdown we had the context of a particular product that we're interested or a shop or with the What did we do at the end? With the workload, we had a particular context that we were looking to see whether somebody should be hired, so workload. So there's always some kind of context. So data in context creates some meaningful information because we will take some of the data that is relevant to that context. And then how does it move to? knowledge, how does it move? When we, from that information, when we get some understanding of the problem, for example, yes, now we understand what, how many people are overworked or actually how many people are taking much longer than the average to do the tasks or when do people buy ice cream? most often during the week or, you know, so we get some understanding that is relevant to the decision. And the decision is, well, should we stock the ice cream or not? When should we order it? What will be the delivery? Or should we hire more people or should we, oopsie.
08:20 Speaker 1.: Right. Breaking down equipment in the rooms, not helpful. Or should we actually train people so we, based on the knowledge and understanding, you could actually make the right decision, right, that is relevant.
08:39 Speaker 2.: So now,
08:42 Speaker 1.: in the exam, how would that potentially happen? Because I know you care about the exam, and so.
08:51 Speaker 2.: I have a question.
08:52 Speaker 1.: Yes.
08:54 Speaker 2.: So when we call knowledge, does it come from that specific data context or the knowledge, it can come from someone's previous experience? Okay,
09:07 Speaker 1.: so that is also a very good question. So the question was, does the knowledge come from, so the data in context is the information. Does it come from the information that we gain some understanding? Or does it also come from previous experience? And this is the problem with words like knowledge, right? So if you do our knowledge management class next year, you'll find that
09:38 Speaker 1.: There are so many ways that we look at knowledge. So

generally, I would say in terms of knowledge, yes, it comes from previous experience. In this context, the knowledge starts from the information we have because we're looking for evidence-based understanding and knowledge. But when people start making decisions, of course, they will also use some judgment, right?
10:04 Speaker 1.: based on experience, but the whole point of business data analytics is that you try to standardize and make things consistent, right? So if it's based on somebody's experience, then we are adding that extra component. So in the context of this topic and this information value chain, we're looking at knowledge as something that we gain from the evidence. Does that answer your question? So really, really good, thoughtful question. And of course, because you've all had some experience, you can understand that knowledge could mean lots of different things. So as I started saying,
10:50 Speaker 1.: Because we all are not that keen on exams, but we'll try to make the exams as least stressful as possible. And so in every class, I'll be telling you how this may come on the exam so that you know exactly. When you get to the exam and you get the questions, no surprise.
11:11 Speaker 1.: So... In terms of the information value chain or translating problems, the way the exam works is we are going to have a short scenario about one page similar to the scenario I'll be using today. It's from the book and I have
11:31 Speaker 2.: kind of
11:33 Speaker 1.: used some New Zealand information because the book is American and some of the things really not. that relevant. And so there will be a description of a situation and a place, so just like what we talked about with Countdown, for example, or today with the university. And from last week, there could be descriptions of a problem there, and you would have to say, okay, well, this is their problem, what kind of question can we ask to address this? problem?
12:12 Speaker 1.: How do we specify it so that it's addressed by data? And from today, we'll go on to the next stage where we say, okay, where do we get this data? What do we know about data they have? So in terms of the information value chain, after today, you'll be able to apply it easier because you would understand what data is there, what's the context. information and so on, decisions, so that could come

12:42 Speaker 2.: up.
12:45 Speaker 1.: What do you think, you know, when we talked about analytics versus data science,
12:53 Speaker 2.: and when
12:55 Speaker 1.: one year, I think about three years ago, we did a scrape of all the job adverts and looked for skills that people expect in analysts and, you know, skills that they expect in data scientists. And of course, there's quite a little bit of an overlap. What do you think is one really common skill that data analysts have with data scientists and what is something that is different?
13:31 Speaker 1.: based on what we talked about last week.
13:34 Speaker 2.: I think data analyst is something working with information from the data and transferring it to the business problems, solving the business problems.
13:46 Speaker 1.: Yeah, so data analysts are expected to have knowledge of the business as well, and that's why we are doing so much of this work on trying to apply to a particular... business context. Data scientists focus more on the analytics techniques, which we'll obviously also cover
14:10 Speaker 2.: in
14:12 Speaker 1.: more detail, and they tend to speak kind of a sciency language, so analysts need to be able to speak to the scientists, but also speak to the business. we've had a lot of feedback from internships on this and we are going to work together on making sure that you guys are able to understand business and talk to them in a way that they will understand so that you don't say things like Oh, today I am going to normalize your data and your data is this and that because they just switch off. So you have to be able to explain a problem the way they understand it. Yes?
15:13 Speaker 1.: No, not at all the same. So
15:15 Speaker 2.: what's the difference, the major difference between these two?
15:18 Speaker 1.: The major difference is that with a business analyst, well again, unfortunately in the ads sometimes they don't really... but the main difference is if you're a business systems analyst, you're really looking at designing and developing technology solutions that may or may not be specifically based on the data. The data is part of it, but there is more to it. There are business processes, there is a technology solution that you have to

introduce in place, and you're looking at how that system will support the business. A data analyst would be focusing on what do we need to do with the data to help the business analyst to create this solution. So I will tell you for example right now we have a major transformation in the university to try and integrate all the systems
16:16 Speaker 2.: and
16:18 Speaker 1.: there is a solution in place to support the main business processes that are not core but all the main support processes. like services, human resource, finance, and so on. And what happens is the business analysts looked at what the processes are and how the solution that we're going to invest in is going to support the processes. So that is all good. There is a reasonable
16:52 Speaker 1.: fit to requirements, but what we had not considered is how ready our data is to actually start migrating to the solution, and what we'll have to do with the data, and so at the moment, we are intensively going through what we are gonna talk about today, which is actually data preparation and cleansing.
17:22 Speaker 1.: would focus on the data and what we need to do with it and what's the format so it's detailed. If you're a business analyst, you should have understanding. of data as well, that's why this is kind of a common course for everybody. But you should also understand processes, systems analysis, requirements, managing projects, and so on. So a business analyst is more than, data analyst is a narrower specialty, if you like.
18:02 Speaker 1.: And at some point, one thing I have to say is, Because we've had a lot of students who've said things like, oh, I'm never going to do that or I'm never going to do that. And it's famous last words. The reality of it is that we have to side shift into different skills and jobs all the time. And especially now with what's happening with AI, one day you might be a business analyst and the next day you have to do something with data because you have an AI enterprise solution coming into it.
18:37 Speaker 1.: We need to really work on the T element and and having us as much as possible in your portfolio of skills But yes good question any other anything else Because you know this is I know this is a starting course so important to get the base, right? So the plan for today is to actually think about

the next stage after we have translated problems into questions, actually figuring out, well, first, where is the data that we need? What kind of data are we going to need and where are we going to get it? And then what do we have to do with it to be able to analyze it?
19:27 Speaker 2.: Uh...
19:29 Speaker 1.: I want to once again say in MB623, you'll be doing this in a lot more depth. But if you're not taking MB623, at least you have to understand that there is something going on before we analyze data. Because one of the early feedbacks we had from employers was, well, your students seem to expect that we are going to give them a lovely, perfect, clean data set, and then they start doing their test to it. So now we have introduced this week where at least you are aware and you are going to have a simple tutorial next week. So at least you are aware that data doesn't come prepackaged like in a supermarket with everything ready, with the labels and formats all fixed up for you. And in fact... If you are going to go for a data analyst job, be prepared that at the beginning everybody has to go through
20:30 Speaker 2.: basically
20:32 Speaker 1.: doing the data cleaning jobs. Because that is the job that nobody likes to do, so they always give it to the last person who is employed, no matter what your position is. So, you know, try to treat that as a, as a fun thing to do, and actually a little bit like investigation, looking for bugs. So in the old days, for business analysts, coding was a requirement. It no longer is, and it would be less and less required with AI. But say 10 years ago, for you to be a business analyst, there was an expectation that you would be able to do some coding.
21:19 Speaker 1.: And a lot of graduates, even if they were employed as analysts, had to do the equivalent of data cleaning, which is testing, you know, finding bugs in people's solutions, and they're like, oh, this is a bit tedious. So in the same way here, you have to look for problems with the data or preparing it. And so today we'll talk about what that is.
21:44 Speaker 1.: But as I said, in a lot more detail, you'll be doing that in the data management course. So just a quick recap of where we are. So we talked about the specify and now obtain the data. So within the specified question, just so that you understand how important it is that you very clear with the

question, I'm not sure if I talk, did I talk about this this week, last week, because we kind of finished with an exercise. Did we talk about different kinds of analytics?
22:23 Speaker 1.: No, okay. Because lots of people looking like not sure. Okay, I thought I didn't get to this bit. So depending on how you word your questions, then it's easy to figure out whether...
22:43 Speaker 1.: This is a candidate for descriptive analytics, which I'm sure I talked about the difference. Also in the old times called business intelligence, hence we have business intelligence in our title. And descriptive and diagnostic analytics are based only on
23:04 Speaker 2.: current
23:06 Speaker 1.: and historical information. So it's based on actual information that we have. So any conclusion. that we make based on that information are conclusions. based on real data, not likelihood, not something that we're predicting in the future. And that is why in terms of what we'll be doing in the first term, all of the questions that you have to be addressed with data would be kind of in the format of what happened in the past or what is happening right now with something or why did it happen. So looking for some kind of patterns or consistencies that give us an indication what could be the reason for some outcome that has happened already. So not something we're predicting. In terms of predictive and prescriptive, this is what you'll be doing in the second half. So based on the data we have from the past, we kind of try to figure out some trends. or forecast what is likely to happen. And those models are all probabilistic because they're based on likelihoods. While the ones we'll be looking at in terms of descriptive,
24:33 Speaker 2.: they're all actual
24:36 Speaker 1.: information that we have. So for obtaining data, we first need to think Well, what data do we actually need? Which I haven't put on this list. But, you know, when you have a question and you think, okay, in order to be able to address this question or to get more understanding, remember the information and understanding, what data are we going to need? And then... The next question that leads from what data we need is, or what data do we have available? Because that is not the same.
25:20 Speaker 1.: And sometimes you might have to,

25:23 Speaker 2.: as
25:26 Speaker 1.: you probably saw in the tutorials, you might have to kind of...
25:31 Speaker 1.: iteratively go back to your question and say, well, actually, we are not able to collect more data. This is the data we have, so we might have to change the question based on what data we have. But, alternatively, if we really, really need to stick with that question, we can say, well, what data do we need to collect, and where are we going to get it? Where is the source? So is there a source where we can purchase it from a third party, like last week we talked about purchasing social media posts, for example, if you wanted feedback from customers, or would we have to actually collect it directly from the source, which is very, very time-consuming and expensive if you had to actually send surveys to customers yourself.
26:28 Speaker 1.: And especially if there are decisions related to collection, you need to think if we have this data, will it actually
26:36 Speaker 2.: adequately
26:39 Speaker 1.: answer the question we had? When they get it, what would be the understanding we get? Would it be enough to make a decision? And I really want to stress this because sometimes in classes, and even though this is kind of a more introductory class, still postgraduate, sometimes in classes we have these linear ways of going through it, right? First you do that, then you do that, then you do that, and life is wonderful. But the reality of it is, first you do that, then you go look at your data, then maybe you go back to your question, then you think, oh, is it worth collecting all of this data? Because this is real life. Everything costs money and time. So you kind of have to think about that.
27:25 Speaker 1.: Is, is it, now the next part which we'll actually focus a lot on today is, is the data of adequate quality. So today we're mostly going to focus on what is dirty data, what is good data, and how can we clean data. What is the equivalent? of cleaning in data analysis. And look for biases, which is
27:59 Speaker 2.: really
28:01 Speaker 1.: important. You know, do we have some kind of disbalance in our data that might be creating a bias because we have oversampling of one type of representative. So all of these important questions to start.
28:21 Speaker 1.: because, and I know most of you are not going to do research projects, but in universities we do a lot of research, it's always based on data,

and sometimes you get the result that you didn't expect, and you think, oh. I, especially if it's a quantitative positivist, you create a hypothesis and you want to prove it. And then it comes back and it's like, oh, that's not what we expect it to get. There's no significant relationship. And sometimes then people start looking at their data realizing, oh, we have a lot of outliers or our sample is really not balanced. The majority of people are within this age range. So, of course, Of course, we're ignoring this other age. And so, yes, you could do that at the end, but much better to do it at the beginning.
29:15 Speaker 1.: So we always let people get on with it so that they learn from their experience and the next time they'll probably look at their data first. But in terms of data analysis, you don't have this kind of luxury like in universities to go back. So please look at your data.
29:38 Speaker 1.: So let's think, let's go back to the grocery example. We're going to real basics now just to figure out what data looks like and what kind of data you can have. And then we'll talk about the preparation. So this is a really quick thinking pause. It's not a major exercise. So let's think about a customer who is buying something. And let's move on from ice cream to, say, cheese. okay so if they're buying it online what data will be required so how many people have purchased anything online this is just testing you whether you're listening because I know everybody has bought something online you know it's not possible that you haven't right so then you should know what data has been required from you and Possibly some data that you didn't realize that it was collected from you. And then think about what data will be stored. Because sometimes some data is required but not saved. And sometimes we have data that is permanently stored. And what data is stored about the product?
31:00 Speaker 1.: And where is that data stored? So some of that might be dependent on what you've done in other courses, so I'll help with that. So let's start first with what data will be required from the customer, because that's the easy one. We all have done it. What data is required? Name, right. Address, because they'll have to deliver it. Anything else? Oh, okay, okay, right. My multiprocessing failed. So...
31:44 Speaker 1.: The product that you would like, right? What else? Contact number, so some other contact details, either number or email. Was it the same? That's the problem when I can't hear chorus. Sometimes demographics

like gender, sometimes they ask about age. Payment method. And when you do the payment method, they will actually ask for your actual credit card details.
32:27 Speaker 2.: So tracking data?
32:29 Speaker 1.: Like your actual... Yes, your previous... Yes, so
32:37 Speaker 2.: this
32:39 Speaker 1.: now is slightly dependent on whether you've ever registered with them before or you're a new... customer just signing in. But as soon as you register once, yes, they will have not only your purchasing history from every time you've ordered or abandoned, even things that you've put in the cut and then abandoned it. And I know some people actually
33:12 Speaker 1.: use that as a way of getting a discount because when you put it in the cart and leave it for a bit, they usually offer you a bit of a discount. But this is all safe, but I think what you were alluding to also is that they're also tracking your... behavior online that goes beyond the actual ordering or putting something. So how long you have looked at a product, what things you have searched for, all of this is being and that would go to your profile. So you can see how some of the data that is stored and collected is not necessarily the data that you consciously and intentionally enter when they ask you for it. Some of it is your behavior online.
34:14 Speaker 1.: And so that is all the streaming, stream data that you get.
34:22 Speaker 2.: What, and
34:24 Speaker 1.: that is all stored to your profile, your account. What about the product? So you said when you're ordering something, then you actually select the product. And what information then we would have about the product? So the quantity, okay, so there are several quantities here, right? First, the quantity you want to purchase, and then the quantity that is available in the inventory. What else? Product number, so always unique identifiers. Type, like... category dairy or whatever they have, which actually would be something. This is really important in the preparation of data because they might have it, in fact, when we had a presentation by food staffs, they said some things were in very kind of broad categories like dairy and some are more granular in other databases. So when you put it together, integration problems.
35:36 Speaker 1.: price of the product and that would be the standard price and any promotion that might apply. Absolutely. So definitely product weight

because that's very important also when you are in the supermarket and you're scanning it. It checks that you've only had one piece and not more.
36:08 Speaker 1.: Oh, yes, yes. So we kind of talked, it's a type of product ID, yes. Yeah, so there will be some description. So, you know, we talked about the category, which will be dairy, but then you have to say cheese, but even that is not clear enough. So it might be the type, EDEM, COBE, whatever. So...
36:43 Speaker 1.: Yes, so there would be a best before date, and hopefully they check with the delivery date, whether it will be past the date, I hope so, yes. I have to test that one, that is something I will test. Yeah, so we also have images, right? Because when you're ordering, you see the image of the product. And in the inventory database, there would also be the supplier because there's more than one EDMT there. You know, there are all kinds of different suppliers.
37:27 Speaker 1.: So a tracking number for your order. Yeah, so that's not specifically for the product, but yes, it is related to your order. So what I'm trying to get from this is that you can see we have two, I'm trying to stay away from terminology, but two entities, two. two things, right, the customer and the product, and each one of them will have a lot of attributes, so characteristics, something that characterize that, and we collect data, so data is always related to characteristics of that entity, and in this particular case,
38:10 Speaker 1.: we can say, okay, some of them overlap, right? Because the product that the customer is ordering also is about this product and its quantity in the inventory, and we have to have the information, and then you are talking about the best before date and the shipping date, so they relate to each other. So I'll talk a little bit about where data is stored. in a minute, but what data will be changed when the transaction is completed based on what we talked about? Some of the data will stay the same, so for example, well, unless you've changed your name, address, contact details, right, this will stay the same. What will be changed in the customer profile once the transaction is completed?
39:11 Speaker 1.: The cart, what would be reset about it? Yeah. So for example, it might be loyalty points that you'll get from that. So I was kind of looking for the specific data. So on your profile or card, because we like to think about the physical, if there is loyalty points that will be increased, your activity list or what you've purchased, which was what we were talking about, this will be added to pass.

39:49 Speaker 1.: So all the information about their transaction will be added to your profile anything else
39:55 Speaker 2.: so
39:57 Speaker 1.: so any Okay, any what we call insights about you if they were calculating your average spent or so on that will be automatically They're recalculated, but that's kind of more information so Let's talk just about the facts, which was the data. Yeah, so anything else? Oh, okay. After, of course, if you were a new customer, then they'll be saving everything, right? Because that's the first time. And somebody said order status. So, yes, you might have something like in progress for the status until it's completed. And so in this example, this is an example of what we always consider to be data. It's fairly, all of. these characteristics are something that we can put in a table just like what we had in the tutorials and it's what we're used to
41:11 Speaker 1.: And it's being stored in a type of database, right? So it's a collection of all of these facts. And all of these facts can be categorized. We did that. So we have labels for them. And then we would have different instances or records. So every row would be. a record and every column would be that attribute or field right so something that characterizes that one
41:44 Speaker 2.: that
41:46 Speaker 1.: each record of that and and of course we are also going to have a metadata that describes what we are going to have so that is why it's called self-describing collection and data can consist of numbers, words, images. So for example, you talked about the image of the cheese, and that is put in the database, and as long as you are able to label it and provide some description of it, so that it can be searched and so on, it is still... structured data, because it's structured, because we can order it, we can put it into tables. And
42:35 Speaker 2.: there
42:39 Speaker 1.: is some hierarchy, as I said, mostly you'll be dealing with that in your data management, but I really wanted you to understand this, because when we get to this part,
42:53 Speaker 2.: in
42:54 Speaker 1.: terms of data analytics, it's important to distinguish between structured and unstructured data, because most of the time what we have worked with has been structured data like this, but actually only about 20% of

enterprise data is structured.
43:17 Speaker 1.: 80% of data is actually not structured. So we keep working on these databases, and of course we try to structure data that is based to core support processes, but we also have a lot of unstructured data.
43:41 Speaker 1.: So you see estimated to be 80% and it requires different ways of
43:48 Speaker 2.: analyzing
43:50 Speaker 1.: it. So you hopefully talk about tax mining next term. It requires more storage. huge advances on that with AI in terms of analysis of unstructured data, and that's where it actually is strong. So we are going to focus a lot on structured data, but we'll also have some sessions on unstructured because it's important. So social media, all of these texts or even within organizations, things like emails, for example. Emails have a lot of information. And if they have to be analyzed, and very often with regulations, for example, we are asked to provide copies of emails.
44:48 Speaker 1.: I mean, I got close to 400 yesterday, you know. what do we do with them yeah because everybody likes to talk to me so um so you know how do we store it because it's a requirement and what do we do with them how long do we keep them for what um what information we might get from them that we can't get from somewhere else. So it's really untapped resource that can be very valuable. So for example, if you have customer posts or customer queries, online, emails they send to service providers, all of that would be unstructured data. So,
45:40 Speaker 1.: We've talked about different data types, so we're clear, and more importantly, we are clear on what structured data looks like. We can order it nicely. We have attributes. And now we need to think about where do we find this data. So if this is a relatively large organization or business, there would be some kind of integrated data source, hopefully, right? But you can see even the university is still working on it. And we're apparently one of the larger employers in Christchurch. So you can see that it's still a challenge. So we may have integrated
46:28 Speaker 2.: information
46:31 Speaker 1.: systems. Have you talked about that in 601? Do we know what an integrated system is? So it's basically a source that it's one single source of

information that many different systems can. many different functions can refer to. But we might actually have separate systems, which is actually what the university has. So even if you have an enterprise resource planning system, which
47:04 Speaker 1.: originally was intended to integrate data, when you buy single modules of different providers, then your data is still not integrated. So to give you an example, if we have an HR module from one provider and a payroll module from another provider, what that means is that For HR to know everybody, for example, say that you decide to do some tutoring, right?
47:38 Speaker 1.: So then you get a contract. And we have all the information about you, your position, your... terms, in terms of how much you get paid, how long you will be doing the tutoring, how many hours per week. So all of that will be in the HR. And then when it actually comes to, for you to get paid.
48:03 Speaker 1.: because that's very important, right? It has to go to payroll. And they have to know, well, who are you? Should they be paying you? What's the maximum they should be paying you based on the contract? And when this information is in two different places, we either have to find a way for them to talk to each other or a different way of integrating it because it's in two different
48:27 Speaker 2.: sources.
48:30 Speaker 1.: This is important because we are talking about data preparation today and most of the time the issue
48:37 Speaker 2.: is
48:39 Speaker 1.: related to integration of different sources so you need to think well okay this information we have in our HR ERP but this information we actually have in the payroll and this information we have into in our customer relationship management and what format it has, how it is presented, which attributes they're storing, because they might not be storing all of them, and they will have missing values. And what happens with, as soon as you get to small, medium enterprises, they they are not able to afford these integrated systems. So sometimes they have their own bespoke databases. They now have some of the cloud database providers holding all of their data.
49:33 Speaker 2.: Or,
49:34 Speaker 1.: dare I say, a lot of people use spreadsheets, too. That's why we still use Excel, because when I ask our employers, including, we just had

somebody finishing an internship into what I thought fairly large employer, I'm not going to mention, and yes, they have a lot of spreadsheets. And then what happens is every branch, because most – you can imagine New Zealand is large in terms of geographic location, but not so large in each branch. And every branch will have their own spreadsheet, and it's just – it's always a challenge with data.
50:22 Speaker 1.: So, so these are the internal sources, different kinds, and I'm talking about obviously structured data at the moment because you can see all the internal ones that I have discussed. Although in the CRM you might have unstructured nodes and also for example in service-based organizations I have found lots of people actually keeping nodes in Word files or in teams folders, all kinds of unstructured data is there as well. But we might actually have the data that we need in external data sources that may be public or may be private.
51:10 Speaker 1.: It needs to be purchased. And so we need to think, for example, I already mentioned social media posts and data. But it could be, for example, government agencies collect a lot of data. This was in the case of universities, for example, because there are public organizations. There's a lot of data that is integrated publicly available. And you might have annual reports, financial statements from any listed companies because if they're listed, it's their duty to report financial data to their investors and stakeholders. So there's wealth of information there which is not structured and actually quite hard to find sometimes. You have to go through pages and pages those reports are in the hundreds of pages sometimes and you have to look for the information and sometimes you have more structured summaries or financial data some of the economists foundations provide that data aggregated over a whole industry or
52:32 Speaker 2.: over a
52:34 Speaker 1.: whole country, so there are quite a lot of databases like that too. So we have internal and external sources. And this is just, as I said, because I was aware that you might or might not have the This is the picture from the book, because I like pictures.
52:57 Speaker 2.: And
52:58 Speaker 1.: it more or less provides exactly what I have been talking about. So you've got your integrated systems that actually also include maps of your processes and might be collecting data on how the process is. performed.

So for example, I am hopeful at the moment that
53:23 Speaker 2.: our new
53:26 Speaker 1.: system with course approvals is collecting the data about what's going on, that we need so many approvals, things, you know, some of you have been having issues, delays, even though we've done our bit, so hopefully that will be analyzed later and we can see, oh, this is where the hold up. has been so when it's automated you could actually so all of that data here this is kind of what this shows is the process is also being collected Then we have the external sources, and then we have, oh, the IoT devices. So any smart devices that you have, you would also be collecting data that is, that is available. So, you know, in, for example, in a smart environment at home, you would have smart bridges.
54:30 Speaker 1.: And apparently there is now AI to tell you what to not order anymore based on how long your broccoli stays in your fridge. So they're actually collecting data, but at work we also have a lot of devices that are smart and collect data all the time as well. So this is our large
55:01 Speaker 1.: picture of data sources. I will finish with the data types, and then we have the break. I remember that this class has a break.
55:10 Speaker 2.: So don't
55:13 Speaker 1.: worry about it. All right, so in the structure data types, we talked about the fact that we can have text or we can have numbers. And because In technology, in academia, we don't like things to be simple. We don't want to just talk about text or numbers, right? That would be too simple. So we have actual names for these types of data. So you could have numerical data, which is actually numbers. So that's
55:47 Speaker 2.: fairly
55:49 Speaker 1.: obvious. So we've talked about all of these numbers. that you can be collecting, and then you can have actually data that is, it's not just that it is text, but it's a type of text. So it's data, because it's structured, it's about categories. So at some kind of data that is represented by text, but it's showing that it's a category of one thing. So for example, say you have
56:27 Speaker 1.: well, the most common one would be if you are trying to categorize people according to age, and there would be categories, right? Ranges that. If it's a range, it's a different kind, but if it is something like what

we do with the generation, right? This is Gen Z, this is Gen X, this is, right? Then it becomes a category. But you also have things like gender, which would be, you know.
57:06 Speaker 1.: how should I put it, a limited number of categories. And they're well defined. So we know we have these categories. And you might have this in terms of transactions, so if you have transactions that are put in the system, you might say, okay, this is a sale transaction, this is a return transaction, this is a compensation transaction, you know, compensating somebody for a bad experience. So there would be all of these different transactions and what happens is if we haven't thought about all the categories
57:43 Speaker 1.: And then all of a sudden we have to pay a customer. We think about which category do we put this in. That's where things also are difficult. So we need to define what are all the possible scenarios that happen. And so one example is, it's quite a few years back now, but.
58:06 Speaker 1.: It was very, very widely publicized. All the schools in New Zealand had a new system which was called NovoPay. And they, some of you, right. So they forgot that there was a category for casual teachers relief. And so they couldn't be paid because there was nowhere to put that. payment.
58:37 Speaker 1.: So you might think this is quite trivial to categorize your data, but if you don't have the structure, it becomes a serious problem. So we need to think about that. So categorical data, as I said, if you have categories that simply divide your data into types that are well defined, but there is no
59:05 Speaker 1.: implied superiority or ranking, right? So if you, if you say, well, we have sales return compensation, or we have male, female, other, you know, these are just categories, none of them is better than the other, right? So then that is called nominal. But sometimes we have categories that actually are different in the way that one has more privileges than the other. So there's some kind of a implied ranking. So when you have things like status with, with airlines, everybody can get it. If you are a gold, there are certain rights to you. they attach to that category that most of us at the lower level bronze don't have. And so then that is called ordinal data. And the reason it's called ordinal is because there's some kind of order, right? Packing order. One goes higher than the other.
01:00:19 Speaker 1.: And in terms, now, do you need to know this for the exam?

Not really, I'm not gonna ask. I mean, this is the sort of thing that we ask undergraduates, but I feel like if you have to talk to people and they start talking to you about ordinal and nominal data and you say you've done analytics, you should know. So I'm not going to test it, but it's important that we talk about this. So in terms of numerical data, again, you know, think about the ranges sometimes that you can set, and these are intervals. So it's not a single number, it's a range of numbers, and so that is called interval. So for example, very, very common when you have ranges of age. So those, instead of putting a single category, they'll say, oh, are you 5 to 18? Because that group is something that we're interested in, 18 to 25. So when you put things in a range, it's interval data. Also very common with something that you really care about, which is your grade. Right, so your letter grade is attached to, uh, an interval, um, which I always have to remind myself, so, because we have the tables with the grade, so we, uh, will mark based on a particular number, but then we look at, um, the range that's attached, uh, to a grade. And, um, a ratio data is always
01:02:04 Speaker 1.: a number that refers to what we call absolute zero. So, you know, if you have
01:02:11 Speaker 2.: a
01:02:13 Speaker 1.: specific weight or specific number, like your salary, and of course you have a zero, which is none, and then your number is referred to it. That's why it's called ratio. So I think that is all I really wanted to talk about, the data types. This is just obviously an example
01:02:35 Speaker 2.: where you
01:02:40 Speaker 1.: could figure out immediately which one is numerical. So even if it's represented by these stars, it's still a number. So, but here, I thought that one is quite interesting because we actually don't know what
01:02:58 Speaker 2.: stands
01:03:00 Speaker 1.: behind it. When they say in-stock or not, I actually thought, oh, this would be a binary yes, no. But then I thought, oh, okay, what they had put in your book is, yes, it could be yes or no, which I thought was the... logical thing. But they're saying, oh, it could be a rule that simply translates an actual number into, yes, it's a positive translate to in stock or
01:03:30 Speaker 2.: zero
01:03:32 Speaker 1.: not in stock. So it could be just a rule that translates an

actual number. So we don't know what this represents. Could be either. All right, let's have Well, you're entitled to 10 minutes, but let's try and keep it a bit less than that, because getting better value. Time for a break. Move. Get up. Get the blood going. Come and talk to me. Ask me questions.
01:05:36 Speaker : Thank you. Thank you.
01:06:44 Speaker : Thank you. Thank you.
01:07:32 Speaker : Thank you. Thank you. Thank you.
01:08:51 Speaker : Thank you. Thank you. A lot of people would be doing it in their own time, and they're okay with that.
01:09:41 Speaker : Thank you. Thank you. Thanks. Thank you.
01:11:04 Speaker : Thank you. Thank you. Thank you.
01:12:27 Speaker : Thank you. Thank
01:12:52 Speaker 2.: you. Thank you.
01:13:50 Speaker 1.: All right, let's start. Breaks are great because it allows us to catch up. It's the only time I get to talk to individuals, and after the class. But this class from this week, I have another class right after, so just letting you know that they will be running back. Okay, so we've talked about data sources. So where we get the data and we talked about what kind of data, right? So now I want to focus on when we do get the data, how do we prepare it for analysis? What are the tasks and what are some of the issues we might have and how do we deal with these issues? So
01:14:47 Speaker 2.: in
01:14:49 Speaker 1.: order to,
01:14:50 Speaker 2.: so
01:14:51 Speaker 1.: What do we mean by preparing it? So first of all, we want to prepare it so that it is of adequate quality. It's good enough that it could, because you know, they say dirty data, dirty solution, right? And so we want to know that all the data is of the right type and of good quality, and we'll talk a little bit more about what quality means.
01:15:20 Speaker 1.: minutes and then we need to make sure that it's complete and that it is actually taken from the right source and that nobody has tampered with it, they haven't done anything with it, so for example, if you've been watching the news this morning and last night, they were talking about, well, it's actually a security incident to a private medical provider, but what

they've done is they've actually tampered with the records, so now some people have the wrong medication in their records and lots and lots of issues, so that is no longer data. that has integrity because we can't trust that it's correct and it's not from the original source anymore so then if this is from the right source and
01:16:22 Speaker 1.: is the right data, we still have to clean it, cleanse it, and so I'll have a separate slide on what dirty data is and how we cleanse it, and finally, preliminary exploratory analysis, which we might have to do next week by the looks of it, we'll see how we go. So, real world data is not clean. It's not consistent. It's not aligned depending on the data sources. Every source would have had different structure, different way of presenting it. And when you put it together, if it's not consistent, we have big problems. So when you first start,
01:17:13 Speaker 2.: always
01:17:14 Speaker 1.: assume that your data is dirty. Even if they have told you that it's not. trust but verify so always look at your data as if it is not prepared check it out and so what are we checking for so I'll go back to the functions but what are we checking for we are checking for missing data
01:17:44 Speaker 1.: So you know how we talked about all the elements that you have, all the attributes to a customer, for example, and sometimes, depending on how the data is collected, people have options to put data or not, right? So they might have not entered some of the data, so then you have an empty field, and so that will be missing data, but sometimes, actually,
01:18:15 Speaker 1.: We get missing data because somebody has verified it, and the people, when they enter the data, have put something completely wrong. So, for example, you've asked them for their profession, and they've just put XYZ or something. Because that happens, because if you don't have good validation rules in the collection, that's what happens. And so if it's been, you know, at some point if it's been validated and they've taken it
01:18:52 Speaker 2.: out, then
01:18:54 Speaker 1.: it becomes missing.
01:18:56 Speaker 2.: Similarly,
01:18:57 Speaker 1.: one of the projects we had, what we had was where they originally were entering addresses manually. And then they introduced kind of automated find an address. process. So you know sometimes when you have to enter your address when you register for things, it doesn't allow you to

manually enter your address and start searching for it. And if your address is not on the list,
01:19:35 Speaker 2.: it's
01:19:36 Speaker 1.: a way of verifying it and making sure it's right. But in this case, This was introduced halfway through, so a lot of the data was manual addresses, and some of them were actually incorrectly entered, they were incomplete. So this is not missing, but they were incomplete, which is, for example, depending on the fields you have, for example, they might have a field for the number of the house and the street, but they didn't put the suburb. or the city, so they put some address there, but it was incomplete, so that would lead to one of the fields being blank, empty, and we have a missing field. So what do we do with missing data?
01:20:29 Speaker 2.: What do
01:20:30 Speaker 1.: we do? If something is missing, how do we go about it? In the situation with the addresses, for example, you may have another source that actually holds your customer. details that might have the detail correctly, and then you would have to obviously somehow consolidate it to make sure that you complete the missing details from another source. And you'll be doing that next week in your tutorial. Another way might be if... if the fields that you have for the address can actually be searched in this database for finding address, and there's only one address that matches it, we could maybe assume that, yeah, okay, this is a place in Rickerton, so we'll put it there.
01:21:30 Speaker 2.: So
01:21:31 Speaker 1.: sometimes you do have other sources that would help you to complete the missing data. But sometimes you don't, and people just haven't, they just don't want to give you the information, right? So
01:21:43 Speaker 2.: what
01:21:44 Speaker 1.: do we do with this missing data? What do you think? Say, for example, you've asked people for their age, and they have chosen not to give it. What are we going to do about those records? So you're going to take them out completely, delete them. Okay, so yes, in general, there are two ways to start, right? We can delete all the records that have missing data, or we can keep them and try to replace the missing data with something. So there are lots of different ways. we can decide on replacing the missing data. It used to be

means, very bad idea. But sometimes you try to calculate something that would make no difference. Or you just put a symbol for missing data and your algorithms can deal with it. But if you decide to eliminate it, you really need to look at well, how important is that field for your analytics? For example, you know, and what proportion of your records have missing data for the age? So say, for example, 50% of them have chosen not to give you the age, but they've given you everything else that is very valuable. Deleting it is going to lose you very valuable information. So you need to... and this is why in your book I like how they call it art not science because there's never a single answer you have to it depends on your context and it depends on how many people have chosen not to
01:23:54 Speaker 2.: to
01:23:56 Speaker 1.: enter it so so the the first port the first point is to try and actually complete the records finding other sources But if that's not possible, then you need to evaluate how important these records are, what is the proportion that has missing records, and then there will be different ways to deal with missing values. They either could be you just... use a symbol for the missing value that would immediately spot it and then in your algorithm it will be able to use that as a missing value or have a calculation that would represent the missing value so what does it depend on? It depends on the proportion of records that have the missing value. It depends on the proportion of fields within the record that has a missing value. So for example, if that person, if you had 12 attributes for a record and they've only filled two, well, that's not a very valuable record then, right? So you probably would want to clean it out. So sometimes people start filling stuff and and they don't bother and it gets saved.
01:25:18 Speaker 1.: So you might as well clean out the ones that have a lot of missing
01:25:23 Speaker 2.: values.
01:25:25 Speaker 1.: But if they're only missing one field and the other information, the data there is still valuable, then you have to have a way to keep it and make a note that it's a missing field. Now, what about inconsistent data? So, inconsistent data is the most common issue. I talked about the formats already. So, what could be inconsistent formats?

01:25:57 Speaker 1.: So, for example, you might have,
01:26:01 Speaker 2.: what's
01:26:04 Speaker 1.: a really common one? I've seen in project. The dates, absolutely, so the American format and other formats, very common. Another very common thing is how we treat unique identifiers. Right? So, or you called it PCU, PQ, you know, all the unique identifiers. So something like customer ID, your student number is a unique identifier because only you have that number. Right? So lots of people might have theoretically the same name as you, but only you are going to have that number. Okay, so how do we treat these numbers?
01:26:52 Speaker 1.: what happens? It's a number, right? So this is why we talked about data and numerical and so on. So if you put the format that it's a number there, what happens? Yes, you have a question?
01:27:27 Speaker 1.: Yeah, I wouldn't use the mobile number as a unique identifier because... you should be generating your unique identifiers in your data because then you are not dependent on the person to provide you with the unique identifier. And you might also want to think how important the mobile number is for your survey. And if it is important, I would probably say put that at the beginning so if they don't give it, they don't fill the survey at all. Then they don't get paid.
01:28:03 Speaker 1.: So, you know, there, I mean, we can talk about these things because this is really common issues, but I definitely wouldn't use a mobile number as a unique identifier. So unique identifiers are usually, it's up to you to decide how it's being generated. It's of sufficient range that you could have the number of responses that. that you expect, and that you are never going to run out of numbers, and remember a mobile number might be unique at that point, but they might change their mobile number, and then your whole record is dead, right? So I would definitely not use that. It should be something that you as the data designer has generated for that record.
01:28:51 Speaker 1.: But a lot of the times, there are numbers. So it's easier when you have a combination of letters and numbers. People
01:29:03 Speaker 2.: don't
01:29:04 Speaker 1.: mix it up because then they realize it's a text. But when it's like your student numbers, which are actual numbers, one really common issue

is that they're put as a format that is numerical. And you might think, oh, well, I can enter it as a number, and it's a number, and whether I call it a number or a text, what difference does it make? But what if you have a situation where you have what we call a leading zero? If it's a number, the zero would be deleted immediately.

01:29:42 Speaker 1.: And the same thing would happen actually to your mobile number. They all have zeros in front. Gone. So when you have a format where one is a number and one is a text, even if you don't have issues with... being inaccurate with the leading zero, when you try to put together something that looks like a number. but one is set in one source as a number and in the other one is set as a text, well, it's not going to want to merge. So we have inconsistent formats. Another common thing specifically now when people are trying to integrate things into common sources is that some of the legacy systems have formats that have been created very long time ago. So they might have a number of characters set out. It is actually much smaller than what the newer one. here so maybe one sets for the product ID six characters as the limit and the other one has 15 and so when you put them together it creates a problem and it depends which one you are merging into right so if you want to merge 12 into six problems, and when you merge the 6 into 12, it might look like it's good, but it would add zeros or something like that, so it's just as bad. Or when we had one project with the student management system, what turned out to be a problem was when it was first created, we never had semesters.

01:31:24 Speaker 1.: And so all of a sudden, we couldn't have the same course code for two different semesters. And, of course, we offer some courses in both semesters, and we actually had to, for a period of time, we had to have a different number, and people kept trying to enroll in this other course, even though it was the same, and it was just crazy stuff. So you need to think about formats in terms of what is allowed.

01:31:56 Speaker 1.: what is the meaning and future proof it in terms of the number in terms of allowing for combinations with letters so if you need to have s1 s2 as as we didn't have at the time it was just a number so we had to have one to three

01:32:15 Speaker 2.: which

01:32:17 Speaker 1.: was our level one course in semester one and one to four which was the same course but in semester two until they fixed it. So you can see how it's got very serious consequences and much better to figure it out. So yes, it could be different data formats. It could be conflicting formats in terms of the type of characters allowed, the number, the type of data. text versus number. So some of these are not obvious. So for example, the date format for American versus other is kind of obvious if you have a really big number, right? So
01:33:02 Speaker 2.: if,
01:33:05 Speaker 1.: but some of these with the numbers versus text are not obvious and you actually do have to check it. And in one of the assignments we had before, we had it and a lot of people missed it out. So bear that in mind if you get your assignments. We might have something like that put there.
01:33:29 Speaker 2.: So...
01:33:30 Speaker 1.: Another thing that could happen, and I'm not sure how many of you with the new Microsoft Outlook stuff have this issue, but it merged two of my identities, the work and other, and then all of a sudden I had all of these duplicate.
01:33:52 Speaker 1.: entries in terms of names but they actually had different details in them and so some of them would have work numbers and some of them would have personal, right? Because sometimes you know somebody at work, but you are also a friend, so you have their personal number in your friends list, and you have the work number in the work list, and it very helpfully said, do you want me to delete the duplicate ones? And I almost said yes.
01:34:23 Speaker 1.: And then I decided to have a look, and I was like, no, no, you can't delete them. You actually have to think about it and merge them rather than delete. So what do we do with duplicate identical entries? So it might be identical in terms of some of the things, but we need to check to make sure that we're not going to miss some information by deleting duplicate entries.
01:34:50 Speaker 1.: And remember when you have, because in your tutorials, And, you know, we, we are being very nice. Your data sets are small. You can actually go through your records. In reality, if you have 10,000 records, you are not going to go through each record checking what's there, right? So you have

to have a way, and I think we are doing this in one of the tubes, to, to check and look up, you know, if you do have duplicates, extract them, see what the detail is there, be able to manipulate it. So I'm sure you're doing a lot more of that in 623. But
01:35:31 Speaker 2.: try
01:35:32 Speaker 1.: not to go with the first thought. Ah, there are duplicate, let's just delete them. Or whatever chat GPT tells you.
01:36:01 Speaker 2.: But let's say you can't guess it and it's critical information about someone's ID or something that you can't guess,
01:36:08 Speaker 1.: then what should
01:36:09 Speaker 2.: you do? Okay,
01:36:12 Speaker 1.: so the question was,
01:36:14 Speaker 2.: if
01:36:15 Speaker 1.: you have, well, it depends on... each of these categories, but I'll, so if you have a record where you have missing or inaccurate data and you can't delete it because there's something important there. Okay, so if it's missing, as I said, there are ways of handling missing data, so you can, you can
01:36:39 Speaker 1.: Basically put a sign that you have attached to missing data So it's it's very easily and all the algorithms will ask you how you have denoted missing data, so It used to be 999, but no more we we have better ways of dealing with that now if it is inaccurate
01:36:58 Speaker 2.: Well,
01:37:00 Speaker 1.: we do have to correct it So we have to go to the source and actually correct it. We have to look for other sources to get that information correct. Because if you're saying this is an important record, say it's a customer, and the address is incorrect, how do we get the correct address?
01:37:21 Speaker 2.: Well, you
01:37:24 Speaker 1.: could contact the customer, but if you have millions and millions, yes, that will be very time consuming. So we... kind of have to decide how important that is. And we either go to the source to collect it, or we make it a missing one. But we're not keeping inaccurate records. So, for example,
01:37:46 Speaker 2.: say,
01:37:49 Speaker 1.: In the loyalty cards, I've noticed because I am very bad and I never put accurate information in loyalty cards. And now when I use it, it

keeps reminding me that my address seems to be out of date. And if I give the address, they are going to give me extra points because they really want to have my real address. But I'm like, yeah, okay, but I don't want your points and I'm not going to give you. So sometimes you have to deal with it. So you basically. put it as a missing and or you try to get the information and see how much it is worth to you so they're giving me points not enough so we'll see maybe they'll offer me something that would eventually make me give the address so that's why it's It depends. So you said, oh, it is valuable. If it is valuable, the rest of the information is accurate. I would just delete the inaccurate if I don't have the accurate and treat it as a missing value.
01:38:53 Speaker 2.: That's the first instance.
01:38:58 Speaker 1.: So how do we know if data is inaccurate? That actually is really hard to know. Data does get outdated, for example, addresses, as I mentioned,
01:39:12 Speaker 2.: they
01:39:14 Speaker 1.: change, and people don't update them, or you were talking about mobile numbers, they also would change, people don't update them, and very often there's no way of knowing. So,
01:39:31 Speaker 2.: some
01:39:33 Speaker 1.: of the errors, can be found based on different rules. For example, what is reasonable for the value in a particular field. So, for example, there was a really fun item in the news about somebody who became the richest person overnight. Did you see that one? Because they got a coffee card worth. trillions by, but unfortunately they can only use that coffee card for that coffee shop, but they could purchase up to trillions. So somebody just had a heavy finger on the zeros and gave that person. So, so sometimes you have errors, there are human errors.
01:40:21 Speaker 1.: used to happen a lot in banks. We used to have people getting more money or overdrafts, like we had one person getting nine million in their account and they decided to skip the country, move to Hong Kong, and they were searching for them. So, of course, now the banks are much better at having these reasonableness validation. So you could find errors in your data based on querying it for is this a reasonable range? But if it is reasonable but inaccurate,

01:40:56 Speaker 2.: sometimes
01:40:58 Speaker 1.: it's really hard to find. But at least we should attempt to query and find things that look like they're out of date. But... Another way of doing it is if we have two sources of truth, we can compare and see whether the new address where products have been delivered to the customer, you know, is the same as what we had in the old. So when you're merging it, you can decide which one is actually the up to date based on the most current transactions. But it's never going to be exactly perfect, right? because that's the point of it. We have to make our best
01:41:43 Speaker 2.: effort now.
01:41:47 Speaker 1.: I have to leave the actual exam question for next. So I'm pretty much finished. We have six minutes. So I'll just go through some important pre-processing steps. So in addition to cleansing your data, you also have to think about how do we consolidate? How do we integrate? So if we have to change some formats so that both...
01:42:15 Speaker 1.: Data sources can be merged, we need to make these. And then we talk about data transformation and reduction. So in terms of reduction,
01:42:28 Speaker 2.: one
01:42:32 Speaker 1.: thing that is important to think about is for your particular problems and what you need, do we need to have all of the attributes? Because we might be collecting a lot of information, but for your particular decision, you might not need some of these attributes and you might want to reduce the dimensions and take some of the attributes out. And that may actually also solve some of your problems in terms of missing data. Because you might say, well, actually,
01:43:06 Speaker 2.: I
01:43:08 Speaker 1.: don't care about the age for this particular problem, so I can take that out of my data set. So these are some decisions to be made very early on. Variable selection is based on your questions. So we'll definitely talk about this next week, but... All of these, a lot of these fields are variables. These are ways to store data that would vary according to the instance or the record. So for example, you have a variable for price, right? Price would vary. And you need to select variables that are informative to what you are trying to learn. So that was actually the exercise I want to do today, which we'll do next week. But

you need to think, what are we trying to have a better understanding of? For example, you might say, we are trying to have a better understanding for the demand of ice cream. What would be a variable, what would be the information we store that would show us the demand for ice cream?
01:44:24 Speaker 1.: What's the information that would show us that?
01:44:27 Speaker 2.: The sales,
01:44:30 Speaker 1.: right? So we can say quantity of sales, that would be our variable, and then you can decide, well, what would be our aggregation? Would it be per day? Would it be per hour? Would it be for day of the week? So then you decide what would be... the information, then you can say, okay, actually, I want to know for the last month, but I also want to have other variables like shop name, because that might make a difference. Day of the week would be another variable that might be related. So things that you, based on your... Who was talking about previous knowledge, right? So some business knowledge you have or thing that you think would be related will be creating as your
01:45:18 Speaker 2.: variables.
01:45:23 Speaker 1.: Yeah, so location, which would be the shop name, right? Because the shop name would be related to location. So you can see how you create your variables depending on the question you... You answered, that's why I spend so much time talking about the questions. Then you need to prepare your data in terms of sampling. I already alluded to the problem with biased or unbalanced data set. If your data set, for example, has information from one store only, one location, or 80% from one store and 20% from another, or you know, something is overrepresented. Or for example, we had one situation where
01:46:11 Speaker 2.: we,
01:46:14 Speaker 1.: one of our Postgraduate students had collected data, but 90% of that was from people 19 to 25, because clearly went to students, right? Now, nothing wrong with asking students, but then you're over-representing one part of the population. So if 80% of your data set is dead, and then 20 is another, you have skewedness or disbalance. So to balance it out, you might need to... reduce the records from the 80%, so you have more, you know, more 50-50 or 40-60, so that you have a more balanced record. So these are all parts of preparing

01:47:02 Speaker 2.: and
01:47:05 Speaker 1.: sampling. You know, if you've oversampled one population, then you need to reduce that to balance it. So this is where we are going to finish as a concluding and for next week I will leave the slide for you to think on the actual case.
01:47:29 Speaker 2.: And you
01:47:31 Speaker 1.: might do some research, if you like, in the meantime. But the quick thinking pause, based on what we've discussed, what makes data good quality data? What are the characteristics of that? We have covered it. We just need to wrap it up. Start.
01:47:52 Speaker 1.: Complete. Relevant. Reliable. Trustworthy. Reliable. Consistent.
01:48:10 Speaker 2.: With
01:48:11 Speaker 1.: the consistent. Accurate. timely, because no point having it two months late, right? So, okay, so I'm fairly convinced that you know it, so here we are, we've got it,
01:48:28 Speaker 2.: the
01:48:29 Speaker 1.: only thing, that we didn't talk is uniqueness, right? So, but we did talk about the way to make sure that we have unique representation. All right, we'll finish here, and what we'll do next week is actually this exercise.
01:48:49 Speaker 2.: I
01:48:50 Speaker 1.: will put the slides, I have put the slides on. I want you to think about the The problem is a very easy one for everybody to relate because you all want to complete, right? So no completion is called attrition, right? So this is a problem for universities. So we want to think within this problem, what is the data? What will be the variables? What are the data sources? All of these things we talked about today will apply to this context. Thank you very much. Oh, only two. minutes late. No problem. This is available on Learn. No need to take photos. Available. It's posted in the lectures folder.
01:49:37 Speaker 1.: Also somebody asked me where the tutorial handouts are. They're in the tutorials folder of Learn. I will have to dash. Next class. So if you want to chat, we have to run.
01:50:07 Speaker 2.: Okay,
01:50:08 Speaker 1.: I'll see it after four minutes. I have to.

01:50:33 Speaker : Thank you. Thank you.
01:51:25 Speaker 1.: I've got about 72 students, so if you don't mind. Keep going, thank you.

View original transcript at Tactiq.

MBIS624-26S1-LecA-Te-s1-low-3

Meeting started: Mar 23, 2026, 10:10:26 PM
Meeting duration: 114 minutes
Meeting participants: Speaker, Speaker 2, Speaker 1

View original transcript at Tactiq.

Transcript

00:00 Speaker : Good morning. Oh, I see you have the limes down today. Thank you.
02:37 Speaker 2.: All right, time to start. I
02:39 Speaker 1.: just feel like, is this the best we can do? Ah, much better. On Learn? Yeah. Let's have a look I
03:04 Speaker 2.: might have kept them hidden sometimes I do that
04:10 Speaker 2.: Sometimes I do a few tweaks to them before but usually a couple of days at least beforehand I unhide them So do remind me that sometimes because I can see them sometimes I don't love to see that you can't
04:31 Speaker 2.: All right, so today for some of you would be a very easy session and for some of you might be a little bit more informative because
04:47 Speaker 1.: this is the introductory
04:49 Speaker 2.: course we do need today. We
04:51 Speaker 1.: are going to talk about...
04:54 Speaker 2.: what is descriptive analytics why we need it so in the sense of intelligence we will talk about how they bring some intelligence but also and this is the part that some of you might find easy and then you can answer all the questions we will talk about some of the statistical measures
05:17 Speaker 2.: that are used in descriptive analytics to make sure that everybody
05:22 Speaker 1.: has got the base, especially
05:25 Speaker 2.: if you're doing 6-3-4 in the second semester. And if you're not, everybody really should know about these basic measures. So let's quickly

remind ourselves about everything we did last week. So last week was a bit more loaded with different topics. So we continued with the SOAR model, but moved on from specifying the question to obtaining the data. And by obtaining data, we talked about not just where you get it, which is the sources of
06:10 Speaker 1.: data, data, but also what
06:16 Speaker 2.: kind of data you can get from different sources and what can
06:20 Speaker 1.: we do with it. And with that,
06:23 Speaker 2.: a lot of challenges around how we put together data from different sources. And I know you've done a
06:31 Speaker 1.: relatively simple exercise, but actually
06:33 Speaker 2.: this week I had quite a few questions on the tutorial, which already shows us that Even though it's basic, it makes you think about how to do some of these tasks. And then we also, I'm not quite sure, I think we finished that slide that I gave you as a homework for the university, yeah? Thank you. I thought that was the case. So we also talked about data quality and you actually came up with most of the categories by yourselves and how we clean the data to ensure the quality before you can do any analytics. So,
07:20 Speaker 2.: The plan for today is just to talk about exploratory descriptive analytics. So when you have prepared your data, you sometimes start an exploration to understand it. So there are two kinds of understanding. There is the understanding of
07:39 Speaker 1.: how
07:40 Speaker 2.: balanced or how good your data is, which was the end part of the preparation of your data, because you might realize that your data is somehow biased through this exploratory analytics. So we'll talk a little bit about that today. And then you would need to do something about that. So this is one type of understanding. The second type of understanding about your data is what does it tell you about whatever business problem or organizational problem you have. So that's the intelligence part. It starts providing us with some insight. when you, when you look at the current state. So just a quick recap on the data quality.
08:30 Speaker 2.: These, these were the aspects of quality, so making sure that the data is accurate, complete, timely, consistent in terms of formats and different data sources, stores, it's valid, so it matches rules for reasonableness,

and it's from the right source, and then we have a single view.
08:59 Speaker 1.: All
09:00 Speaker 2.: of these are still really big challenges, so no matter how much we talk about improvements of
09:07 Speaker 1.: tools, The data itself is still a problem
09:14 Speaker 2.: because a lot of the times it's been collected over time in lots of different stores It's inconsistent And so any I have not encountered any organizations and in fact there was a statistic From a couple of years ago where of course itself reported
09:38 Speaker 1.: So we have to always challenge self
09:40 Speaker 2.: -reporting, but only 9% of the businesses said that their data is clean.
09:48 Speaker 1.: And
09:49 Speaker 2.: of course, that simply
09:50 Speaker 1.: means that they thought that.
09:51 Speaker 2.: So today, you see, in terms of the preparation, we'll talk about some of these preliminary exploratory analysis to evaluate your data. but also seeing
10:05 Speaker 1.: what data you have to
10:07 Speaker 2.: answer your initial question. Remember, because we specified a question. So do we have all of it and is it in a good shape to be able to answer this question? And this is where we want to start. I want you to think so in this context of this problem, which is very common for all universities. And the reason I selected was two. One, it's just something we can all relate to. because we're all part of a university, right? The second thing is that in that document, there is some actual New Zealand information, so we could, after that, have a look and discuss, but please don't look to start, because we're trying to maybe do better.
10:54 Speaker 2.: And the third reason is that it is of similar... level of difficulty as to what you might get
11:08 Speaker 1.: in
11:10 Speaker 2.: the exam, in the test, because we call everything test now. So I know I've been getting emails already about all the courses and about the tests, and so in this course it's going to be, I still don't have a date, waiting for

booking, but most likely on Friday afternoon of week six. At least that's what I've requested, because that's when I'm free. And it's very important that
11:39 Speaker 1.: I
11:42 Speaker 2.: am there in the test, because if you've got any questions, it's a little bit of a support. And in the test, I think I promised a cheat sheet, didn't I? Yeah, so one sheet. You can start making notes. And what you get is a description, for example,
12:04 Speaker 1.: us with a university.
12:06 Speaker 2.: with more detail on the problems the universities face and how they're being measured in terms of performance and different types of students they get, mature students, you know, how the students get in because
12:22 Speaker 1.: some
12:24 Speaker 2.: of you might not be aware of the New Zealand system. So, for example, some students get through NCEA results, which is the school leaving results, and some of them would come. when they're older than 25, then they don't have to have the exams. And of course, we have international students like a lot of you who would have to demonstrate equivalency, you know, either the age or some kind of exam results.
12:56 Speaker 2.: One of the big success measures for universities is completion. Now, it's not completion in terms of a course, although of course we care about that too. It's completion in terms of a qualification because it's really not that beneficial if you've done a few courses but you were planning to do a degree and somehow you didn't manage it. So that's considered to be
13:27 Speaker 1.: not
13:29 Speaker 2.: a good performance. Of course there could be lots of different reasons why people don't complete. For example, I'm probably an outlier but I have over the years of working here done quite a few courses in the university and because you're not allowed to take a single course for interest you always have to potted towards a qualification. So I have a very heavy pointed but not completed degree, BA degree, because basically just doing lots of different language and philosophy courses for fun. So, so I mean. The assumption in this measure is obviously that people, students come in to complete a qualification, right? So, and the measure is the rate of completion or the rate of drop-off. And so what I want you to... now in a team, because that's where we ended last, and

we will be going through today through this example for the different parts. So for this part, all I want you to think about is what are some questions, as many as you can generate, that we can answer with data that will give us some insights into this problem
14:54 Speaker 1.: of
14:55 Speaker 2.: attrition and potential reasons or understanding of the situation around it, right? So anything you think might be relevant as data, we don't have to have the answers, just what data we will need to understand this problem better and to
15:21 Speaker 2.: Think about supporting the goal of retention. What data do we need? What, where are we going to get this data? So at this point, don't feel in any way constrained. Just think, oh, if I could know about something, this is what I would like to know. And then we'll think, where do we get this data? And finally, when you do say where we get this data, then we'll see what challenges are related to that. Is the question, are the questions clear?
15:56 Speaker 2.: So three questions, two minutes each, six minutes. You can start now. You can talk to each other and you can talk to me.
16:59 Speaker 2.: No, it's all right. I was just going to get it out.
17:03 Speaker 1.: Give you more space.
17:32 Speaker 2.: Well, you have to come up with some questions,
17:36 Speaker 1.: but the goal of the
17:39 Speaker 2.: university is obviously to improve retention, or to reduce attrition. This is the goal. So now you have to think about, to support this goal, what are some questions we can ask?
18:01 Speaker 2.: So that's a predictive kind of question. There is something probabilistic, right, that we want to predict. So at this point, we are looking at the current. So descriptive is about what is happening right now or what has happened. So
18:18 Speaker 1.: what do we look at in
18:20 Speaker 2.: the current and historic data about what has happened? Because everybody likes to do predictions, but actually we have... Okay, so I don't always recap everything, but I just want to
18:42 Speaker 1.: quickly, because this was, I think, in week one
18:48 Speaker 2.: when I talked about the different kinds. So descriptive

analytics is about current and historical data, and we look at what has been happening on what... what is happening, so we're not predicting, it's not probabilistic, it's certain. We're looking for any information about the current or past situation. And the other question I had was, what is the question? So remember,
19:20 Speaker 1.: I
19:22 Speaker 2.: give you my problem. My problem is that I need to
19:29 Speaker 1.: ensure student
19:32 Speaker 2.: retention,
19:33 Speaker 1.: right?
19:35 Speaker 2.: This is my problem. This is how the business people come to you. They don't give you the data questions. And I'm asking you to generate the questions. You know, what can we ask to figure out the problem and see what might be things that are related to it?
20:08 Speaker 2.: Well, try not to look at the study. No, it's on the problem because that's what they did. But you might do something better or different. We'll have a look at the study later and see whether they were useful or not.
20:56 Speaker 2.: I see
20:56 Speaker 1.: some people are ready.
20:59 Speaker 2.: So if I've come to you and said, you know, one of our measures of success is student retention, what can we do about it? Because that's kind of how conversations start with analytics. What would be some of the questions that you'll be asking? In no particular order. Let's generate questions and then see what the answer to that question will tell us. And also whether actually we can get the answer with data or it's a bit more subjective. So let's see what you came up with.
22:00 Speaker 2.: When does retention happen? Mostly happen. Mostly happen. Okay, so what do you mean by that,
22:09 Speaker 1.: when?
22:10 Speaker 2.: Do you mean in which year we had it better? Okay, no, no, no, kind of, tell me, like, what would be the data? So, if we have a question, and this is to everybody, if we have a question and we say, oh. All right, so we have a when question, and it's related to retention, right? So now
22:47 Speaker 1.: we

22:48 Speaker 2.: need, I have several questions immediately. What are some questions that we, how do we get this information? What do we mean by that? If we get this data,
23:00 Speaker 1.: what data would we be looking for?
23:09 Speaker 2.: Okay, so the first thing actually is we clearly have to get some data on retention, right? And the retention rate is
23:22 Speaker 1.: the percentage of people who stayed. So
23:26 Speaker 2.: we need to have information about how many left. how many stayed and how many enrolled,
23:33 Speaker 1.: right?
23:34 Speaker 2.: So to get this, we need number of enrolled. Remember that number of enrolled for that year couldn't give us
23:47 Speaker 1.: retention. We actually have to look
23:51 Speaker 2.: over a period of time
23:52 Speaker 1.: for that degree.
23:54 Speaker 2.: And then for that same cohort, number of discontinued or
24:04 Speaker 1.: left,
24:05 Speaker 2.: and hopefully the difference will give us the retention, right? Okay, so that will give us, now what about the when? What do we mean by when? When it happens. No, no, I just want, yeah, what do you mean by that?
24:28 Speaker 2.: Okay, so you see, these are decisions we need to make. So somebody says year, somebody says semester, somebody says week. So this is actually the level of granularity that you will be setting in terms of your intervals, in terms of your period. And you have to think about how, what, which week.
24:53 Speaker 2.: help you know why do we want to know so what would that tell us you know and if it is each week would they how are we going to get that information because
25:06 Speaker 1.: there
25:08 Speaker 2.: is a bit challenging because we are never going to know which week because once people enroll unless they withdraw completely From the courses it's not actually going to so it would be useful to see how many people will draw from courses But it's not going to give us the retention because some of them might stay on and fail

25:35 Speaker 1.: So if you look at the weeks,
25:38 Speaker 2.: you're going to get an information about people who consciously, before they fail, decided to leave and withdraw. And
25:49 Speaker 1.: that
25:52 Speaker 2.: is based on an assumption. And you always have to think, what is the assumption? And the assumption is that every student who decides to drop out of university actually withdraws. from the courses. But most of them, if the first two weeks have gone, they're not gonna get refund.
26:13 Speaker 2.: They don't bother. They are on our books until the end, and they fail because they haven't done anything, but we don't know until the very end. So it's not to say that the week as an interval is not going to give us some data, but it will be only about certain people who very consciously have decided to do it and then they've made the effort to
26:46 Speaker 1.: withdraw.
26:48 Speaker 2.: So now let's talk about the semester. The semester, if we look at the end of each semester, That would give us better information because we'll be able to see all the courses that they have failed and the courses they're enrolling in for the next semester. So that would actually give us some actual information on dropping.
27:15 Speaker 1.: So we can go, so we have
27:17 Speaker 2.: semester. or we can have ear, as somebody said. Okay, so the thing with the ear is, ear is also helpful,
27:29 Speaker 1.: but
27:30 Speaker 2.: what's the assumption with that?
27:33 Speaker 1.: Basically, basically, when I say ear, attrition, we need to define attrition. Attrition means they are leaving university. So we have to give some interval so that we understand that the ear is actually left. Yes.
27:48 Speaker 2.: Yes, so actually we, you're right, we need some time to understand, so that's why weeks is difficult. And what you were saying is, well after a semester they might have not left, but then after two semesters. So it will give us, after a year is actually, there would have been something I would have thought as well. But the issue with the year is how then you define year as a period in academic
28:19 Speaker 1.: life. Because this assumes a little

28:24 Speaker 2.: bit like a school year that you start in February and you finish in October. And a lot of students actually come in July. And so then, when you compare years, it becomes a bit more difficult. But in the old days, when everybody was coming in February, and in fact, I don't think they even had semesters, they had full year courses, they would have been the best measure,
28:57 Speaker 1.: because.
28:59 Speaker 2.: Everybody comes and goes at particular times. But because now it's
29:04 Speaker 1.: a bit different, and
29:06 Speaker 2.: actually we have quite a bit of influx in July,
29:10 Speaker 1.: a
29:13 Speaker 2.: semester might be, but year will still give us something. It's just that it will be hard to,
29:18 Speaker 1.: yeah,
29:20 Speaker 2.: it will be hard to capture some of them. But it will still capture everybody who leaves. So it's still an option. okay so there was the you see how a simple question that you had required actually quite a few additional question like how do we measure retention and what do we mean by when and and what's how do we set that these are decisions that
29:49 Speaker 1.: you would have to
29:51 Speaker 2.: make in conversation with your domain expert So with the person who will be making the decisions. Because you see, otherwise you might say,
30:02 Speaker 1.: okay, let's do a week because
30:03 Speaker 2.: the more granular, the better. I mean, generally with data, that's always a good thing. But in this particular case,
30:10 Speaker 1.: hard
30:12 Speaker 2.: to find the answer. What other questions did you have? You see, not scary.
30:19 Speaker 1.: How is it measured, like, if you took a class this semester but didn't take another semester, are you considered as a current student? Or are you going to count twice? Or is it just one time you enrolled? Or how does that affect the data?
30:35 Speaker 2.: Yeah, that was my first question, actually. And even in that

document, it's not very clear. So when do we actually qualify a person
30:44 Speaker 1.: as
30:46 Speaker 2.: dropped off, right? Because you might have after a semester somebody failing all of their courses. That actually is very, very common.
30:59 Speaker 1.: And
31:00 Speaker 2.: in many cases, they go on, they pick themselves up, and they actually do really well. So it doesn't mean that they're going to drop off completely. You know, it just simply means that Maybe they just didn't put much effort into their study to start with you know They may be enjoyed the independent life a bit more or whatever it is So so this this is a really important question. How do we?
31:33 Speaker 2.: identify who is retained and who is dropped off. So an easier way to do that would be we have very easy way of Measuring completion. So, because we don't know necessarily who discontinued and when they discontinued and so on, but we do know who has graduated when they started. So the only question then is how long do we give them
32:11 Speaker 1.: to,
32:15 Speaker 2.: and do we just look for completion or do we
32:18 Speaker 1.: still look for currently enrolled if they. That is true. That is true, but that is
32:37 Speaker 2.: a different question. So that's a where question. So the question you had was how or what is the measure of retention?
32:50 Speaker 1.: And so... We can say,
32:57 Speaker 2.: well, completion and probably within a period that we will consider reasonable. So whatever, say six years or whatever they have said. But that would actually make a difference. So again,
33:14 Speaker 1.: you have to discuss.
33:16 Speaker 2.: Completion within a
33:19 Speaker 1.: period.
33:22 Speaker 2.: Then you had the other question. Well, you see, you were starting off with an answer. And that could be a good one. You can say, oh, lots of things might be happening. But always try and then transfer this to a question. because otherwise, you know, I don't know what data you want me to do. So,

33:45 Speaker 1.: we
33:46 Speaker 2.: had a specific example, and that's actually also in the document, that some people might go to another university, and...
33:57 Speaker 1.: It's
33:57 Speaker 2.: not that they're not completing, it's that they're not completing in this university. So then, there are two things here that you would have to ask. First of all, do we care whether they complete? So this is the organization versus the country level, right? The macro level versus the individual. Because for the university, if we are measured in terms of how many complete here, we wouldn't care whether they go to another university. They have left this one. But for a government perspective, because they are subsidizing education for a purpose, yes, it would matter. So again, you have to think is this from the perspective of the university, saying this is how we are measured, or is this from the perspective of the government? And if it is from the perspective of the university, we might use that
35:00 Speaker 1.: to justify some
35:03 Speaker 2.: of the lack of completion. So Here we can say, well, where do they go,
35:15 Speaker 1.: right?
35:19 Speaker 2.: And your question was,
35:23 Speaker 1.: how many go to another
35:31 Speaker 2.: uni?
35:33 Speaker 1.: So therefore, from a
35:37 Speaker 2.: government perspective, they're not actually dropping off.
35:42 Speaker 1.: You had something. For the domestic and international, the passing market criteria is not the same. For international, if you were getting less than D-, then you are failing that course. But for the domestic, it's not the same. So it can be another major. You're failing, of course,
36:03 Speaker 2.: less than B minus. Or do you mean for your postgraduate? Yeah, yeah, yeah. Okay, so again, this depends on postgraduate versus undergraduate, and then international
36:13 Speaker 1.: versus domestic, because
36:15 Speaker 2.: internationals at undergraduate level, it doesn't matter what grade. I mean, you're thinking about your B because of your internships or

master's level. That's a different thing. But very good questions. We can say, well. What are the types?
36:36 Speaker 1.: What are the,
36:38 Speaker 2.: well, okay, I didn't want, no, I'm not going to say. Yeah, but I don't want to say, we would have to find a measure, so,
36:48 Speaker 1.: of
36:49 Speaker 2.: students who,
36:53 Speaker 1.: who drop off. Sorry, you. Yeah, I
37:06 Speaker 2.: like that because, you see, I'm trying to kind of go with those words because they're very helpful when you think you have sussed it all out and then you remember. Okay, I've talked about where, I've talked about when, I've talked, oh, I forgot about who. So, so, some of these what ones you write would be who,
37:34 Speaker 1.: who, I just
37:36 Speaker 2.: don't know, I've forgotten how to write who. Who are these people? But you have your revenge, because you have to write on the test, and I have to read it. Ah, okay, so now we have, okay, this is a really good one because you're starting to think about what are the causes, right? So this broadly is the last question, which is why,
38:09 Speaker 1.: okay? Why are they leaving?
38:12 Speaker 2.: Now, that's a really hard question, but. very very relevant okay so these are all any other questions you have Okay, so this is like all degrees, right? So because remember that some degrees are not within departments, and I really hope that there will be more of that
38:47 Speaker 1.: happening because jobs
38:51 Speaker 2.: are not related to departments. They're more interdisciplinary, but yes.
38:59 Speaker 1.: So what degrees? have higher or lower retention
39:12 Speaker 2.: rates. Anything else? Performance of the student. Okay, so performance, I'm going to write, and then we are going to make this. So what is, was,
39:41 Speaker 1.: the performance of which students?
39:44 Speaker 2.: The ones that are dropping off.
39:47 Speaker 1.: Okay, any other questions? Oh, okay.

40:14 Speaker 2.: Yeah, okay. This is a really good question. And this is where I would call this This is a question that we then have to think about what is the scope of the problem. Because this is something that whether you're a business analyst or a data analyst, you would really have to work on, and that is managing the scope. Because scope creep is real. It happens a lot because people start solving one problem, and then they... attack another problem. So what you have demonstrated, thinking about the connections, which is great. This is exactly what we want, some relevant thinking. But we also have to watch it that we don't go beyond the scope. So if our scope is to understand retention and how it happens and how well we're doing, and they say, oh, what can we do? do to make them come back? Okay, well, those people have left out of scope, but I can say, okay, but within the scope of our problem,
41:34 Speaker 1.: by
41:37 Speaker 2.: understanding why they're leaving, we would have an understanding about what are some of the things that we
41:47 Speaker 1.: can do better.
41:49 Speaker 2.: to make people stay in the future. But those who have gone, you know, they've gone. So, okay, so hopefully this gives you an idea how... when you have a problem, there's never one or two questions. There are lots of questions that are related to data. So now, this was just generating ideas about the questions, and now we are going to do the hard bit about thinking, where are we going to get the data, and what is that going to tell us? And then, in a bit, when I talk about descriptive analytics, we'll talk about how we do it. But right now, we're only on obtaining the data. So where are we gonna get all of this data? So
42:38 Speaker 1.: people
42:41 Speaker 2.: dropping off,
42:42 Speaker 1.: or
42:44 Speaker 2.: not completing, every university will have that data. So if we're just doing it for our university, it's not a problem, the data is there. So it is within a university, probably student management system, but also potentially could be triangulated with the financial system because, you know, stop of
43:07 Speaker 1.: payment might
43:09 Speaker 2.: be another indication, right? So we may need to... kind of

check it whether it's consistent. So what is the measure completion within a period? And we can definitely get that data again from the same systems. Now, where do they go?
43:29 Speaker 2.: When people leave you, they usually don't say where they go, right? So this is an interesting question, but how would we get the answer?
43:56 Speaker 1.: Yeah, right.
43:58 Speaker 2.: So this is where the university level information is not sufficient, right? Because where they go, they might go, for example, they might leave the country,
44:10 Speaker 1.: right?
44:11 Speaker 2.: So then completely, right? So they might actually leave New Zealand. They might Leave education. So when people leave education, what are the options?
44:27 Speaker 1.: They work.
44:30 Speaker 2.: Or they have some reason why they can't work, and so then they would probably have some kind of social benefits, right? So they will be supported. They might not, but we're just looking for any ways where we can get data, right?
44:52 Speaker 1.: They might go to other
44:57 Speaker 2.: education. And that includes, yes, it's great to think about universities, but I was reading a report because, you know, with AI, everybody is slightly freaking out. And they said that the highest jump, which was something like more than 600%, of applications for particular training was for plumbers in UK last year. Of course, AI is not going to do the plumbing. So it's really quite interesting. So some people might decide, well, actually, I don't want to do a degree. I want to do some trade or vocation.
45:52 Speaker 2.: This is actually something that I would like. doing more in my life. So they might drop out of the university and go for a polytech training or some other vocational training or another university. So lots
46:09 Speaker 1.: of options for other
46:11 Speaker 2.: education and training. Anything else that we can think about that people could do? So let's stay with that. So if they have left New Zealand, We could access this data if somebody lets us, right? So this data would be with the border systems, right? Because they show you have to say if you plan to

stay in New Zealand in the next 12 months, if you're leaving, if you've come back. So all of that information is there. We could actually cross-reference those people and see if they've left.
47:03 Speaker 2.: Oh, absolutely. They might have passed away. So I didn't think about that one, mainly on the assumption of a particular age. But yes,
47:17 Speaker 1.: yes.
47:18 Speaker 2.: I thought it might be some illness, and that's why I put social benefit. But yes, it's the extreme, right? So please
47:29 Speaker 1.: stay well.
47:33 Speaker 2.: Okay, so yes, that one we can definitely get information on. Okay, so, and if they've left New Zealand, we can get information. Now, if they've started work, we can get information from,
47:50 Speaker 1.: well,
47:51 Speaker 2.: the tax office knows everything, right? We're assuming legal jobs. So, so... The information is there. We will discuss the challenges about obtaining it. If there are social benefits, the Ministry of Social Development would have the information for everybody who gets social benefits.
48:18 Speaker 2.: And if they've gone in other education training. Several ways we can get it. If they're using financial support from StudyLink or something like that, there will be information on that or from the institutions themselves because they have to report, we have to report. everybody and how they're doing and so on. So you can see how actually for all of these reasons, the information is there. The challenges, there are several challenges, right? They're all in. different systems, and they will have different levels of security and authority about who can access it. So this is actually personal information, and so it will be very difficult for, say,
49:12 Speaker 1.: you
49:13 Speaker 2.: as one university employee or university management to be getting any of that information. This is a real challenge that sometimes data exists, but you don't have access to it. And so, of course, if this is at government level, they can require the data for their reports and purposes. But for... single university if we were looking at it the only thing possible here would be to try and go to
50:00 Speaker 1.: the

50:01 Speaker 2.: source of truth and the source of truth is The person who's left, right, the student. So if we have, because, you know, when you enroll, you have to provide permanent contacts and so on. So we can say, well, we could contact, well, of course, we are not going to know if they've passed away. But you can actually try to contact people and you might get. a small percentage of the response and then figure out from that percentage, if it's representative sample,
50:42 Speaker 1.: what are some of the reasons.
50:43 Speaker 2.: But you're not going to get the full exact truth. That is not to say that you should always give up. You know, like people say, oh, well, we can't get this information. So always think, who is actually the single source of truth here, and how can we get it? That's why
51:04 Speaker 1.: we
51:06 Speaker 2.: get all of these questionnaires and surveys, and you need to
51:10 Speaker 1.: really appreciate how valuable that
51:12 Speaker 2.: is, and never give it for free. Or unless you think it's going to help somebody, then that's different. So, of course, you could have universities do, although we very often are put in a kind of a situational position where supposedly we compete with each other. Actually, we don't. We do work together on things.
51:44 Speaker 2.: If they've gone to another university or training, that would probably be the one that would be the easiest to access. Because you can just contact the other universities and say, hey, let's do this together, let's figure it out. And so if they're in another university, that would probably be the easiest from a university point of view to find information on.
52:06 Speaker 2.: So now who are the students who drop off or what are the types? So several things were mentioned, right? Postgraduate versus undergraduate,
52:27 Speaker 2.: international versus
52:34 Speaker 1.: domestic.
52:39 Speaker 2.: And any other things we can have, actually, lots of information that we get about who these students are, because we collect all that data. All the demographics. So we can have
52:53 Speaker 1.: age.

52:55 Speaker 2.: So we can see when they started, for example, was it straight after NCEA or after 25? We can get their permanent address so we can see whether they're local for this university or they've come from somewhere else.
53:19 Speaker 2.: Do we collect that? We ask you for that? Oh, okay, I don't have access to it. All right, so that's great. I am very pleased that there is one thing that I don't have access to, because sometimes it worries me how we have access to information that we shouldn't have. So, okay, so if there are household incomes or types of financing, that's probably in the finance system, so we would have that information as well. So, you know,
53:49 Speaker 2.: Any of this information that characterizes the people might give us an idea if we, as we'll talk in a minute, if one of them seems to have a higher frequency, that might give us an idea about reasons. So lots and lots of information,
54:08 Speaker 1.: demographics.
54:13 Speaker 2.: And then, while they're leaving, if we have a lot of this information, we will be looking to see if there are any patterns in the data that imply that we can infer that something happens more often that relates to dropping off.
54:38 Speaker 2.: What degrees have high or low retention rates? So again, we can, we can, it's kind of related also to the type of student. We could look at all the degrees, that's university information. And what was the performance of students who drop off? Now, what do we mean by performance?
55:04 Speaker 2.: So it could be a pass rate. It could be actual GPA. So you see it's different. That's why it could be pass rate. Because, you know, if you say pass rate, this is really giving me lots of good stuff when we talk about descriptors in a minute. Because sometimes when we talk about statistics and people just... because it seems so theoretical. But these measures matter because if you do
55:32 Speaker 1.: pass rate,
55:33 Speaker 2.: you're kind of smoothing them all into two categories. But if you look at the GPA and you say, oh, they've passed, but most courses C minus, you know, versus, oh, they not only passed, they had A's, you know, so there will be difference.
55:51 Speaker 1.: So you can see how

55:53 Speaker 2.: So pass rate or GPA. Okay, so that's available in the university. We have that. We have all of that information. What other performance we might need to think about? Because remember, these are people who are leaving the university.
56:14 Speaker 1.: Attendance
56:15 Speaker 2.: is actually very strongly correlated factor, but we don't
56:25 Speaker 1.: keep that data.
56:29 Speaker 2.: And, yep, actually for some, for some categories of students we have. duties to keep the data but we kind of try to do it in a more casual way so we don't record it unless we see a problem so you're right this will be useful but we don't have it unless you make classes compulsory and then you take a role and when you have 400 people impossible to really be sure Like in this class, I can kind of see now roughly who comes in. By week five, I'll know your names. Yeah,
57:18 Speaker 2.: you could see whether they have history, whether they've changed their majors. So that's related to what degrees, right? That would be kind of here, but this one, you mean change of enrollment. So does it show that they were kind of not happy already with their degree? So that might be something interesting. Okay, but another thing that we actually do have information about, depending on the type of student. Oh, okay, so the thing, okay, this is really useful.
58:10 Speaker 2.: The issue with that is that we
58:13 Speaker 1.: will,
58:14 Speaker 2.: if we could know for every person who's dropped off, see their posts, yes, that would be useful, but because it is personal. social media. It's not like getting aggregate, you know, it's not like when people talk about the university, is there a positive, negative sentiment? Then we don't know who they are, we just have an aggregate kind of
58:40 Speaker 2.: sentiment for this one because you're you're saying we want to see if that particular student there would be private but yeah good idea if we could but I hope we never will be able to because then to me that goes too far into invading people privacy
59:00 Speaker 2.: Their ability to pay. Yeah, so we did talk about that in the financials, household, all of that, yes, when we talked about demographics, so

financials. Another measure of performance, because I'm sort of really not doing well with time as usual, another measure of performance would actually be their performance prior to the university.
59:26 Speaker 2.: Because for them to come to university,
59:29 Speaker 1.: what did I say? NCA,
59:35 Speaker 2.: over 25, or international, right? So for all of these, we would actually have some data on their prior. academic performance. So if it's an NCEA exam, you know, how did they do on the exams? If it's international, we have your previous exams. So clearly you're all postgraduate, so it will be your transcripts. But if you're undergraduate, we would have some kind of equivalent to university entrance exams or school leaving exams or something like that. So we would have
01:00:15 Speaker 2.: performance prior to
01:00:17 Speaker 1.: university.
01:00:26 Speaker 2.: And you can see how beyond where do they go, actually for all of these questions, we do have data in the university. I don't know how much we do on analytics. I try to get somebody from the university to come and talk to us. And they might still, but they said, oh, they have to be very careful about sensitive data and so on.
01:00:55 Speaker 2.: So I thought, okay, I don't have to be careful because I don't have the data. We can ask the questions. Okay, so what will be the challenges in
01:01:11 Speaker 1.: once
01:01:13 Speaker 2.: we, because I said the data is there, right? Beyond the government kind of where they go, the data is there, the performance and so on. The issue is that, for example, the NCA results would be in our student management system where the records are and then your performance is actually in a different system where we report the grades and the results and we get the transcript and this is what most of us can view because we have
01:01:51 Speaker 1.: to
01:01:51 Speaker 2.: approve courses and so on but we can't view your NCA or any of this prior, it's in a different system and so
01:02:00 Speaker 1.: It
01:02:01 Speaker 2.: would be, the good thing is you all have student IDs. So

that's your unique identifier. We would be able to. integrate the data per record, which is a student ID, but it would be, I am absolutely sure, inconsistent in the way it's put in. Some of it is in notes, some of it is structured, so it would require quite a lot of cleaning and putting it together, and some of it...
01:02:37 Speaker 2.: about majors, also they will be incomplete because some people select majors, some people don't, then they change their minds. So there will be some incomplete data and some people choose not to share some demographics and
01:03:00 Speaker 2.: that would also show as variance over the years, because in previous years I don't think people had as much choice, but then the systems were not good enough to capture when somebody doesn't enter a field. Now the systems are better to capture if you haven't entered something, but because of that we have to say this is actually optional. So some information would be optional and some of it will not. So we'll have some missing data, we'll have some inconsistent data. So you can see how even within the same organization this would create preparation and cleaning exercise. So what we are going to do today is continue from this because this is all from last week, right? where are we going to do data, is actually, as I said,
01:03:57 Speaker 1.: perform
01:03:58 Speaker 2.: descriptive analytics to answer some of these questions. Descriptive analytics, this is why also used to be called business intelligence. really, really powerful because they're based on real information. It's not probabilistic. It's not predictive. They give us real insights. The problem is that we have to know what we are looking for. So these questions that we had. And so it's always about what happened or what's happening. And we always apply some kind of
01:04:36 Speaker 2.: Characterization, you see, you already were doing it. We characterize, we start grouping people, post-graduate, undergraduate, or this age, that age, international, domestic, right? Or people who come through the NCAA route versus people who've come in a different way. So characterization, organization, so you put them. into these categories, and then you sort them somehow to see how many we have, calculate, summarize. So all of these are features of descriptive analytics. Trying to actually make sense of the data so that it has some meaning for decisions.

01:05:23 Speaker 2.: So where do we find So we talked about this specific example for the university, but in general in businesses. very often we have these what we call generic functions. And so I personally prefer to think about processes, because they go across functions,
01:05:48 Speaker 1.: but data is
01:05:51 Speaker 2.: very often organized around functions. Each function has their own data source. So I'll just go through quickly some of these so that we have Again, base understanding. So in the human resource function, so whether this is a system, it's called human resource system, or it's actually whatever data store they use in the human resource function. What does human resource as a function do? So I know some of you are from a business background, but some of you are not. So let's think about it. Human resource management is about managing
01:06:33 Speaker 1.: employees and people,
01:06:35 Speaker 2.: right? So what do we mean by managing people? How do we manage them? Beat them with the stick to work harder. Right, so there are certain, you're right, so there are certain rules or conditions of your employment that you need to fulfill, such as number of hours, for example, which depending on your contract, you could easily go over and so on. So we have that. So it actually monitors whether people, so there's some performance management, right? Whether they are performing the way they meant, so they might have targets. But also ensuring that the conditions are appropriate so that people can do their job.
01:07:34 Speaker 2.: And also that, you know, more and more people are talking about upskilling, right? And a lot of the times we are expected to upskill on the job in our own time, but actually it is the role of human resource to support training and upskilling as well. So this is why the human resource is not just about performance and conditions of employment, it's also about developing that resource, making sure that they're up to the task and that they're satisfied with the job. So there's a lot more to human resource than performance management. And so providing benefits, sometimes also there are parts of the employment, trying to think what benefits we have.
01:08:35 Speaker 2.: Difficult one. But in some businesses, for example, they have free gym memberships or actually the union members have here

discounted health insurance and different ways to incentivize employees. And finally, we have a major process with the human resource, which is recruitment. Recruitment is managed by human
01:09:08 Speaker 1.: resource, and this is actually
01:09:10 Speaker 2.: the process of... finding and recruiting
01:09:15 Speaker 1.: new employees.
01:09:18 Speaker 2.: So this goes from creating the position description and conditions, finding candidates, evaluating, supporting the process of evaluating candidates to the actual signing and onboarding. So this is all part of the human resource function. And Any, because you know we keep talking about transactions as if this is always financial transaction, but any transaction in terms of information transaction, for example, if somebody has ever put in an application for a job at the university, that actually is kept
01:09:59 Speaker 1.: in
01:10:01 Speaker 2.: the human resource records. This is actually the one we have serious trouble cleaning the data because it's in all sorts of. places and shapes but
01:10:12 Speaker 1.: the data is there
01:10:13 Speaker 2.: and how many candidates there have been for particular positions, where they came from, who signed on, all the information about these people is actually kept for a while, and so we do have that, and then we might have also obligations to delete some of this data, so that
01:10:41 Speaker 1.: also needs to be
01:10:44 Speaker 2.: done over periods of time, but we do have a lot of these details, and if you have a human resource ERP or some kind of integrated source, then all of that would be there, including the payroll also goes
01:11:05 Speaker 1.: in
01:11:06 Speaker 2.: HR, so IRD, tax. Additional, if people have taken extra work, all of that would be kept
01:11:17 Speaker 1.: there.
01:11:17 Speaker 2.: So just like what we were talking about, about students, when you have employees,
01:11:22 Speaker 1.: all information related to employees
01:11:25 Speaker 2.: will be in the human resource. And then we have external,

right, because employees internal, external is customers. So similar to human resource management, but with customers, you providing particular services, and you would have certain obligations to customers in all interactions, including transactions that are sales, but transactions that are information exchange. For example, if this is a bank and somebody has asked for
01:12:00 Speaker 1.: pre
01:12:02 Speaker 2.: -approval or assessment of their credit or anything like that, that would be recorded in the customer relationship management. And in fact, now banks even record their web streams
01:12:18 Speaker 1.: because
01:12:20 Speaker 2.: They would have your IP address. I'm not sure if you've noticed when you use online banking. Some of the banks now, if you use online banking from a different device,
01:12:32 Speaker 1.: even
01:12:35 Speaker 2.: if it is within New Zealand, because before it used to be if you go overseas, which fair enough. But now, even if it is within New Zealand, if I use a different device, it would say This is not where we expect you to access your account from, so they will send a netcode. So they have all of that information about the customers, usually in the CRM. And so any behavioral information, so when we talked about groceries, if you are ordering online, if you are using your card, what time, all. these would be there and any
01:13:21 Speaker 2.: discounts that you seem to be sensitive to or particular privileges that they will be giving you and now especially dynamic pricing that a lot of businesses are kind of tinkering with where they offer different specials to different people well that would be in your profile
01:13:46 Speaker 2.: Oh, I forgot about the break. Let's have a break. Yep. Let's have a break. You should be more
01:13:57 Speaker 1.: assertive.
01:13:59 Speaker 2.: Tell me when it's time for... I never think about breaks. Try
01:14:11 Speaker 1.: to move.
01:14:14 Speaker 2.: Yes, let's do a bit of that. Yeah, I know. Athena, right? Yeah, yeah. I know. We haven't – because you were late, those people I keep on a list, and then the tutor is going to mark them later. Yeah.
01:14:34 Speaker 1.: So I was not able to complete it. I was having an error in

that.
01:14:40 Speaker 2.: What error were you having?
01:14:43 Speaker 1.: But you
01:14:46 Speaker 2.: only were asked
01:14:47 Speaker 1.: to do the PBIX file, right?
01:14:50 Speaker 2.: The XLOOKUP is in the Excel file?
01:14:54 Speaker 1.: Oh, okay.
01:14:59 Speaker 2.: Okay, but have you submitted it now? Oh, you're still having problems. So I want to just do it. Submit it even if you have problems, because at some point we'll...
01:15:12 Speaker 1.: or
01:15:13 Speaker 2.: I can submit it? No, just submit what you have. Yeah, yeah, I mean, there's so much extension. This is not reason for extension. Having problems is not reason. Just submit. Remember, it's a pass-fail.
01:15:24 Speaker 1.: Okay. How many tutorials do we have?
01:15:28 Speaker 2.: So, this term we would have at least four. I'm not completely sure about next term because we have a visitor, but we would definitely have at least another three. So... you are given percentages for the best six. So we will definitely have more than six. So you are allowed to have one blip or
01:15:54 Speaker 1.: two. But
01:15:55 Speaker 2.: submit. You know, problem or no problem, submit.
01:16:11 Speaker 1.: Let's have a look. Let me just blank this.
01:16:22 Speaker 2.: I don't actually have anything to do with the, this is all central, but I do have a way of seeing whether they are there. They're
01:16:37 Speaker 1.: there, I can see it. And
01:16:40 Speaker 2.: they're linked to this class. and this is the one that's currently being recorded. Where
01:16:53 Speaker 1.: do you go to look? Because you
01:16:57 Speaker 2.: see, I have no control over that system, so you might need to report it to
01:17:03 Speaker 1.: the,
01:17:11 Speaker 2.: you don't have to go to term one, that's not where the recordings are.

01:17:16 Speaker 1.: Okay, so that's different, okay. Right, so you actually have to go to, if you go further down. You see, that's where you access the video. Yeah,
01:17:30 Speaker 2.: no problem.
01:17:31 Speaker 1.: It's for every class. Okay.
01:17:39 Speaker 2.: No multiple choice questions because this is postgraduate class. So it'll be kind of a similar thing to what we do in class. You get a bit of a description of a business situation or an organizational situation. And any questions you have will be questions we've asked in class. So you'll be asked to maybe generate questions and then based on the questions to say what kind
01:18:01 Speaker 1.: of date.
01:18:04 Speaker 2.: No, there are right and wrong answers, but there's no one. one right answer. There will be many right answers. And as you can see, sometimes, this is why I try to do this in class and we take a lot of time, is when you give an answer, you have to make sure you explain what you mean. Because sometimes, for example, if people say, when does it happen? And I'm like, okay, and in terms of data, what do you mean? Do you mean semester, year, right? And if you said
01:18:36 Speaker 1.: week, I would say, okay, weak would be,
01:18:39 Speaker 2.: you know, I'll give you some marks, but say, okay, you haven't explained
01:18:42 Speaker 1.: why weak, right? So how many words with a particular question you can? Oh, I
01:18:51 Speaker 2.: don't normally put limits
01:18:55 Speaker 1.: towards.
01:18:57 Speaker 2.: You can, you can, you can. But you also look at the marks that.
01:19:03 Speaker 1.: the question has and
01:19:05 Speaker 2.: if you have low number of marks it's not requiring a whole page right so so try to answer the question and try to answer it
01:19:16 Speaker 1.: clearly because
01:19:18 Speaker 2.: you see you can see how this didn't require a lot of
01:19:20 Speaker 1.: words but it did require thinking
01:19:23 Speaker 2.: yeah yeah so well the brainstorming

01:19:26 Speaker 1.: was because we're in a group when when you're by yourself
01:19:29 Speaker 2.: you know you just have to think right And make sure that you articulate what you mean. Yeah. But it's never about number of words. It's never about how many.
01:19:41 Speaker 1.: I've had people who have
01:19:44 Speaker 2.: perfect answers with very short answers. And people who write a lot and still don't answer the questions. So try to take the time. I always make exams kind of to. I factor in a lot of time that you should have to think about it and to actually read the scenario and think about it, not just... It's not a speed writing
01:20:14 Speaker 1.: test. So the test material would be to week five, right? Yeah, week five. Yeah, yeah. And how many questions would that be? Because I'm very new to the excellent method and
01:20:27 Speaker 2.: everything. Okay, I'll
01:20:30 Speaker 1.: give you a practice exam. I'll try to
01:20:36 Speaker 2.: give it next week. Yeah, yeah, yeah. We'll do that. We'll do that.
01:21:06 Speaker 2.: Okay, I had one question during the break, which reminds me. reminded me that You know in this class there may be questions that are not specific to the course because you're still starting and some of it is beyond that so and It was actually really helpful because I also had a couple of emails similar to that and I couldn't understand so people are asking why they can't access the lecture recordings and when we so I saw what you're trying to do so I'm just quickly going to show you that in order to access your recordings, you actually have to go here, not in the, because this is the part that I control and download, and this is essentially a controlled one. I have no management of that beyond all of me saying, yes, I'm happy for it to be recorded, live streamed, et cetera. So this is where you access for all your courses. When you go to your course page, you have to go through the
01:22:12 Speaker 1.: echo video to access the recordings.
01:22:17 Speaker 2.: This sometimes shows you how we... make assumptions that you know about some things. So feel free to continue asking me questions like that. But do explain also what you are doing. Because you see, when I get an email that says I can't access the video recordings, I go have a look and I'm

like, oh, it's there. So it's not a problem. And then I say maybe it's a problem with your account. Because I assume that you
01:22:45 Speaker 1.: can't access versus
01:22:46 Speaker 2.: I don't know how. It's different.
01:23:02 Speaker 1.: Do you have all
01:23:04 Speaker 2.: class rep? Yeah, we can have a class rep. Who would like to be the class rep?
01:23:14 Speaker 1.: There you go. You see? But
01:23:16 Speaker 2.: if you ever have any doubts, the best way, really, because even if you have a class rep, the class rep will come to me, right? So please, this is a postgraduate class. Please do come and ask me because that is the fastest way. But yes, if anyone would like to be the class rep and they didn't want to say in front of... Everybody, come to me after the class, and then I'll put the name and the contacts on Learn, and then you can also go through your class rep.
01:24:00 Speaker 2.: No, because I don't
01:24:02 Speaker 1.: have it yet.
01:24:03 Speaker 2.: You mean for the second term? Oh, for the tutors? Sure, sure. I will put her email address. I will ask her first. Because technically the job is to take tutorials. Not to do personal tutorials. So I'll ask her if she's okay with me sharing the...
01:24:33 Speaker 2.: Yeah, she's one of the LEARN participants, so you can actually contact her, yeah, but I don't want to say that.
01:24:40 Speaker 1.: She is on LEARN,
01:24:42 Speaker 2.: so if you know the name, which I gave you in week one, you know how to contact her. But for me to post the actual email address, I will ask, because it's kind of personal. So remember, the best way to ask questions about your tutorials is to go to the tutorials.
01:25:10 Speaker 1.: Because,
01:25:12 Speaker 2.: no, no, I know this sounds very kind of slightly condescending, but
01:25:17 Speaker 1.: you
01:25:18 Speaker 2.: see there are quite a lot of you.
01:25:20 Speaker 1.: And if,

01:25:22 Speaker 2.: say, 20 people decide to email with the same question, that means 20 times 5. 100 minutes of that person's time,
01:25:33 Speaker 1.: while if
01:25:34 Speaker 2.: all 20 of you are in class during the tutorial, only one of them needs to ask the question, and then 20 people get the answer. So that is why it is better, and that's why we have the tutorial sessions. All right, I will, as usual, speed up a bit. So we talked about HRM, CRM, because, you know, NIT, we like to use abbreviations. And now supply chain management.
01:26:14 Speaker 2.: During COVID, it was practically in every paper. So supply chain
01:26:21 Speaker 1.: is
01:26:22 Speaker 2.: called a chain because it's all the activities, sequence of activities related to
01:26:32 Speaker 1.: procuring and
01:26:35 Speaker 2.: sourcing of
01:26:36 Speaker 1.: the...
01:26:38 Speaker 2.: raw materials or import materials for any products that a company would be manufacturing. So mostly for manufacturing, but also for intermediaries that are buying and selling, right? Because they still have to buy wholesale and then distribute. So that is also anything related to
01:27:06 Speaker 2.: the movement of products as a supply chain transaction and then all of the information around the suppliers, in this case also called vendors, so the suppliers, their information, the types of products that they provide, what they prefer in terms of
01:27:35 Speaker 1.: payment method, arrangements
01:27:37 Speaker 2.: around invoicing, all of that will be in the supply chain management. And what is great actually
01:27:44 Speaker 1.: is that usually it's B2B.
01:27:46 Speaker 2.: You know, you are dealing one business with another business, which allows for more integration. And if you've been doing six or three, I'm sure you have talked a lot about B2B
01:27:58 Speaker 1.: e-commerce.
01:28:05 Speaker 2.: Finally, financial reporting systems, because ultimately all of these functions would report some financial measures and the accounting

systems will have all the informations on incoming revenue, outgoing costs, and will be distributed across departments, across activities, across products,
01:28:36 Speaker 1.: lots
01:28:37 Speaker 2.: of information on cash flows. So, going to go through now the 20 minutes, that's okay. So, through the, some of the main techniques and some of these are techniques. Some of them are measures. So we'll talk a little bit about the difference. So one is about how we measure something to represent new information. And the other one is how do we represent information visually. So in these techniques, there's a bit above. So we might have techniques that broadly are called counts. So something that we count. So, for example, in all of your questions with the university, we had things like, well,
01:29:37 Speaker 1.: how
01:29:39 Speaker 2.: many have discontinued, for example? How many have dropped off? How many came through the NCAA route? How many
01:29:50 Speaker 1.: would...
01:29:53 Speaker 2.: How many are international? How many are domestic? How many are postgraduate? How many? So these are accounts of different categories and it helps us to get an understanding.
01:30:10 Speaker 1.: But accounts
01:30:12 Speaker 2.: by themselves They always have to be in relation to something else, right? Because if I know, for example, that 500 domestic students have dropped off, without knowing how many we have altogether, it's not a meaningful answer unless I am comparing count with count. So I can say, well, in those they have dropped off, 500 are domestic. 20 are international, so then it gives us a comparison, right? So then that's useful. So think about when you count something, how would, how would that provide you with some insight?
01:30:56 Speaker 2.: So you see here how, for example, here, the example is with some product
01:31:02 Speaker 1.: that was sold,
01:31:03 Speaker 2.: a number of products, in a particular period by region and by store. So this is a way of organizing and characterizing it according to a store and a region. So we would have the counts of products, but we would have the regions and the stores, and so then we would be able to compare how different

stores are doing in terms of of the sales.
01:31:29 Speaker 2.: So very basic and very useful way of describing your data as long as you think about how we are going to characterize it and
01:31:41 Speaker 1.: categorize it.
01:31:43 Speaker 2.: So then we might actually perform some kind of calculations or further manipulation of the data to represent it with
01:31:55 Speaker 1.: totals. So
01:31:58 Speaker 2.: you might say, okay, we want to calculate the total sales. Revenue per region, per store. So instead of giving us a number of transactions or a number of goods, it will give us financial information. We might have things like averages. So remember in one of the classes we talked about. the workload, right? And you said, oh, we could figure out what is the average time that people take on a task and see how long it takes. So averages can be very useful.
01:32:46 Speaker 2.: Where it is not useful
01:32:48 Speaker 1.: is if
01:32:51 Speaker 2.: you have Try not to use distributions. If you have quite a significant variance and
01:33:01 Speaker 1.: very
01:33:03 Speaker 2.: different numbers, for example, if you have two people that are taking three days to do a task, so... If we count that by three working days, 24 working hours to do a task. And then we have five people who take 15 minutes.
01:33:25 Speaker 1.: When
01:33:25 Speaker 2.: you do the average,
01:33:26 Speaker 1.: it would smooth it, right?
01:33:30 Speaker 2.: And I think definitely one of the tutorials I saw that where there were lots of averages and it kind of looks good. But when you zoom in, you realize, oh, There's quite a bit of variation here. So the average is not a good representation. So the averages are good if you're comparing things, the same things, or if your data is relatively normally distributed.
01:34:00 Speaker 1.: So kind of a bell shape.
01:34:04 Speaker 2.: So Then we also have subtotals within the total, so you might have, okay, this is the total revenue of sales, and then this is the subtotal

for perishable products, and this is the subtotal for fresh,
01:34:23 Speaker 1.: for
01:34:24 Speaker 2.: non-perishables. And so, again, allows us for some comparisons. Then in terms of variations, we have variance. We have minimum, maximum, median, and standard deviation. So I actually do have some separate slides on these things because we talked about how some of you haven't got the book yet. And of course, there's lots of information about that publicly available. So minimum, obviously, the lowest. value, maximum, the highest value. So it gives us some idea about the variation.
01:35:30 Speaker 2.: That's a very good question. So I'll go through quickly through the measures, and we'll go back to our example. So there we have the visual description, the visuals, and you've already used, I think, bar charts. I'm not sure whether you've used bar charts or histograms. So one is more for continuous data and the other one is for categorical and other visualizations. So I'll just quickly speed up so we can go back to our example and answer your question.
01:36:01 Speaker 2.: So the average or also mean, and don't worry about the equations. Because you don't have to use them because all the tools use them for you. But I did want to show you that basically with an average, what you're doing is you are adding all the values together and then dividing by the number of values. And so this is quite typical. So say with your grades, sometimes we'll have things like
01:36:43 Speaker 1.: people who have so we have a lot of people who have then we have 60 and then we have
01:37:09 Speaker 2.: So within this range, obviously, just for illustrative purposes, you can see how if we get an average where we add them all together, we'll probably get
01:37:25 Speaker 1.: roughly around 80 but you
01:37:31 Speaker 2.: can see how or even less than that but you can see how actually a lot of them are in the lower end or the higher end so you have a great variation. So the average 80 is not going to be that representative. So sometimes we actually want to see, well, what is the median, which is the number in the middle, literally the number in the middle. So you sort your data, and then you look at what is your middle number, and then in this case we had,

oh, I made it symmetric.
01:38:11 Speaker 2.: Okay, so, and I actually did the middle, because I can't help myself. So, if it's asymmetric, it would have been this number, and you see how important this is, because of the huge gap you've got here. So... So the median is, is literally the, the number in the middle. So 62, which will be very different from 80, which will be the average. So. It is important to think about that. And then you look at the mode, which is the most frequent. Now, in some cases, this could be very relevant if you find that particular value comes up more often than other.
01:39:02 Speaker 1.: So, for example, if you find that,
01:39:04 Speaker 2.: okay, out of all the people who
01:39:08 Speaker 1.: are
01:39:09 Speaker 2.: dropping out, we have a greater number of Cs, for example. So then we could actually get the mode. So these are the measures that show us how centered our data. Where is the center? Is it the average? If it's kind of normal distribution? Is it, you know, where is the middle one? Different value and the mode. tells us what's the frequency. Then we have measures of dispersion,
01:39:44 Speaker 1.: how spread.
01:39:47 Speaker 2.: You know, what's the variance? And so the maximum and minimum is something that we're all familiar with, right? Because we give a range. We say it starts from this and it ends there. So we can say, okay, the grades in this course, the lowest was 73 and the highest was 98. And so you immediately have an idea about how dispersed your data sample is. And then you
01:40:15 Speaker 1.: look. at the
01:40:18 Speaker 2.: variance in terms of, okay, well,
01:40:22 Speaker 1.: if we
01:40:25 Speaker 2.: look at it in relation to the average, how far it is, or we can have the standard deviation again. These are all measures that try to look at the distance from your center.
01:40:45 Speaker 2.: And now, would you have to use some of this? No, but I wanted you to understand the importance because last week we talked about
01:40:56 Speaker 1.: preparation

01:40:58 Speaker 2.: and looking at how skewed or how unbalanced your sample might be. And so you should be looking to see percentages of representation of one particular category. For example, if you have demographics and you find that most of your sample is within one category, so you could actually do frequencies and
01:41:24 Speaker 1.: say all
01:41:26 Speaker 2.: 80 out of the hundred are within the range 20 to 25. So these descriptives would give you an idea about how good your data is. Not answering your questions necessarily, but looking at how good your data is and whether you might want to balance it a bit more. Also great, the dispersion is great in showing us...
01:41:55 Speaker 2.: our outlier situation. Because how we deal with outliers in data matters as to what results we get. And we decide whether we want to keep outliers or we want to eliminate some of the outliers if they don't provide good information. So, so you can see
01:42:16 Speaker 1.: how if you
01:42:20 Speaker 2.: I don't think you have used the box and whiskers plots yet, but it's a visual representation of all of these measures. So the mean is your average and the median is the exact middle. So it's exactly where half of your data is on both sides. And then if you decide to split your data into quartiles, then it's
01:42:48 Speaker 1.: four.
01:42:51 Speaker 2.: quartiles, and actually the 50% of the data goes into the box and whiskers plot, and then the other two quartiles are on the outside, there are the whiskers, and anything outside that is considered to be an outlier. And
01:43:14 Speaker 2.: Why do we care about these things? Because it shows us
01:43:19 Speaker 1.: which of the
01:43:22 Speaker 2.: measures would
01:43:24 Speaker 1.: be relatively meaningful.
01:43:27 Speaker 2.: So, for example, you might think, okay, the median and the mean, not that different in this
01:43:33 Speaker 1.: sample. Or,
01:43:38 Speaker 2.: you can see here, so this one is kind of a normal in terms

of centrality, but we have two things. We have skewness and we have ketosis. So the skewness is about how your skewed your data is on any side, right? So this one is a little bit more negative and this one is more positive,
01:44:05 Speaker 1.: right?
01:44:06 Speaker 2.: And the ketosis is whether we have kind of a high peak or more smooth and this this would obviously matter because if you have here so this one for example is
01:44:20 Speaker 1.: relatively
01:44:23 Speaker 2.: normal and then your whatever your averages are here would be useful so it just provides you with some visual representation and with some measures of how balanced your Data is if you have skewed data You might think about okay. What does this mean? It's actually quite common to have skewed data
01:44:47 Speaker 1.: because
01:44:50 Speaker 2.: Not all of it is distributed. For example, any kind of traffic or demand or service demand data is usually not normal
01:45:02 Speaker 1.: distributed.
01:45:03 Speaker 2.: Think about it. If somebody comes
01:45:06 Speaker 1.: to,
01:45:09 Speaker 2.: let me give you an example with marking, because that's something I love all the time. Marking doesn't happen every day a little bit at a time, right? It comes in bursts, because you get an assessment, four courses get assessments about the same time, and then with one resource, the load goes up, and then after, so then you get backlogged for a bit, and then everybody keeps waiting, and so it's not kind of smooth, and so demand data usually It's not normally distributed. Similarly, they say alcohol consumption samples also not normally distributed. So let's now go back to your question. Because you are asking me, is it going to give us the reasons these measures or
01:46:14 Speaker 2.: Do we just describe it? And the answer is
01:46:17 Speaker 1.: off.
01:46:20 Speaker 2.: We describe it, and we start getting some ideas about the reasons, but we don't say this is the reason. Next week, we'll talk a little bit about correlation and regression as descriptive analytic, and then you'll see you might then draw some conclusions on correlations. But

01:46:40 Speaker 1.: if we go back to our example and
01:46:47 Speaker 2.: say, well, where do they go,
01:46:49 Speaker 1.: right?
01:46:51 Speaker 2.: If we did have that information, we can provide it as percentages of the sample of dropped off. We can provide it as bar charts, right, because different categories and visually show us how they're different. And we can, but we can't do actual here for the reason. There's no averages or anything like that because there are just different categories
01:47:22 Speaker 1.: of
01:47:23 Speaker 2.: reasons that we can, how many. So, but we can definitely have counts, right? How many have left for this and how many have left for that. And so how many go to another uni? So these are all counts,
01:47:41 Speaker 1.: frequencies.
01:47:43 Speaker 2.: But if we want to say what are the types of students, again, these are categories. We can do counts. But we can also start doing ratios. We can start saying what percentage to try and figure out what is the highest. So either percentage or you could do even a mode. What's the most frequent? degree that comes up, does it happen more with undergraduates or postgraduates? So I wanted to show you actually the report that was created, and some people are looking at it already, but I said this is a little bit.
01:48:26 Speaker 2.: Okay, so the first one they have created is just totaling it, right? So they're doing it over a number of years. And you can see they have selected five years because this is what is considered for them the period of time in which people need to complete at the most. So remember I asked how do we decide the period? Is it five? Is it six?
01:48:54 Speaker 2.: And so
01:48:55 Speaker 1.: they've
01:48:59 Speaker 2.: put a total of people who have and they have explained why they have chosen this population.
01:49:13 Speaker 1.: And
01:49:14 Speaker 2.: so here we are, where did the existing students go? Now, because this is a government report, they clearly have that information that we were talking about. So, so you see how they have presented it as a percentage of exited population,

01:49:31 Speaker 1.: and
01:49:33 Speaker 2.: as a percentage of the total cohort. Look what difference does it make if
01:49:40 Speaker 1.: you have
01:49:42 Speaker 2.: relatively small numbers to the total population, so all the students who enrolled, it's actually quite, it, it might look like it's not big difference, right? 5%, 1, 1%, 1, 1, 2, right? Very similar. But when you have them as percentage of the ones who dropped off, you can see immediately that The two highest groups are, they've gone to get
01:50:13 Speaker 1.: a job,
01:50:15 Speaker 2.: 32%, and they've gone to another university, 27%.
01:50:26 Speaker 1.: And
01:50:27 Speaker 2.: then you have other degree study, and this is what I meant by grouping things, putting them into categories. So we can't put it just at another university, but we can actually put a bigger group, which is... any education or training, so they're continuing with education and training. And then you can see that within
01:50:56 Speaker 1.: that group, we
01:50:59 Speaker 2.: actually have 14, 16,
01:51:02 Speaker 1.: 18, so we have
01:51:05 Speaker 2.: 45%. So that actually is bigger than the percentage of people going for a job. So it depends how you present information, how granular it is. You know, is it for another university or is it overall as an education?
01:51:25 Speaker 2.: Now, you are saying, well, do we put a reason why they go, right? So you can see how here you can start making some conclusions about what the reason is but actually we're not sure from that because going to another university the reason might be that this university is awful at educating people so they had bad experience but
01:51:53 Speaker 1.: it could be that
01:51:57 Speaker 2.: Well, it's more expensive if you live away from home and maybe they went back to their home place so they can get more support from family and live in the family home or whatever it is. So it doesn't actually give us a reason. It just gives us a pattern as to what people do. And then you can

actually look at the timing, which was what you were asking. When does it happen?
01:52:28 Speaker 2.: And here are the counts by themselves, and then also presented as percentages. And so what do we see here? Thousand, two, and then small. So... Two things we can immediately see in this
01:52:51 Speaker 1.: data is
01:52:52 Speaker 2.: that the majority of them live within the first two semesters. So if we were using the year,
01:52:59 Speaker 1.: which
01:53:00 Speaker 2.: equivalent to two semesters,
01:53:03 Speaker 1.: it would actually give us the greater.
01:53:05 Speaker 2.: But you can see here that actually the majority of them, the worst one seems to be the second semester, right? So if this was looked at in terms of let's try and see how we can improve the experience or make it easier. We might say, okay, actually, we can, most of them do try to have another semester. Let's look what happens in the first semester and provide additional support to people who are under risk because the second semester is crucial.
01:53:36 Speaker 1.: After that,
01:53:37 Speaker 2.: not much difference with the third one. And then if you actually look at So you see how all of them more or less follow the same pattern. The second semester is double the first, apart from the vocational one. So because you see here, the second double for another degree, sub-degree, so the educational ones about the same. But the ones who actually go to work.
01:54:13 Speaker 2.: although this is the sub-degree, so some kind of vocational training, most of them after one semester.
01:54:20 Speaker 1.: So they realize, okay, university,
01:54:23 Speaker 2.: not for me. So you can see, yeah, yeah, come in, otherwise I'll be here forever. So this is an example how the descriptive analytics can start giving you some ideas about what's going on, so that you can hypothesize what the reasons might be, and then you can test the hypothesis.

View original transcript at Tactiq.

MBIS624-26S1-LecA-Te-s1-low-4

Meeting started: Mar 23, 2026, 10:10:33 PM
Meeting duration: 113 minutes
Meeting participants: Speaker 1, Speaker 2, Speaker

View original transcript at Tactiq.

Transcript

00:00 Speaker 1.: Thank you But, you know, it should be engagement, because otherwise, what are we doing? We're just giving marks for showing marks. Well, that's true, because if someone's got a lot of students, they're probably not going to perform as well anyway, so... Yeah, yeah. I have one chair that, though, if anybody who's different, I have a self-esteem, wouldn't talk to them, wouldn't go to them if I put up their hand, wouldn't answer these questions, which is completely snorkeling for the whole semester. I'm obviously building. I mean, you know, I try to be raised. You know, my intention is to be raised. She was like a therapy teacher, too. So she wasn't like mathematics for a year, so it's like hard science. Tell you what, I haven't taught any measure, but it's practically bimodal.
01:10 Speaker 1.: Thank you. It's very easy. It's like a lecture note and nothing goes in.
02:23 Speaker 2.: For a minute there, when I saw people leaving, I was like, what's happening? Oh, that was quite fun. All right, while I am
02:42 Speaker 1.: getting ready,
02:44 Speaker 2.: there's a little reading for you.
03:36 Speaker 2.: All right, how is life? In this class, you don't have to say, fine, thank you. You can say whatever it is. You need two more? So I was catching up with the tutor
04:05 Speaker 1.: and
04:08 Speaker 2.: I do understand that some people started late with the tutorial work and we will be giving you the assignment in the final week to do

over the break in the first week of the next term I know this feels very long but you know time flies so I wanted to make sure that everybody's caught up this is only a one-off we're not going to repeat this because you know I have to justify the extra work and so on so if you would
04:47 Speaker 2.: Like any help with the tutorials that you've had up to now because there's only one left Before the assignment and it's like I looked at it. It really is very easy Please do go to Erskine 248 now some people said all they have clashes now remember that This is not a two-hour session where you have to be there for two hours. This is a two-hour period where the tutor will be available. You can go at any time for five minutes, ten minutes, whatever time you need. There's not going to be any lecture or tutorial. He'll just be sitting there,
05:36 Speaker 1.: lonely,
05:38 Speaker 2.: hoping for you to go and ask him questions. Now, the second
05:43 Speaker 1.: question I had, this
05:47 Speaker 2.: might be related to the fact that people are asking about class reps, was is there going to be Power BI on the exam, on the test? Sorry, test. We have no exams in this class.
06:00 Speaker 1.: So the
06:02 Speaker 2.: answer is no. Because the test is based on the lectures, and I have mentioned that already, but I thought, okay, I'll repeat it. And please do email me with questions like that, because the tutors actually genuinely do not know. I don't test my tests with the tutors. I test them with other members of staff to see how long it takes them to do it and so on. So the tutors actually don't know the answers. questions so if it's anything to do with the course please email
06:37 Speaker 1.: me okay so in
06:39 Speaker 2.: relation to that because I know you stress about assessments today's reading what you got is is kind of a an example of
06:54 Speaker 1.: the description
06:57 Speaker 2.: that you might get a scenario that you might get for the exam. So you can use that not only in terms of what we are going to do today, but you can use it as a scenario to practice different questions on and think about it. And also, I will release a practice test next week where you can actually

practice It's going to be like a real test with marks and everything. And you can practice answering the questions. So that when in week six we see each other, the day before the test, we can go over the practice test. You can tell me anything that you are not sure about. So trust me, by the time we get to the test, you'll be ready. Now, talking about the test, we finally have a room.
07:55 Speaker 2.: received email confirmation, but I called just before I came to make sure because I was like, I can't go into this class and tell them that I'm still waiting. So we do have a room in forestry. It is for that Friday of
08:12 Speaker 1.: week
08:14 Speaker 2.: six. And I'm pretty sure it was 4 to 6.30, but the
08:21 Speaker 1.: test itself is only two hours.
08:23 Speaker 2.: So we just get a bit of extra time
08:25 Speaker 1.: to
08:28 Speaker 2.: collect scripts and so on. As soon as I receive the email confirmation, I will post all the details on Learn, which will be sometime after 5 o'clock today, because I'm teaching until 5. But after 5, I will post something on Learn with the exact details.
08:55 Speaker 2.: So it's on Friday, what
09:00 Speaker 1.: is it?
09:04 Speaker 2.: Friday 27th. Oh, that's a problem, because you have to be here for a test. Yes,
09:17 Speaker 1.: four o'clock. So,
09:25 Speaker 2.: I don't know, because tests are something that
09:28 Speaker 1.: you have to be available for.
09:38 Speaker 2.: Well, I have been saying that it's going to be on a Friday afternoon of week six, and it is on Friday afternoon of week six. So it is what it is. It is a test week, week six. It was in the course outline, and we have the lecture on Thursday, so there was no way it was going to be before that.
10:05 Speaker 1.: So please
10:07 Speaker 2.: make some arrangements. Are you traveling? Okay, email me with your flight arrangements, and we'll see what we can do. Oh, that's the wrong one. I got the wrong one.
10:34 Speaker 1.: Hold on.
10:55 Speaker 2.: All right, so today we are going to more or less finish with

11:02 Speaker 1.: any completely
11:06 Speaker 2.: new material. Next week we will also talk about visualization, which there would be some new material, but a lot of it you would have... had done already
11:20 Speaker 1.: in
11:22 Speaker 2.: tutorials. So, a much lighter session next week.
11:29 Speaker 1.: So, what
11:32 Speaker 2.: did we do last week? Yeah, so we talked about descriptive analytics, which is based on what is happening, and we had that very long session about student retentions and thinking about data sources and what we do with data and what answers it is going to give us. And actually, without realizing, we started talking about diagnostic analytics, because diagnostic analytics... is also based on current and historic data.
12:16 Speaker 2.: And... I just want to give you a little bit of intro what we are going to focus on today. So it tries to, diagnostic analytics aims to look for root causes or underlying reasons why something might be happening,
12:41 Speaker 1.: right? So you
12:44 Speaker 2.: would have noticed last week that when we were looking at the descriptive data, we were actually already formulating kind of a hypothesis and suggestions about what could be going on and then we were looking at the data to formulate some of these ideas about what could be the reasons why people are dropping off. So this is exactly what diagnostic analytics is. We'll cover a few techniques and specifically
13:19 Speaker 2.: a little bit more quantitatively looking at correlation and regression. So before we start, as I said, just Just to kind of give you some reassurance that you have sufficient number of examples to practice in addition to the student one, we have one here from the book. It's about a town in the US and what they've done.
13:51 Speaker 2.: because I was hoping that you'll be reading, but very few people were while I was getting ready.
13:58 Speaker 1.: So what
13:59 Speaker 2.: is this scenario about? This is also a way for you to start thinking how when you get... A description of a scenario in the exam, how do you read it? So remember the SOAR model, you immediately start thinking

about the problems. You know, what were the problems? What was this related to? So that would be the first kind of starting point, and then you try to think about what did they do, how did they use the data in this case, what were some questions that can be answered by the data they were collecting that they didn't have before.
14:49 Speaker 1.: And
14:49 Speaker 2.: then for you, practice, you know, what different ways you would be suggesting that they can use that data. So I'll just give you another minute because all of a sudden people started reading. Oh, you have only one? Oh. Does anyone need extra? Okay. No problem. We have. And there are, okay, sure. I'll come. I'll come. I'll come. Don't
15:24 Speaker 1.: worry.
15:29 Speaker 2.: Any spares going? I think there are a couple of spares over there.
15:35 Speaker 1.: Yeah. Plenty.
15:39 Speaker 2.: Right, anyone needing extra?
15:44 Speaker 1.: Okay.
15:51 Speaker 2.: That's for you.
15:53 Speaker 1.: I'll get one.
16:13 Speaker 2.: So again, all of these slides are available on Learn for you. Are they not?
16:18 Speaker 1.: Yeah, they are. Okay.
16:20 Speaker 2.: Just when I see people taking photos, I
16:22 Speaker 1.: worry. And I
16:25 Speaker 2.: immediately express my worry.
18:22 Speaker 2.: All right, so normally in exams, there is at least 10, 15 minutes incorporated for reading, but we don't have this luxury in the class, so we have to read a little bit faster. What is this scenario about? What is the problem that they were trying to solve?
18:48 Speaker 1.: They have a lot of data now after deploying the new system. They have
18:55 Speaker 2.: a lot of data now after they've deployed the system, but what was the problem before that? Why did they deploy it? Right, right. So they really were trying to reduce use of water and not actually used by consumers

necessarily, but they were trying to figure out whether they had wastage of water, right? Leakage or any other wastes. And they had no way of knowing because they said the meter were originally measuring it
19:43 Speaker 2.: monthly, right? So you get a monthly reading. And this is exactly what we were saying before, that the level of granularity and using averages sometimes, because when you get a monthly reading, it's quite hard to understand. I mean, you would know whether it was more than the previous month, for example. But if you have regular leaks. across a whole town,
20:17 Speaker 1.: this
20:18 Speaker 2.: is not going to be that useful, right, because it just tells you, oh, in this location, this month we have a little bit more, but over a whole month it kind of smooths it over. It's very difficult for them to identify, but more importantly, right, they're not just trying to identify leaks, they're trying to prevent
20:44 Speaker 2.: waste of water, right? Because when you have the hourly reading, they could actually detect issues
20:54 Speaker 1.: quickly and
20:56 Speaker 2.: prevent further And they gave an example where there was an industrial dishwasher going on or something going on for the weekend, and it could have used a lot of water and energy for a whole weekend, but because of the hourly reading, they were able to detect it and quickly stop that.
21:24 Speaker 2.: Of course, in this situation, we already are given the solution in terms of how they collect data. In the exam, you might actually be just given the problem with the monthly reading, and then you kind of have to think, well, what would be a way to do it? Or based on this scenario where they already have the solutions, I'm... I might say, okay, what kind of analytics are they applying to be able to solve their problem? So they have talked a little bit about that already.
22:01 Speaker 1.: So
22:02 Speaker 2.: let's see as a kind of practice reading, what kind of analytics have
22:09 Speaker 1.: they used?
22:12 Speaker 2.: What is their descriptive analytics that they have used to address this issue? And what have they found? What were the results?

22:44 Speaker 2.: So yes, and this is where the diagnostics comes today because they're looking for anomalies or outliers, something that is different from the expectation or something that is different from what they would consider normal use for their particular user. And so this is only possible because they have the values over different periods, so they're comparing the current value with previous values at the same time, or same time of the day, right? And it is able to detect when, so it's comparing them, right? And it's able to detect if it's different, and the solution is, if it is obviously,
23:41 Speaker 2.: town property, they can respond directly, but if it is a private customer, then they're notifying them so that they can check it out. And in addition to that, they are saying, so customers can also use self-service, so they can set their own alerts.
24:05 Speaker 2.: and they can check it on the portal. So they're also empowering the customers to be using this data for their own personal management of their water use. And what was one kind of surprising
24:26 Speaker 1.: discovery?
24:29 Speaker 2.: for them. It's kind of further down. One surprising discovery was that actually they were thinking, and this is what happens with the monthly readings, you don't know exactly where the wastage is coming from, so the town was spending money to encourage water efficiencies, and they were offering rebates on
24:55 Speaker 2.: installing particular types of toilets that use less water and rain barrels and so different ways that they can reduce the use of water. But they discovered that actually
25:10 Speaker 1.: people were not
25:12 Speaker 2.: really using that much. So this was not the right direction, if you like, to address it. It wasn't the use, it was
25:22 Speaker 1.: just these wastes.
25:24 Speaker 2.: they were creating the higher water
25:28 Speaker 1.: meters. So
25:30 Speaker 2.: then they said, okay, well, it was helpful because then they know how they can prioritize their actions, right? And it also helps them to react quickly. Another thing that is going on here, which is not related to today's class, but it is related to previous, and remember in the test, I'll be asking

questions about different different topics, and they're talking about having to link two disparate databases. And they have a billing database and the meter reading database, and they need to make that, bring that together to make it presentable. So we might want to think, okay, I probably would have given you a bit more information, because you probably don't know what happens in a database for billing. database for billing would... would be based on the billing period, which is probably still a month, right? So we'll have the water consumption for the month and the rate that that customer is paying water for and how much they have to pay, right? So we'll have all the customer details and so on. And then you would have another database now that is reading it every hour, and it will have the customer customer as well, the address,
27:06 Speaker 1.: the
27:07 Speaker 2.: water use by hour. And so you can see that it's quite a lot of duplication apart from the accounts that maybe they're directly debiting and so on. So apart from the payment details, there is quite a lot of duplication. And what is the risk?
27:29 Speaker 2.: What were some of the issues where we have these disparate, not single source of truth? Well, yes, we can have, the whole thing is about data not being clean, right? So let's try and think. I mean, first of all, we have the customer details repeated in two different databases. So the likelihood, if somebody changes some of their details, they have to change it in two different places, so it actually increases the risk of getting it wrong, or, you know, even if we were talking about the addresses, right, even if you get the address slightly wrong, spelling mistake in one of them, that would not be able to fix it, and I actually took a picture, but I didn't, on my phone the other day, I was at the bus stop, and you know how they have these machines showing when the bus is coming? And for the same one, there were three different
28:43 Speaker 1.: times.
28:48 Speaker 2.: And this happens a lot at the university stop, around 8 o'clock when I usually leave, because... I checked why this is happening, because you know, I'm just really... nosy, and the way they do it is they're showing, they change the way they're showing based on a GPS, right, so the real time, but they still have the time that is based on some old timetable, and so you see it actually is incorrect because it's showing it comes in two minutes, and then it's

showing it comes in five minutes, and if you're hoping for the one in five minutes, good luck, because as soon as the two minutes one passes, the other one disappears. So this shows you also the risk of duplicate records. If you have a tiny thing different, if you try to merge those databases and think about a customer record, it might say, well, this customer
29:52 Speaker 2.: has two names, and this one has three, and if the ID, if the unique identifier is not linking them right, it will think that it's two different records. So it is happening a lot. So you could have had a question about, you know, what are potential risks with having these two databases, and you would need to discuss it. So it's not something we haven't talked about, but it's not there for you to pick it out and
30:21 Speaker 2.: So some of the questions would be in the description and some of the answers to the questions would be based on what you've learned in different classes. All right, any
30:37 Speaker 1.: questions?
30:39 Speaker 2.: Because I'm now going into this test preparation mode. So the format would be, you would be getting kind of a short one-page scenario description like this, and then about two to three questions with lots of ABC to make sure that you know what you are meant to talk about. All right, so diagnostics, as I said, is really...
31:16 Speaker 2.: In many ways, you can use the descriptive analytics measures, but you use them for the purpose of identifying underlying reasons. So a lot of what we have talked about last week in terms of descriptives could give you some ideas about underlying reasons, and today I'll cover a few more techniques. So there are two...
31:42 Speaker 2.: There are two types of approaches. So the first one that somebody already mentioned with this scenario was that we look for outliers or anomalies,
31:55 Speaker 1.: right? But So
31:59 Speaker 2.: last week, for example, we talked about how you see the outliers, right? When we had the picture with the box and whiskers plots, or if you just have your spread, you could decide which are the outliers and identify what is different about them. But anomalies is a bit different. It is actually what was happening in this scenario where you are comparing a value to the

expected. And it's different. So.
32:33 Speaker 2.: So an outlier is similar to an anomaly, but it is actually kind of an observation point that is different from the overall distribution or from the average, and it's a bit too far out of what is expected in terms of the spread of the observations. While anomaly is based on an expectation that may be based on data, maybe based on a business rule. So it could go beyond simply using data. So the second
33:16 Speaker 2.: So the first one was looking for anomalies and outliers. The second direction in terms of diagnostics is to look for relationships, which a lot of you were doing last week without realizing some relationships between different
33:34 Speaker 1.: variables.
33:37 Speaker 2.: Now, did we talk about what variable is? So variables are used to store data about the various values. For example, it could be something like price of a product. If it varies, that would be a variable.
34:00 Speaker 2.: It could be a text like a customer name would be a variable. It could be a town name. So anything that stores values that varies across observations. It's a variable. And so we look for relationships between those variables. So for example, we were looking last week, you were kind of suggesting that there may be relationship between student performance and dropping off. And that one is a variable, because they either drop off or not, right? It's a binary, and so we were looking to see whether there was some relationship between the rate of dropping off and student performance, and so we can actually do either some statistics to look for these relationships, or actually drill down, start looking at the details of your descriptives according to categories, which is exactly what we did last week.
35:09 Speaker 1.: So first
35:12 Speaker 2.: of all, on anomalies and expectations, just so that you can see how it's not simply looking at your scatter plot and picking the outliers. In marketing, you might actually be looking at the increase in sales, and you might be checking against your competitors. So you could
35:39 Speaker 2.: So, normally, it doesn't have to be always negative. So, for example, with the water usage, we
35:45 Speaker 1.: were looking for bad

35:49 Speaker 2.: results, right, where people are using way more than they should. It could be positive. In this case, the sales grew twice at
35:59 Speaker 1.: the
36:00 Speaker 2.: rate of competitors. So, it's happy news, but we might not know what was behind that, you know. Why did we get this good results? So then we're looking for possible explanations based on something that is different also in the data. And in this case, we had a suggestion which would have to be checked. with the data, that a new product was introduced, and that actually has sold really well, so it contributes to this increase in sales. Now, I really want to emphasize that
36:46 Speaker 1.: these
36:49 Speaker 2.: possible explanations
36:50 Speaker 1.: are
36:52 Speaker 2.: based on data, you would have to look for data and see, okay, what happened in this period that seems to be tracking higher in terms of sales? And then you would see that for that particular product, so this would be a drill down approach, right? So you'll be looking at the sales for each product and you identify that this particular one is really high and contributes to the sales compared to other products.
37:20 Speaker 2.: So it doesn't just come magically, this explanation. It is based on data. So that will be a drawdown approach on product categories. This one with the turnover rate, I think one of your tutorials was related to turnover. I'm not sure. Might be the one that's coming next week.
37:42 Speaker 2.: But basically it's the rate of people leaving the job, so this is something that is quite important in human resources and in operations management. And so the expectation is... for this business, they had expectations that it will be stable, more or less the same, right? And what they discovered is that it's 25% higher than what was expected. So then they will be looking for the, you know, what could have caused that, you know? And you'll be looking at the data and think, okay, because we had a similar example in class with the workloads, right? So you might look whether the workload has increased, you might look at whether some management has changed, so maybe the managers are causing some stress, or what else is happening? And after you investigate all of these, they have identified that actually possibly

based on the data from the exit interviews, other businesses are in need of skills and are offering higher salaries. So this is actually an external source of data where you're looking at, you know, what is the going rate for these
39:17 Speaker 2.: And so similarly, okay, with accounting, a lot of it is about compliance. So compliance with standards and compliance with regulations, and here we have an example where there was a rule about how particular expenditure needs to be allocated, and they found that actually they have put it as expense instead of
39:50 Speaker 2.: capitalized. So you could actually also look for in the data for things that you wouldn't expect to see in a particular category. Very, very common with auditing to look for anomalies and outliers. It's not always about what's the higher, what's the lower, it could be comparing texts for types of expenditure or any other anomalies that you wouldn't expect that category. And in finance, actually a lot of unexpected things usually happen, but so anyway.
40:32 Speaker 2.: They had expectations about the price of bonds as it relates to the treasury yields and so then they actually fell.
40:44 Speaker 1.: And
40:44 Speaker 2.: so they weren't expecting that and they were looking to what was the underlying reason for this poor performance. So they found that there was a change in the grade, the risk rate, and obviously that had an influence. So very often with diagnostics, it's about
41:06 Speaker 1.: drilling
41:08 Speaker 2.: down and thinking, well, you know, what is different about the data and what seems to have contributed to this result. So looking for these underlying changes that have led to the change that was unexpected. So very much the same techniques as what we have discussed in descriptives, but with the purpose of looking for causes.
41:47 Speaker 2.: But in addition to these standard descriptives, we could also have some statistical measures and analysis. You could have statistic descriptives measures which we covered last week in terms of minimum and maximum. This is also what you've been doing in tutorials with your profitable products. And so similar to what was in the table when you discover what is profitable or least profitable or how it has changed, you can see how it contributes to the overall performance.

42:33 Speaker 2.: Okay, in terms of statistical, this could be new to some people, and it could be very boring to others who have done statistics. But in this class, we really need to ensure that everybody knows it. One thing that I find still a little bit baffling is that very often people use correlation and regression interchangeably. This is actually not the same thing, so that's why I've made sure to have this table so that you are very clear on the difference. Yes, there are related concepts, but correlation
43:12 Speaker 1.: is
43:15 Speaker 2.: a Coefficient that shows us how two variables are related, whether they have a relationship at all, whether it's a positive or negative relationship. And so
43:34 Speaker 1.: you
43:36 Speaker 2.: may have... So I've got some additional slides because I know that you might not be getting the bulk. And so it's an association relationship. So if one goes up, do we have an associated change in the other variable? So for example, if... one variable goes up and the other one goes up, we have a positive relationship. And if one goes up and the other one goes down, we have negative relationship. So that is what correlation does.
44:15 Speaker 2.: Correlation is not in any way causation. It simply shows some relationship between variables, and if you go on the web, I was planning to do this, but I always go over time, there are some really funny examples of correlations, you know, that
44:38 Speaker 1.: people have used.
44:42 Speaker 2.: So here is no relationship situation. So for example, you have the completion rates and the student height. Okay, because we talked about, I know it's funny, but a lot of them are funny, where people find correlations, and I find this also in research, they find correlations between some random variables, and they say, oh, here, here, this must be causation. I'm like, no, just, just. Random correlations, so here we have, so you do have to question correlations, you know, if you find that two variables are related, you need to find also whether this makes sense from the business point of view to be actually relevant. So we have here clearly the student height had no impact whatsoever, they're not related to completion rates, you'll be very happy to know.

45:42 Speaker 2.: And then we have completion rates in terms of the whole degree and course failures. Now clearly, course failures contribute to whether you complete your degree or not, so this one.
46:04 Speaker 2.: is a no-brainer, and it is definitely confirmed. So it's a perfectly negatively correlated, which means that when you increase the course failures, the completion goes down. Or when you increase the completion, the course failures go down. So I've picked things that are obvious, but
46:30 Speaker 2.: also to show you that when you see correlations, it needs to also make sense. And then we have perfectly positive correlations where when one increases, the other one also increases. So these are, because this is from another book, American SAT scores, but if you don't know what the SAT score is, any score on a test you can think about. So they are tracking the scores on tests, and then they're looking how that's related to the completion rate.
47:06 Speaker 2.: very similar to what we were discussing last week, that you could actually test whether the NCA results are correlated to people dropping off or completing. So in this case, because its completion is positively correlated, if you had taken the other one dropping off, you would have negative correlation. Any questions on correlations? So correlation always is a symmetric relationship between two variables, and there's no causation implied, right? It simply shows that the two are related, and when one goes up, the other one also goes up.
47:56 Speaker 2.: So then we have also regression. In regression, now regression can be used both for
48:06 Speaker 1.: diagnostics,
48:09 Speaker 2.: to look for patterns and relationships in the existing current data and historical data, or for prediction. So right now, we're only going to talk about it in terms of diagnostics. You will be talking about how you use it for prediction next
48:28 Speaker 1.: term. So with
48:33 Speaker 2.: regression, there is an implied causation. And so you have an outcome or target variable, so something that is going to be changed, and import variables that actually contribute. So you are saying this contributes to that, so it's no longer symmetric, it's one directional, right? And while with the correlation could go both ways. technically speaking, with regression we are

saying, well, we actually want to assess how
49:13 Speaker 1.: some
49:16 Speaker 2.: imports are contributing and are related to a targeted output. And so here we have, we want to see out of different factors, pay, training, convenience of working hours, flexibility, ability to work remotely. Okay, which of these factors are associated with employee turnover? So which of them actually? So the turnover is the output that is being targeted, and then we look at different.
49:53 Speaker 2.: factors that might influence it. Now, because I have been very, very naughty, I'll just do the last one and then we'll have a break, because I keep forgetting in different classes whether we're having a break or whether we... Moving on. So finally, you could also test our hypothesis, right? So you could say, we think that one of the reasons is that we had this new product, so you might have not seen it in the data. You know, you could have business ideas, and then you could actually test it. So for example, here in the... ton of carry with the water, they actually had some hypothesis that were not confirmed, right? So they had hypothesis that some people were using too much water, it was unreasonable, but they discovered that actually that was not the case. And the wastage was based purely on leaks or accidental
51:00 Speaker 1.: use of commercial
51:04 Speaker 2.: water using appliances, right? So sometimes you may have hypotheses based on what makes sense for the business, and sometimes they sound very plausible. And what this scenario shows is that we always do need to back it up with data, otherwise you'll be taking the wrong business solution approach. Such as they were putting so much money on trying to encourage people to use less water when they already were doing that.
51:43 Speaker 2.: All right, so let's have a break. Actually, on time, until 11, you can come talk to me so that I'm not bored, and then we'll continue with the class. Or at least get up and move. Otherwise, what's the point of having a break?
53:36 Speaker : Thank you. Yeah, yeah. The problem is that there are headlines. Are different from the time-taking guys, you know? So the time-taking guys are still shooting heads.
55:06 Speaker : Thank you. Thank you. Thank you.

56:23 Speaker : Thank you. Thank you.
57:10 Speaker : Thank you. Thank you.
58:12 Speaker : Thank you. There's so much going on.
59:13 Speaker 2.: And this one is also a type of practice, right? And
59:17 Speaker 1.: I will also say
59:20 Speaker 2.: next week when we finish. And all the classes and I know what we've done, I will also release the
59:25 Speaker 1.: practice. Okay. And the test date, it still says TB8 to be announced? Yeah,
59:34 Speaker 2.: because as I said, I just got it confirmed, so I will put it at 5 o'clock. You must have not been here when you started. At 5 o'clock, I will post something about the test. It's not going to be in your timetable because I've booked
59:47 Speaker 1.: the room. Thank you.
01:00:23 Speaker 1.: Thank you.
01:00:59 Speaker : Thank you. Thank you.
01:01:38 Speaker : Thank you. All right.
01:02:08 Speaker 2.: This break's very, very useful.
01:02:13 Speaker 1.: So
01:02:14 Speaker 2.: I had quite a few questions that I had answered, which is a little bit of a worry, but remember that the lectures are... The lectures are recorded, so you can watch it over and over so that you know that the answers were given. So a common question repeated that I got was, are we going to have a practice test? What is the answer? Yes, because next week when we finish the class, I will release a practice test. And this scenario today also you can use as a practice test. questions we've done in class for discussion
01:03:01 Speaker 1.: question
01:03:04 Speaker 2.: number two is is the test going to be online no it is going to be in a room and you will be writing in a booklet by hand, and I will have to read it. I know. I know. That is why I will make sure that the questions are such that you don't have to write a lot so that I don't have to read a lot. Yes, we will cope with it. Because at the moment, there is no option for us to do it on computers.
01:03:43 Speaker 2.: because there's no way to infigilate it, and also we don't

have big enough labs for this class. So we'll have to be on paper. And then a question that probably was related to the fact that I didn't make it clear. Sometimes the reason people also don't know is because you really need to listen to me. So this was more of a clarification. And I think that was something I didn't
01:04:19 Speaker 1.: specifically say.
01:04:21 Speaker 2.: It was like, are we going to have one question, one scenario, one question, another scenario? No, there's going to be only one scenario to read, and all of the questions are going to be based on that one scenario. So you would only have one scenario to read. So I didn't think about actually making that very clear, but it would become clear with the... practice test as well. So please don't wait for the practice test to start practicing and getting ready, because as I said, we've done the student one, we've done the, you have this one, and you can practice all the questions we've done in class, because the questions you are going to get in the test will be very similar to something we've done in class, if not exactly the same. I don't believe in surprises, so you'll see.
01:05:25 Speaker 2.: Okay, so a little bit more, because I showed you a bit more about correlations, and now a little bit more about regression. So regression, I said... you have a target or an output that you want to achieve and you look at the different factors that will contribute and influence that output, right? So the output or target is called dependent variable because it depends on these other factors and the variables that contribute or influence the output are called independent.
01:06:11 Speaker 2.: So they help to explain the dependent variable. And now in various books, it's M, B, A, B, B, beta, all kinds of words, letters. So you don't need to memorize
01:06:35 Speaker 2.: the letters or the formula But you do need to think about how we actually get these regressions to see whether a relationship and whether a contribution is supported.
01:06:54 Speaker 1.: So we
01:06:55 Speaker 2.: are trying in a way, this is quite important, and we were talking about this in the break. Regression can be used to predict a future value, but But in the context that we have now with diagnostics, what we're

trying to see is whether something has contributed to the outcome based on the current, so actual information. And so what you would get is, and you'll be doing this in the next term.
01:07:34 Speaker 2.: when you do conformatory and predictive analysis for regressions, but what you have is you have your observations and you try to fit a line to show this linear, if you have basic linear regression, to show this linear relationship. And so you have...
01:07:56 Speaker 2.: So you have two elements in addition to your two variables. One is the
01:08:04 Speaker 1.: slope, and
01:08:05 Speaker 2.: the other one is the intercept. So the intercept is actually the value of your dependent variable when your independent is zero. So that's why it's the intercept, where it starts.
01:08:26 Speaker 2.: And then for the slope, it actually shows the degree of how steep the change is. And the way it's calculated is for every unit of change in X, what is the, you know, how much do we change Y, right? So, but I don't need you to remember all of that. I just want you to really think because sometimes you get the results. and you don't know what they mean. And so I want you to kind of think that obviously if you have a more steep line, then it's stronger.
01:09:10 Speaker 1.: In that case, Y is the target, right?
01:09:13 Speaker 2.: What we are targeting. Yeah, Y is the target. And X
01:09:15 Speaker 1.: is basically what contributes. Yes. Yes. We are measuring. Yes. The steepness shows that if it is steep, then probably the impact is higher.
01:09:26 Speaker 2.: Yes, yes, yes. Yeah, that's what I was going to say, that if it's steeper, it's a stronger relationship, and if it's kind of a, and so you could actually start getting some ideas about what this regression is showing you.
01:09:45 Speaker 1.: Now,
01:09:46 Speaker 2.: Also another very interesting conversation in the break was about hypothesis testing and how do we set hypothesis, right? And remember that in this particular case, we're only looking for hypothesis not as prediction for the future, but as explanation for what has happened in the past, right? So
01:10:10 Speaker 1.: we
01:10:11 Speaker 2.: can... So the question was, well, how do we know which hypothesis to test? There are two ways we form hypothesis. The first one is if

you've got some business insight as to what might be some contributions, for example, the business have said, well, you know, we introduced this new product line, let's see how much it contributes to the overall sale, and then you go look at the data and
01:10:45 Speaker 2.: it tests your hypothesis whether that particular product has led to the unexpected return.
01:10:54 Speaker 1.: But
01:10:55 Speaker 2.: another way is if you really don't have an idea, to do this drill down descriptives and check what do we know from the data and what seems to indicate a relationship and then you can test that relationship statistically.
01:11:16 Speaker 2.: So there are different approaches you can take to hypothesis forming and hypothesis testing. Well, ultimately, this is why it's called business intelligence and not statistics. You try to look at the data and you try to use business sense and it's an iterative process, looking at the data, forming hypotheses, looking for more to make sure that you have good understanding of what the underlying causes are.
01:11:50 Speaker 2.: Okay, and again, I really want to emphasize, because I'm also not teaching in term two, and you'll be looking at regression in term two as well, as a predictive model, and so you need to think about the fact that...
01:12:12 Speaker 2.: With diagnostics, they're all deterministic models, and they're deterministic because we know the data, and if you are testing your regression, it is based on actual data and actual contribution of a factor to a target, not forecasted or predicted contribution. So I see confusion.
01:12:38 Speaker 2.: And confusion shows that you're thinking, which is really good. So this is our actual data,
01:12:46 Speaker 1.: right?
01:12:47 Speaker 2.: And when we fit the regression to the actual data, this relationship, this regression is real. When we say, oh, let's now think about the future when we go on, do we continue this relationship? That is now speculation and prediction. But this is real, that's why it's called deterministic. And when you do correlations specific to something that's happened, it's deterministic because it's already happened. When you say we now speculate that this correlation will be the same in the future, that is probabilistic, right? So

very important distinction because I get quite
01:13:37 Speaker 2.: when people just throw regressions and correlations in all sorts of ways, and you do need to remember how you are actually using it. So in diagnostics, we're using it to find these relationships based on the real data, their current relationships, and that's why they're called deterministic. We're not judging likelihoods of something to happen.
01:14:05 Speaker 2.: And so all events are known beforehand, and so it is actual current data. All right, so now a little bit of a discussion point, another practice for the test, because that seems to be all we care about, it's a bit like a test gym.
01:14:26 Speaker 2.: This could have been a longer scenario where I explained to you how the banks work, but because this is something we can actually sort of relate to, I've used it as a shorter discussion. So we have a context for a loan provider, a bank, and they
01:14:52 Speaker 2.: One of their products is mortgages, and mortgages are loans that are used to borrow money against actual physical property, right? And so if you are the borrower,
01:15:08 Speaker 1.: you
01:15:09 Speaker 2.: agree to repay the loan. You have specific terms in terms of the period of time, in terms of the interest rate, and how much you are going to repay. And if you meet all of these terms, then always well, successful repayment, but if you fail to do it, that is called default on a loan. And so then the bank obviously has to use the property as a security, as a compensation. So this is the context. It's a business situation. Now I want you first to, before we start with the analytics, just to make sure that you're aware, what is potentially an outcome that we want to, from the bank point of view, to explain? to know why it happens,
01:16:17 Speaker 2.: why people are defaulting on mortgages, because what do you think banks want? not to default, right? This is quite important because sometimes I talk to people and they think banks are quite happy for people to default. No, because they're in business to provide this service and ideally they would like people to repay the loan. You pay a lot in interest so they make money out of that and they don't want people to default. So in a way, the default is the target variable. It's a little bit like, this is why I said, when you get something, it will be very similar to something else we've done. So it's a little bit

like when we talk about students dropping off. We don't want them to drop off. Well, this is obviously
01:17:20 Speaker 2.: worse for the people, but it's a similar concept, right? Because you have a default on the loan, people dropping off from what they had signed up to do. So try to always think about that when you see a new scenario, what are they trying to achieve, right? So the target is the default, and we want to see what
01:17:47 Speaker 1.: what would be
01:17:48 Speaker 2.: some input attributes, which probably would, if you get the test, there will be some indication in the scenario, but this is a context that we can all relate to. What are some input attributes or factors that you think could
01:18:08 Speaker 1.: contribute
01:18:10 Speaker 2.: to people defaulting? Okay, so in this case, they're taking a loan. It's very specific, so we don't have variation there. It's specific for a house. Mortgages
01:18:30 Speaker 1.: are for property.
01:18:36 Speaker 2.: The value of the house would affect the... Okay, how, why do we think that? So if the value of the house would determine
01:18:49 Speaker 1.: how
01:18:51 Speaker 2.: much the bank would be willing to give, but, so you're right that it has an impact, but it has an impact on the value of the loan, once that has been
01:19:07 Speaker 1.: set.
01:19:10 Speaker 2.: how would the value of the house contribute to people defaulting? So if the value of the house decreases, your loan is still the same though, how does it influence you paying your monthly installments? That is true, and that might influence whether you make a profit or whether the bank would make a profit if you default, but how would it influence you paying your set amount every month?
01:20:00 Speaker 2.: Well, it's easier if we discuss it, right? Because, no, this is really good because people start thinking about different things and you need to think, what does this influence? And you see, because this is a concern for buyers, right? They're concerned, oh, what if I get a loan and then the value drops? Yes, it is a concern, but it actually doesn't affect your paying the

installments because how much you are meant to pay.
01:20:30 Speaker 2.: It doesn't change with the value of the house. All other debts that you might have. So if you take on further debts,
01:20:51 Speaker 1.: yes,
01:20:54 Speaker 2.: that would influence it,
01:20:55 Speaker 1.: but
01:20:58 Speaker 2.: not if you had existing debts before the deal. So you might say, okay, if they have an existing debt, that would influence. So a lot of the things that you are talking about are relevant in when they set up how much to give you. Because they will take that into account, right? They'll say, oh, they already have an existing debt.
01:21:24 Speaker 1.: So you
01:21:26 Speaker 2.: need to think that, and I really like this conversation because you come up with lots of great ideas. They're not directly related to the default. They're related to, like you think as a consumer, what are the things that would make it hard for me to get a loan? to get the loan, but this is after you got the loan, how do you pay it off? And so the main contributor is really what do we use to pay the loan?
01:21:56 Speaker 2.: Income, right? So what is the thing that would affect it most would be any changes in the income. So then what changes in income? Getting fired. Oh, yes, yes, yes. That's very hard to predict, but it has immediate effect. So getting fired, and so how would then, how are we going to put that as a factor to kind of relate to default, getting fired? You know, how are we going to look for that? How are we going to sort of think, is this person going to get fired?
01:22:39 Speaker 1.: So
01:22:40 Speaker 2.: unemployment rate
01:22:42 Speaker 1.: would
01:22:42 Speaker 2.: influence defaults, right? So if we have, so that's more of a macro variable. For the individual customer, they actually look at the type of industry and how safe it is in terms of having a job.
01:23:03 Speaker 2.: So for example, there is a category whether you are self-employed versus employed. And people who are self-employed are considered to be at higher risk because well, you don't have the security of the employer,

but certain industries are actually considered to be more risky than others because in certain industries there are more jobs. So even if you get fired or you lose your job through no vote, if you're in an industry where there are lots of other jobs, you could find another job, right? So the risk
01:23:44 Speaker 1.: of
01:23:45 Speaker 2.: being an employer is actually based on the industry and your personal history as well. Okay, true, true. So the thing is that they, okay, I see a little mix up here between do they give you a loan, right? So these are the things that influence whether they give you a loan versus once they've given you a loan, what is it that might make you default? So your previous three months before they give you a loan,
01:24:26 Speaker 2.: is something they will consider before they give it to you. But it could be that you say, okay, your spending pattern would influence the default, right? Because if you're an impulsive spender and you have lots of credit cards that you max out, so your prior thing might... might make you, it would contribute to a potential default.
01:25:01 Speaker 2.: Again, so this is really good one because it shows that some of the factors are specific to the person. So what job they're in, how long they have been in that job, so the job security. And some of them are outside their control. So we talked about general unemployment rate would affect default rates. the high level, and as you were saying, sorry, what was it? I don't want to say it wrong.
01:25:39 Speaker 2.: So the change in the official cash rate and the change, so the fixed rate is actually kind of following to the change in the official cash rate and the borrowing rate from overseas markets, because we are a small country and we can't rely on our own.
01:26:08 Speaker 2.: Right, so anything that could affect clearly at the individual level, anything that can affect your income. is actually a risk factor, right? So they could see, then the banks can actually see, well, how many times this was through a medical issue, how many times this was through unemployment, and they could actually see which industries we have more unemployment in, and which, actually one thing that I find quite interesting is that age has a very high
01:26:52 Speaker 2.: Why is that? Do you think it's better to be younger or older? Younger, because you have longer to pay off and also easier to... to find

different jobs, right, to reskill or whatever. Of course we're talking in general, right, so each factor would have its own influence and it may be compensated. So for example, you might say, okay, well, you're older, so that's riskier for job security and time you have to pay, but you have
01:27:38 Speaker 2.: more history where you've been paying things off, so you know, safe that way. So they will be, all of these factors are treated as contributing, and then ultimately, when you try to see which one contributes, they'll be checking all of them and kind of balancing them, if that makes sense.
01:28:05 Speaker 2.: So for descriptives,
01:28:08 Speaker 1.: we could
01:28:08 Speaker 2.: look for patterns in terms of what are the, you know, what data are we going to have? We're going to have tons of data from previous loans. And we will have... actual results, we would know who has defaulted and who has not, who has paid off and who has defaulted, right? So we would be able to actually see, well, out of those who have defaulted as descriptives, you know, what is
01:28:45 Speaker 1.: the top
01:28:47 Speaker 2.: two industries, right? And then that could give us sort of ideas for hypothesis, for diagnostic to see what is the contribution between the type of industry and a default.
01:29:07 Speaker 2.: Because maybe they're the two top industries, but the industry has a relatively small impact
01:29:13 Speaker 1.: based
01:29:15 Speaker 2.: on the overall data,
01:29:17 Speaker 1.: right?
01:29:17 Speaker 2.: So then they might say, okay, let's look at the
01:29:21 Speaker 1.: age
01:29:23 Speaker 2.: range and in different categories for the age, where are the defaults mostly falling, right? What's happening there? And things like, do they have kids? How many? That would probably also has influence. So all of these things, then we can first look at descriptives.
01:30:04 Speaker 1.: that the borrower is taking the loan. Right, so,
01:30:07 Speaker 2.: okay, so a mortgage is always given against the value of a property. You're right that in most cases when you're buying it, but it could be

that you have a property that you bought and you have
01:30:22 Speaker 1.: it outright, and
01:30:25 Speaker 2.: you need some money, all of a sudden, you might not be buying it, you might have had it for 10 years. with a paid off mortgage, and now you need money for some reason, and you can take a loan against the property. So I think that sometimes, this is what I call people thinking at an instance level that we can relate to,
01:30:52 Speaker 1.: because we...
01:30:54 Speaker 2.: we would take a mortgage if we have to buy, because we don't have this kind of money. But some people have inheritance money, some people have safety enough to buy a house first, they don't take a mortgage, but later on in life.
01:31:10 Speaker 2.: things happen and they need more money and they might need to borrow against the house. So the mortgage as a instrument, as a product, simply says that you are borrowing against a property. Whether you are buying this property and paying a deposit or whether you have the property already and you're borrowing against it is kind of... It may be relevant. It might be a factor. Yeah, it might be a factor. So you might say, okay, do you have a business if you're self-employed? This is why they're considering them to be more risky because they very often borrow against the house to keep the business going.
01:32:11 Speaker 2.: Yeah, yeah, so you could, again, you could check your existing data and say how many were borrowing it for investment property, how many for the house, how many already had a house and were borrowing against it to help with the business, and then you could say, okay, well, this particular reason is more risky. So, for example, that is already the case. Investment properties are considered to be more risky.
01:32:41 Speaker 2.: And so it's treated as a riskier option and they're asking for higher deposits, right, for that reason. Yes.
01:33:16 Speaker 2.: You mean from the customer who's got a loan, if they've signed a contract? How could they be unwilling? I'm sorry, I'm just not following.
01:33:36 Speaker 2.: Right, so this is then, because it's very hard to predict People have signed a contract, right? So they have said they're willing to pay, right? So how are we going to look in the data whether people are willing to

pay or not? Remember, this is descriptive or diagnostic, so it has to have happened already. So you need to think, if people are not paying off their loan and they had an investment property, what else was going on? Like, what would they... What were they using the money for? What did they need the money for? I think this is a better way of looking at it because when you start with these kind of more psychological variables like willingness, it's very hard to judge that. Were people unwilling or were they unable?
01:34:32 Speaker 1.: Your past history. Yeah,
01:34:35 Speaker 2.: yeah. So always good to find an objective measure rather than things that, because then you start getting things like, oh, I really like you and you seem to be willing.
01:34:48 Speaker 2.: Yep, let's try and be a bit rational. It's bad enough already with statistics. People fudge statistics all the time. Right, so hopefully now this kind of makes you think about how do we form these hypotheses about relationships and all of them then have to be confirmed based on the data we have. So in this case, you could use regression to say, based on the data we have, what is the relationship between having investment property and default, right? And then it would confirm your idea that people with investment property is less willing or less whatever, right? more default. So that would be the way to go with it. And I think not enough is done about testing existing data. We always think about forecasting future, but actually a lot is there to be learned from the actual data. And to be fair, financial institutions, because I've worked with the banks for close to 15 years, they really use the data for that. because lots of compliance.
01:36:08 Speaker 2.: All right, well, four minutes. I don't know how I'm going. I wanted to talk specifically this week and next week about different applications, because this is business intelligence, right? So different applications in different disciplines, and marketing is a really big area where data is used a lot for different purposes. So when we talk about marketing, we normally think of
01:36:39 Speaker 1.: selling
01:36:43 Speaker 2.: or promotion,
01:36:47 Speaker 1.: right?
01:36:47 Speaker 2.: So we either think they are advertising to us a lot to
01:36:53 Speaker 1.: make

01:36:53 Speaker 2.: it convincing that we want it or they sell it. But actually marketing also is related to all activities that create a product that is valuable to customers and it's affordable to the business. And all the activities, so it's actually quite a lot broader concept than what we experience, which is the advertising and promotion. And so it's also related to understanding, because in order for something to be valuable to
01:37:34 Speaker 1.: customers.
01:37:38 Speaker 2.: we need to understand what their needs are because value is related to how well something meets my needs, right? It's quite subjective. Something that would be valuable to me might not be valuable to you. And so this is a really important also area in marketing, understanding their customers, understanding the target audience for a particular product, and what are the elements of the product that are important to the customer.
01:38:17 Speaker 1.: So
01:38:18 Speaker 2.: what are examples? For example, you might have a group of customers that is what we call more price sensitive, right? So people who... Maybe on restricted budget or care more about the price because I have found that that's not always the case. There are people who have a lot of money and still make decisions mainly on price. So this is kind of a personal value for different customers. At the other end, I very often see people who don't have their own money like teenagers who are completely not price sensitive because the parents can pay anything.
01:39:02 Speaker 2.: This is just how that. segment of customers feels towards that value. Similarly, there's quite a lot of conversations now about sustainability as value, right? When we buy something, do we care about how it has been produced, what is the ethics behind it, what's the quality of the products that has been used, how much consideration has been given to the environment. So we all say that we care about it, but when you have to buy something, would you pay extra for that? Or you say, well, I care about it, but I'm very price sensitive. So these are the things that marketing professionals have to figure out for their company for their product and for their customers. And then there's no point having a great product that meets these needs if you don't
01:40:08 Speaker 1.: make
01:40:09 Speaker 2.: customers aware that this product exists. So this is where

the promotion awareness comes and then any activities they performed to make us want to buy it
01:40:25 Speaker 1.: now. So,
01:40:28 Speaker 2.: sometimes we resist. Okay, so the famous four Ps, in fact, very often talked about on the news for Sometimes no reason, but refer to product. So marketing looks after, you know, is the product offered to customers, does it meet their needs?
01:40:55 Speaker 2.: Price, is it at the price? So remember, it's always externally looking, right, to the customer. So is it at the price that the customer would be willing to pay? Is it offered in a place that the customers would be buying it? You know, what is the best place to sell it? And finally, the final P, which we all know about, which is promotion communication, so communication and techniques to convince us to buy. That we need it, and we need to buy it.
01:41:37 Speaker 1.: So,
01:41:38 Speaker 2.: how are these four elements related to each other? So, first of all, you need to think that in order to have a product that meets customers' needs and wants, well, this is clearly very closely related to the price. Because the more needs and wants you are satisfying and you're making the product better and better, the higher the price. So if your customers are very price sensitive, you have to say, okay, well, actually the price is sort of constrained. This is how much they can pay. So what are the products and needs that we can fit within? this price. But if they were not price sensitive, they're willing to pay anything, then,
01:42:36 Speaker 1.: oopsie,
01:42:39 Speaker 2.: Then you can say, okay, well, we are not going to care as much about the price. They would increase, but we will increase the qualities and attributes of the product to meet the need of this segment. So very, very closely related. In the same way, the distribution channel actually does affect both the needs of the customer. And the price, because for some distribution channels, say if you are selling, I was reading this scenario about Lego, and they were saying, well, they have all of these agreements with departmental stores and so on, and this adds a cost compared to having a warehouse and selling it online. So, right, so this contributes to a price of a product.
01:43:36 Speaker 2.: there are customers,

01:43:38 Speaker 1.: customers
01:43:41 Speaker 2.: that predominantly go to shops, then we have to pay that price. So you can see how all of that is related, and clearly the promotion is related to all of three aspects, because you have to tell people about the product, about the place, and how there is a competitive price.
01:44:05 Speaker 2.: We can't look after one aspect without thinking about the other. So now, how is marketing related to other business functions? Because this is where I'm starting to try to kind of make us think about how businesses work and how these data questions sometimes relate to different parts of the business. And people in these different functions would have very different views about how you use data.
01:44:34 Speaker 1.: So if
01:44:36 Speaker 2.: marketing is is looking after
01:44:40 Speaker 1.: designing
01:44:44 Speaker 2.: or collecting the needs of the customers from a product. Production is actually about delivering on this. specification and on this promise to the customers. They have to make the product and they have to physically design it to meet these needs. And I had quite an interesting, a little bit of contrasting kind of outlier example with a local business. It's an engineering company, and they produce electronics products. And I was talking to the engineers about something else, and in passing, they said, oh, and these marketing people, we really don't know what they're doing. And I was like, oh, well, they're the ones who talk to the customers, you know, so they really are. are doing very important jobs, and they were saying yes, but they come to us and they ask us to make these products that are really expensive to make, and then they want us to be able to make it at a cost to sell it. for much lower price. So this is really important because when marketing is collecting the needs and desires, they actually do need to talk to production about what is possible. So this is where obviously there was some lack of communication and actually last year we had another project for my level 300 class and that was based on another manufacturing company,
01:46:35 Speaker 2.: it was not quite so drastic, but they were saying, oh, well, the marketing guys have just managed to sell so much product and we don't have it. And they have promised a delivery date in two days, so now we have to

have 24-hour production and we have to hire more people. So you see, when they're able to sell or promise something or or take some needs, they also need to kind of talk to each other to make sure that production can deliver. Otherwise that is going to affect reputation and then obviously promotion and so on. So in terms of operation, production, that is where the actual transformation of inputs into outputs happen. So it doesn't have to be always manufacturing, it could be a service.
01:47:33 Speaker 2.: where something, for example, in education, you might say, okay, you get people coming in, they're promised something, and then we have to deliver, right? And you come out. hopefully meeting your expectations. So this is where you need to make sure that it's all aligned, otherwise we're not meeting expectations. So production doesn't have to be just making physical products, it could be a service that is offered. And so within that we have control of the quality of the service or product, and Sourcing raw materials, if it's a physical product, planning of how things are going to actually run, so that we can deliver very, very important functions. And then what does accounting do? Anybody from an accounting background?
01:48:38 Speaker 2.: Oh, interesting. A little. Okay, so actually we're not going to gang up on the single person who was brave to say they were in accounting, but what do we think accounting does? What do we associate accounting with?
01:49:01 Speaker 2.: profit, loss, right? So balance sheets, right? And we very often blame the accountants because sometimes they make the financial decisions, right? But the reality of it is that as I've worked in this department for a while, I had to work with a lot of accounting colleagues. I've been more educated because
01:49:28 Speaker 2.: I've only done one accounting class in my engineering degree and I remember just put this into this column, put that in that column, you know. But actually it's more than that. So accounting really is about keeping track of financial information. So they're constantly recording and monitoring the financial health and information. And whether we liked it or not, this is really important. It's a little bit like what we were talking about, the loans. You have to keep track, you have to keep with it. within the budget, and so accounting actually supports pricing by saying whether we have the budget to produce this product that you have promised. So all of these functions are

actually related, so when you're looking for data, and I know I have to finish, when you're looking for data from different functions,
01:50:30 Speaker 2.: What I find is that particular department might be using the data from, say, the customers. And they say, well, this is the product we really need to make because this is what the customers are saying is important. And your job as analysts would be to be that Switzerland person, neutral to a particular function, and say, all right, let's go and talk to production and see what they need to make this product. So let's get the data for that. And finally say, ooh, let's go and talk to accounting and see how have we been doing before with spending money on similar products. Is this a realistic?
01:51:17 Speaker 2.: assessment and estimate of
01:51:19 Speaker 1.: what it
01:51:20 Speaker 2.: will cost us so that you actually are getting good quality balanced data. All right, I'll finish here. I had a bit more, but this was the program minimum, as I call it. Thank you very much. I'll see you next week. Our last actual, like, new material lecture.
01:51:58 Speaker 1.: If people
01:51:58 Speaker 2.: were late, if somebody was late and didn't get a copy of the scenario, I have a few left. Yeah, we're only doing the process and evidence.
01:52:54 Speaker : Thank you. Thank you.
01:53:30 Speaker : or you told me. Okay, this is I usually, when people ask me, I approve, and I tell you to turn. So when you tell me after the fact, it's kind of my top lock, I told you. So you can email me again and say, sorry, I need to ask you, can you mark it? Sometimes you don't understand me, I assume you're not. Okay, I have said that you turn. Okay. All right. Thank you. Thank you.

View original transcript at Tactiq.

MBIS624-26S1-LecA-Te-s1-low-5

Meeting started: Mar 23, 2026, 10:10:41 PM
Meeting duration: 114 minutes
Meeting participants: Speaker, Speaker 2, Speaker 1

View original transcript at Tactiq.

Transcript

00:00 Speaker : Thank you. Thank you. Thank you. Thank you.
02:16 Speaker 2.: Happy days. I reckon week five is the hardest one. Things get better after that.
02:34 Speaker 1.: You're
02:35 Speaker 2.: all thinking about your tests. I hear you have quite a few next week.
02:44 Speaker 1.: So I had two
02:47 Speaker 2.: emails asking me the same thing.
02:50 Speaker 1.: So I assume more people...
02:58 Speaker 2.: are asking the same. And so I've put it on a slide today just to confirm because I think it's just a slight low-level anxiety creeping in and you want to confirm something I've said already.
03:14 Speaker 1.: So here is my USB.
03:20 Speaker 2.: Somebody has realized that they've taken it. Great.
03:24 Speaker 1.: So So the question was,
03:29 Speaker 2.: are you allowed to cheat sheet? And what's the answer? Yes, yes. So please don't email me individually asking me the same thing. You are allowed. Okay, but then somebody asked, oh, what exactly are we allowed? So you're allowed one side of an A4. You could, what was the question? How dense the notes can be? Knock yourself out. Do whatever you like.
04:01 Speaker 1.: You
04:02 Speaker 2.: could type it in size 6 if you like, as long as you can read it. I really don't mind.

04:10 Speaker 1.: Do
04:10 Speaker 2.: whatever you like on this one side
04:12 Speaker 1.: of
04:13 Speaker 2.: A4. So genuinely doesn't matter. So I am also going to show a bit. the information again because
04:27 Speaker 1.: I'm aware that not
04:29 Speaker 2.: everybody is here today and today is the last class that I will be introducing any material but to be fair it's actually the topic of visualization is not really introducing new material it's more like
04:51 Speaker 2.: rehashing what we've done already. So it's a very easy session. We have lots of time. There are lots of slides mainly. showing you different examples, but it really is not anything significantly novel. So first to start with, of course, the most important topic for you. Oh, you can't see. Nobody is saying anything.
05:29 Speaker 1.: All right, so first of
05:31 Speaker 2.: all, next week there are no tutorials. Because we, one, have a test, and two, we'll be releasing the assignment. Yes, yes, yes. There will be a lecture on Thursday. But as I said, it wouldn't be anything, I will not be doing anything new. What will be happening in the lecture is tomorrow, because today is like fully booked until 8 o'clock, but tomorrow, I will release a practice test for you. So something for you to practice.
06:08 Speaker 1.: And
06:08 Speaker 2.: next Thursday, we will go over the practice test. Now, please don't wait until next Thursday to try and do the practice test, because it will be the day before. And I have this theory that when it's the closer you are to an event, the more likely it's for things to be stressful.
06:30 Speaker 2.: This is based on analytics. So what I would recommend is that you try to answer the questions in the practice test. You revise all the material. And then when we come here next week at this time, we will just have questions and answers session. I'll sort of give you some hints about things that.
07:00 Speaker 2.: Perhaps you could do better if you've answered your question. So the more work you do, the more useful it's going to be. That's what I'm saying.
07:10 Speaker 1.: Again, even

07:13 Speaker 2.: though I mentioned it. Quite a few times, people still ask me what's going to be on it. So it covers the first five weeks of lectures. And there will be two to three applied questions to a scenario. So very similar to things we have done in class. So we've done a lot of scenarios. The loans last week. the town of Kerry, before that, the students, so there will be some description of a context for a page, and then all the questions would, two or three questions would be
07:51 Speaker 1.: on the,
07:55 Speaker 2.: related to that scenario.
07:57 Speaker 1.: But,
07:58 Speaker 2.: clearly it will be something that we have done in class before, so what have we done? Well, different analytics techniques. We've talked about different issues with obtaining data, specifying questions. So revise everything and think, how would you apply it to any scenario we've had before? So all the questions that are on the test, something we have done before in class.
08:27 Speaker 2.: You know, try not to stress about it. And by all means, use notes for things that you think would be
08:35 Speaker 1.: helpful. I
08:38 Speaker 2.: find that the more notes people have, I used to have a lot of open book exams, and people generally did worse on open book exams because they thought that they have lots of time to read during the test. And so they were wasting their time. time reading rather than writing answers to the question, and this is why now I'm kind of saying, okay, try to organize your thoughts on this A4 sheet so that you have something like notes, but at the same time you've done a bit of work before that.
09:17 Speaker 1.: Any more questions
09:19 Speaker 2.: to try and
09:21 Speaker 1.: one,
09:23 Speaker 2.: make you feel better and two, reduce the number of repetitive emails. So any questions, most welcome.
09:46 Speaker 2.: Yeah, you need to have it on you because we check you against your photo ID. So you have to have it with you in the test. Yep. Yeah, the one that has your
10:00 Speaker 1.: photo on.

10:03 Speaker 2.: Any other? I need a little bit of your help
10:19 Speaker 1.: before
10:20 Speaker 2.: we start, because I couldn't quite remember where we ended last week. Did we actually cover the marketing KPIs? No, where did we end? The product, yeah, there's four Ps. I thought so. So I've put it here as something we've done, but
10:41 Speaker 1.: I will also
10:43 Speaker 2.: go through the KPIs because... This is important from a business analytics point of view to think about, you know, we've talked a lot about measures and target outcomes and variables, and they usually are related to key performance indicators, things that are important to business, and there are some specific unique ones related to different functions. So we've done it up to the marketing main.
11:19 Speaker 1.: we talked a lot about
11:21 Speaker 2.: diagnostic analytics how it is different to descriptive analytics how it is similar so it is similar because it's still deterministic it uses current and historical information it kind of looks backwards because we're looking from now and to the past to see
11:40 Speaker 1.: what's been happening and
11:42 Speaker 2.: it's really Some of the techniques that we use for description of information can help us to start forming hypotheses and use drill-down analysis. And we also talked about statistical analysis to test hypotheses
12:06 Speaker 1.: and relationships, so
12:07 Speaker 2.: correlation, regression, and how we form hypotheses, and we had a few examples. So in terms of marketing, definitely I remember us talking about what is marketing and the role of it and how it relates to other functions. And we talked about the four P's. But we didn't talk about the marketing KPIs. So
12:30 Speaker 1.: I will. I don't have it.
12:36 Speaker 2.: I will definitely cover that today. And then after that, we'll talk a little bit about financial or accounting analytics so that you can see. what other KPIs exist, so very short, and then the main topic for today, as I said, something you've been doing a lot already in tutorials, visualization.
13:01 Speaker 1.: So how do we

13:03 Speaker 2.: represent results, how do we report results to
13:09 Speaker 1.: have the
13:10 Speaker 2.: greatest impact and to actually convey what you are trying to say as a story. So different techniques for visualization, and you have, as I said, used a lot of them already. So now, just a quick recap first from next week on the marketing,
13:38 Speaker 1.: hopefully.
14:15 Speaker 2.: All right, so I think we probably were, did we do the types of marketing
14:21 Speaker 1.: questions for descriptive and diagnostic? I'll
14:24 Speaker 2.: just, it's better to repeat than miss it.
14:29 Speaker 1.: So while we have discussed
14:34 Speaker 2.: that in terms of descriptive, it's always questions that kind of take the format of
14:43 Speaker 1.: what
14:44 Speaker 2.: has happened, what's been happening, what is happening, right? So current and the past. And for a diagnostic is
14:52 Speaker 1.: why did
14:55 Speaker 2.: it happen or why is it happening? So we look for underlying reasons. There are some specific questions within the product meds.
15:04 Speaker 1.: Right? So
15:05 Speaker 2.: for product, if we want to think about what is happening or what has happened, we would like to know things about product sales. So remember in descriptives we had counts, we had
15:21 Speaker 1.: statistical
15:23 Speaker 2.: measures like averages mean, we have ratios, calculations.
15:27 Speaker 1.: So this is what you'll be looking
15:29 Speaker 2.: at. If you have a scenario that that talks about the specific context, think about what kind of counts can we get from the data, and how would these counts help us to answer some business questions. So here it is, how many units were sold in a period, and this is a count question, and what that would help us is for different products, we might be able then to compare how well it is performing, right, or how much revenue. than we're getting because if we have the price or the average price and then we can get revenue

for each region and we can also get the cost.
16:15 Speaker 1.: So that would, the cost of the product or the cost of
16:20 Speaker 2.: specific marketing campaign.
16:24 Speaker 1.: And so,
16:28 Speaker 2.: Then you could have things like here, this is actually something I will talk in a minute about KPIs, but for promotions, things that are important are not just how much... The promotion has cost because obviously there is the cost of the design, there is the cost of the distribution, but also if a discount has been offered, that is also a cost, right? So if we only look at,
16:57 Speaker 1.: say, how...
17:00 Speaker 2.: what is the increase in sales after a promotion, that
17:06 Speaker 1.: might give us a
17:08 Speaker 2.: realistic view of how successful that promotion was,
17:14 Speaker 1.: because it would simply show us that,
17:16 Speaker 2.: okay, we have increased revenue, but if the cost was equally high, then we haven't really made any profit. So this is why it's important to also think about the return on marketing campaigns, which
17:35 Speaker 1.: is actually also one of the KPIs. And for places,
17:42 Speaker 2.: what are things that are important as measures descriptively? So place was the distribution channel, and we talked about how it can be online, it can be in a physical store, and we would like what makes a place successful.
18:01 Speaker 1.: Well,
18:03 Speaker 2.: our customers come to that place. right so
18:07 Speaker 1.: we have visitors but
18:09 Speaker 2.: also they purchase in that place because maybe they come but in that area this is This is not an area where this product is going to sell, so we need to think, you know, how successful was this place for this product? So you would have things like, well, how many units were sold in one particular area, like a region, or in one particular shop, or online, if this was our distribution channel.
18:41 Speaker 2.: And in terms of visitors, it's harder to know that in a physical shop. shop, but in an online shop, it's very easy because all of that information is collected when people visit, so we can see how many views or visits we have, so we can measure the traffic to a website, so that's about a place,

19:06 Speaker 1.: and what
19:08 Speaker 2.: was important about price from a marketing point of view, do you remember? No, not in terms of the actual questions, but in general. What is important? What is the, what does price stand for in the marketing mix? No, the advertisement is in the promotion, right? So pricing is about the price of the product. But what's important? What's the goal for the price?
19:49 Speaker 1.: Cost of? Right, so yes, the
19:55 Speaker 2.: cost, but the cost by itself doesn't, so the price is really about
20:00 Speaker 1.: setting
20:02 Speaker 2.: a price of the product that is attractive to the customers, because remember, marketing is outward focused towards the customers, attractive to the customers, but At the same time, yes, we have to make profit. So this is what we were discussing, how sometimes marketing, you know, they need to connect to accounting. So this is kind of where you are going with the cost. We do need to also consider the cost for the product and set the price so that it's attractive. but also makes profit.
20:43 Speaker 1.: So this is actually, I definitely know we talked
20:46 Speaker 2.: about that last week. And so then if you remember what these four elements are, we have distractions here. Little kids walking and nobody's paying attention. This is a perfect example of the power of marketing.
21:08 Speaker 1.: Really, I give up.
21:13 Speaker 2.: So, I mean, even I am distracted. The only other distraction could be, you know, in the spring you see ducks, the ducklings. They're also very cute. All
21:26 Speaker 1.: right, so back to work,
21:29 Speaker 2.: guys. You know, they are going to work, so we have to also do work. So sometimes you need to remember the high-level fundamentals, and then you can generate lots of questions. Sorry, you had a question.
21:50 Speaker 1.: Yeah, so you see,
21:54 Speaker 2.: this is – actually a really good question, because when we talk about descriptives versus diagnostic, right, and we say diagnostic is when we discuss the reasons, and descriptives is when you present the results. The reality of it is that it very often happens that if you present, and we'll talk about visualization, if you present your descriptives

22:21 Speaker 1.: in a
22:23 Speaker 2.: meaningful way, as we were doing that with the student retention example, you start forming ideas about the reasons, right? So for example, if we group it in certain way and we were saying,
22:38 Speaker 1.: okay, 30
22:40 Speaker 2.: % were Or 60% had poor performance at school, right, before university. Now, this is a pure descriptive, because we're simply showing what percentage. We actually are not saying, oh, the reason for people to
23:01 Speaker 1.: drop
23:03 Speaker 2.: off is because they had poor performance. But you start forming ideas, and you can
23:08 Speaker 1.: say, okay, quite a large proportion here.
23:12 Speaker 2.: of the group is this percentage. So it helps us to form ideas about the reasons and with the drill down analysis we can continue to check that form hypothesis and then test it. Yes, so really good question. Sometimes it's quite seamless how you go from descriptors to
23:34 Speaker 1.: diagnostics.
23:37 Speaker 2.: But sometimes descriptives are not always related to reasons. Sometimes there are simply ways of us monitoring performance. So for example, you might say, this is what,
23:51 Speaker 1.: well, we get this all the time.
23:55 Speaker 2.: People would say, okay, this is what we were aiming for in terms of
23:59 Speaker 1.: revenue,
24:01 Speaker 2.: and this is where we are.
24:04 Speaker 1.: So it just helps us to
24:06 Speaker 2.: see how we are doing. Or for example, Before results were released on Learn, we used to have our own system, and you used to get things like, okay, this is your grade, and your position in the class is number five, and you're in the top 10%, or something like that. I mean, there are no diagnostics there, right? It just gives you an idea about how you are performing compared to the rest. So descriptives can help us to see some patterns and form ideas about reasons, but they could also be simply about saying, look, look how we are doing. This is another story. Look how successful we have been.

25:04 Speaker 2.: satisfaction has grown, or you might say, oh, every year satisfaction has grown, but our revenue has come down. So, we're doing well, but we're not charging enough. So, so, descriptives go well beyond looking for reasons for things. You first have to see how we're doing, and then look for reasons. Does that make sense? So, this is why it's important that in different contexts you think about what are some important measures that
25:35 Speaker 1.: we want to
25:36 Speaker 2.: monitor with descriptives and this is why I'm saying you know what's important about place what's important about price so if if the whole purpose behind price is to be attractive but also viable for the business we need to look okay what's the highest product and how much profit the how the highest price product and how much profit do we make from it because it might be actually a very high price but maybe it's quite a hard one to make and maybe we're not selling a lot so then maybe it's not that profitable so price by itself You know, will give us some ideas, but then we might need to look at other measures. This is really important also. I mean, I know I care more about real life, but
26:33 Speaker 1.: in the exam,
26:34 Speaker 2.: you might have to suggest measures like this. You know, what should they be looking? If there are some indication of a problem, or not even a problem, a business that's functioning in some ways, and you might say, these are some measures or ways for... them to see how they're doing.
26:56 Speaker 2.: So, and then in terms of diagnostics, then you actually look specifically, even if you are doing descriptive analytics and you are drilling down into the categories, you start looking for reasons.
27:11 Speaker 1.: So, again,
27:15 Speaker 2.: for price,
27:16 Speaker 1.: if we have
27:17 Speaker 2.: found here that
27:22 Speaker 1.: that we've had some change in
27:24 Speaker 2.: the pattern of a particular, you know, price of a product, we can say,
27:31 Speaker 1.: okay, what was, we see that we have,
27:36 Speaker 2.: the price has gone down overall, how has that, you know,

affected something or what was the reason for this price to come down because we were offering a lot of discounts. Why did we do that? You know, how is that related?
27:58 Speaker 2.: So, similarly, very often you look to external sources to benchmark and compare. So, for example, you might say, well, how here you might say, how do our prices compare
28:12 Speaker 1.: to competitors? And then you might find
28:15 Speaker 2.: that the competitors are charging a lot less. And you might say, OK, what are the contributing factors to our price? Why are we charging so much more? And you might find. that you're using more quality, higher quality parts for your product, and then that could actually inform some promotion campaigns.
28:38 Speaker 1.: Because you can say, okay, well, we do need
28:41 Speaker 2.: to explain why we're charging so much more than competitors. But diagnostics are always about understanding, understanding based on the measures you've already discovered in your descriptives, what are the underlying causes, and especially if you have changes, you know, what has brought this change, or what has brought this result.
29:10 Speaker 1.: So,
29:12 Speaker 2.: I think descriptives we've talked a lot about already. These are some examples of questions. Again, to give you ideas, if you, if you want to practice questions. So this is around specifying questions that are addressed by
29:31 Speaker 1.: data.
29:33 Speaker 2.: So for example, how many people visited our website. This is simply traffic data for the website, you could see how many people have come to the homepage, or you could see how long they have stayed,
29:48 Speaker 1.: and how many people have viewed a particular
29:52 Speaker 2.: product, so all of these are measures that would give us an idea about
29:58 Speaker 1.: whether
30:00 Speaker 2.: we are generating enough leads, right? People are interested. But it doesn't mean they're buying, right? So this is where I want you to go with the important performance indicators so that you can see how, what information they actually give us. So for example,

30:25 Speaker 2.: I already talked about revenue, right? Revenue is something that people like to talk a lot. But revenue by itself, and this is something that I'm not sure if I shared with you, went to business leaders. breakfast, did I talk about that? And there was a couple of speakers talked, obviously AI is a big thing, but a couple of them also talked about the issue we have in New Zealand with productivity and how we work hard, but the outputs are not that high, so we work hard but we don't produce as much.
31:14 Speaker 2.: But the other one that was very interesting and related to this
31:18 Speaker 1.: was the revenue question. So they said very
31:21 Speaker 2.: often people look to grow revenue and they don't think whether this extra revenue actually generates more profit. Because you sometimes generate more revenue by getting more people to work, right? So your overall profit is not better. So you should be looking to generate revenue by being more efficient or more effective and more productive so that with the same resource you are generating more revenue.
32:00 Speaker 1.: So this is why
32:02 Speaker 2.: revenue as a measure is
32:04 Speaker 1.: useful. It
32:06 Speaker 2.: helps us to see whether we are selling. This is a very marketing kind of measure, but we also need to think about, okay, when we generate more revenue, what is the cost of that?
32:20 Speaker 1.: So that,
32:21 Speaker 2.: is it really worthwhile? So, for example, they were suggesting that actually another approach could be that you keep the revenue the same, but you try to be more efficient, so you reduce your cost. And that is not by slashing people, but by helping them to be more productive. And this, of course, is the promise of AI. We'll see how that works. Also, we have, this is a typical marketing measure, and that is leads. So leads... is a measure that shows how many, so it's a count, how many customers have, potential customers have expressed interest in a product.
33:10 Speaker 1.: So if say this was a bank
33:13 Speaker 2.: and we talked about loans last week, you know, how many queries we've had about banks in different branches, they've made appointments to talk about it, or they have clicked, for example, I'm not sure if

you've seen on the website, they have these calculators.
33:35 Speaker 2.: So where you can calculate if you wanted to take a loan, how much. So they actually would collect that information and treat that also as a potential customer, somebody who is interested in it. So these are potential customers. They're not real customers. They're not all going to take a loan. Maybe you do the calculation, you realize, I'm so far off this, you know, I need another 10 years of savings to get to the point where I can
34:10 Speaker 1.: ask for a loan. So you are a
34:13 Speaker 2.: lead, but you are not going to convert into a real customer. Or you check their calculator. Then you check some other bank's calculator and realize, oh, they're offering a better deal. So you are a lead, but you've gone somewhere else.
34:29 Speaker 1.: Lead is dead, lead is dead. Yes lead
34:33 Speaker 2.: simply shows that we have attracted people and so this is why I showed you the customer journey and the touch points because from a marketing point of view this is are we attracting people we are attracting them but not converting right so it's simply a measure of attraction And for a similar kind of... measure we would have things like for websites traffic so how many people are visiting how many people are viewing particular pages so they're attracted to this information how long they're staying shows us a bit more about are they serious or they just clicked on something
35:22 Speaker 1.: and
35:24 Speaker 2.: this is actually a really important one when they used to do web design we talked a lot about that Bounce rate.
35:33 Speaker 1.: So the
35:35 Speaker 2.: bounce rate shows
35:38 Speaker 1.: us the
35:39 Speaker 2.: proportion of visitors to our homepage that abandoned it by simply looking at the homepage. So these are visitors who come to the web. to our homepage and then abandon the website. So they don't look at anything else.
35:59 Speaker 1.: So what
36:03 Speaker 2.: does that show
36:04 Speaker 1.: us from

36:06 Speaker 2.: a business point of view? It shows that something brought them to our page.
36:12 Speaker 1.: So clearly
36:14 Speaker 2.: some kind of
36:15 Speaker 1.: promotion or, you know.
36:19 Speaker 2.: in terms of special or advertising has brought them, but they were not sufficiently interested to look at anything else.
36:31 Speaker 1.: So, in a minute,
36:37 Speaker 2.: no, actually let's talk about this now.
36:40 Speaker 1.: If you have, say,
36:44 Speaker 2.: a high bounce rate, what additional information would you be looking for to understand better why that is happening? Sorry? Okay, so the targeting, which audience we are targeting or targeting what? What is the data that we'll be looking at? The demographics. If we have the demographics of people coming, we might not, because remember there are no customers yet. But there is something about demographics, just think. Okay, so yes the device and what how would that help us if we say? 70% were
37:38 Speaker 1.: Clicking
37:40 Speaker 2.: from a mobile.
37:48 Speaker : Yep Yeah,
37:52 Speaker 2.: okay, so actually
37:53 Speaker 1.: In
37:54 Speaker 2.: addition to the this is why I was asking because I was trying to figure out what you were Getting at so what you actually are getting at is instead of just looking at a device We actually want to know the source that's brought them in so how did they come and they actually do have the data whether it's a particular link or particular promotion that you've used
38:20 Speaker 2.: even if it's a link that is on another website or something. As you said, you were looking on
38:26 Speaker 1.: your mobile and you
38:28 Speaker 2.: touched, you know, you were swiping and you touched by mistake. So they can see that. And then we can actually see, all right, this is all coming from this ad that is shown during a game or something. 90% of people who come from that source bounce off.

38:49 Speaker 1.: So what would that tell us? They are not actual buyers.
38:55 Speaker 2.: It was probably not a good idea to put the ad on that game, right? Because they're clearly not our target audience. So this comes back to your... Did you have a bit more thinking about the demographics? We can't have demographics of people who are bouncing off because they're not our... customers yet. However, we could have demographics for people playing a particular game because this is a third party provider. If they want to put our ad there and we pay for it, they could give us the demographics,
39:38 Speaker 1.: right? So the other
39:41 Speaker 2.: thing is then how do we decide whether these are good demographics or not? We actually do need to understand who are our profitable customers, who are the customers who actually buy regularly, and we're looking for bad demographic. And this is why we are constantly looking at the customers, because
40:03 Speaker 1.: you
40:05 Speaker 2.: try to identify who are your target audience. You know, do we have... more women buying this product? Are they all certain age? What kind of occupation? And a lot of these sites that show the ads, they actually would have the demographics for people who visit those sites. And you try to target sites that match your target audience. So if you see that you have a very high bounce rate from a particular source, clearly that was not the right. demographic for the target audience. So you were kind of right with the demographic, you just had to think about,
40:48 Speaker 1.: well, what
40:49 Speaker 2.: do we have? What is another indicator, what is something else that we can get from a high bounce rate? So one is obviously these people, I mean, they're clearly not interested, right?
41:07 Speaker 1.: But what else can we... Yeah, so
41:17 Speaker 2.: maybe if we find that they actually are coming from sources that have specifically targeted the right people, but we still have a high bounce rate, maybe they stay a bit longer on the homepage, so they're not... as you're saying,
41:40 Speaker 2.: clicking the, we obviously are all doing the
41:43 Speaker 1.: same thing, clicking the

41:44 Speaker 2.: top right corner quickly, right? So maybe they are viewing it,
41:49 Speaker 1.: and we can see the bounce rate in
41:52 Speaker 2.: conjunction with the viewing time. So then that could also indicate something for us to consider as our homepage. attractive and usable. Maybe they're looking at it, but they're not finding what they need, or they come, you know, we've hidden some important information. So this would actually be an indication that we need to do a little bit of usability analysis with customers to see what is the first thing they look at when they come to our page.
42:29 Speaker 1.: And hopefully,
42:31 Speaker 2.: Oh my goodness. All right, and hopefully when I show you some heat maps later for visualization, you could see how they actually do generate information on what people are looking at most and where they hover for longest and so on. Okay, then we have the cart abandonment rate. Now, what does that show us? This is the proportion of people who've actually put items in the cart, but then they don't go through with. the purchase. Now, what does that show us? People who put items in the cart, but they abandon it.
43:14 Speaker 1.: Yeah, so it could be that the
43:17 Speaker 2.: process is a bit difficult. They have to, you know, you put it in the cart, and then you have to put too much information. They can't be bothered. So maybe there is too high effort to pay. What else? When, I mean, we've all done it. When do we abandon the cart?
43:38 Speaker 2.: Yeah, you see the total price. Exactly. You see the price. Maybe you know how most of them actually don't want to show you the postage until you've gone through the cart. So we can see, all right, what is it that... So this shows that people are serious. You're not going to put it in the cart if you didn't
44:02 Speaker 1.: really...
44:04 Speaker 2.: Think about getting it at least. I mean, nobody goes and puts stuff in cards for no reason. So you obviously would like to have, I wouldn't say need, because we don't always buy things because we need them. But you would like to buy it, right? But then you abandon it. So this is a lot more serious than people just visiting the homepage. So these are stronger leads.
44:31 Speaker 2.: But we are losing them. So then we should be looking to see when do we lose them? Is it when we show the total price with the postage, for

example, so the cost is too high?
44:43 Speaker 1.: Or, you
44:45 Speaker 2.: know, what else is going on? Is it that they have started entering their name and then they realize, oh, I have to enter all that information, I can't be bothered. Or some of them actually say you have to register before you can pay, and then I'm usually forget. I don't want you to have my information
45:03 Speaker 1.: forever. So we could again
45:05 Speaker 2.: do descriptives, drill down, and say, okay, out of all of these people abandoning it, how many did it,
45:14 Speaker 1.: when
45:15 Speaker 2.: they showed the cost, how many, when they had to fill the details, how long did they stay, did they actually get? Did they try to go to the checkout process or did they just leave the items there in the cart? And so this is actually why some sites, when you leave stuff in the cart,
45:37 Speaker 1.: they email
45:40 Speaker 2.: you if you were already registered and say, hey, we see that you are interested in this, we'll give you 10, 15% off. Because then they know that it must be the cost. So they're trying to convert you, and this is the final one for marketing, very important, the conversion rate. So this is the proportion
46:02 Speaker 1.: of people who move
46:04 Speaker 2.: from being leads to actually going through a transaction. So they're the ones who
46:11 Speaker 1.: buy.
46:15 Speaker 2.: And then we have, you know, new visitors because you could see that through the IP addresses. If you have new IP addresses visiting, how many new people we are attracting. You could also see new visitors from a particular source. If you're using an ad, you can see how many have come from that source that we didn't have before. And that way you could see how successful that was.
46:41 Speaker 2.: When we looked at horizontal and vertical analysis reports,
46:49 Speaker 1.: we
46:50 Speaker 2.: had one where it had year-on-year growth. So what it shows is
46:57 Speaker 1.: the change. So, so in say sales,

47:03 Speaker 2.: if sales is your measure and you have the sales for 2024 and 2025, the year over year growth would show the difference from 24 to 25. It could actually be negative.
47:18 Speaker 1.: So although
47:18 Speaker 2.: it's called growth, it might be negative. So you get that change so that you could see,
47:28 Speaker 1.: well,
47:30 Speaker 2.: how are we tracking over time? And that was,
47:34 Speaker 1.: was it
47:35 Speaker 2.: horizontal or vertical?
47:39 Speaker 1.: Horizontal.
47:42 Speaker 2.: So actually both horizontal and vertical do it over time, but the horizontal was doing as a change while the vertical was using the base and then going for each component compared to the base.
47:59 Speaker 1.: So, okay, finally
48:01 Speaker 2.: one that we have done something very similar already to
48:08 Speaker 1.: churn.
48:10 Speaker 2.: As a marketing
48:12 Speaker 1.: term, it's a measure for
48:14 Speaker 2.: existing customers who were converted. So they have purchased products or services from us before. And they have stopped using our sites or they have stopped buying products and services.
48:33 Speaker 1.: So
48:34 Speaker 2.: what is a similar concept that we have? discussed before. Actually the student one is a similar one, right? They come, they start it, and they've dropped off. So it's It's actually related, the reason this is more important in a way than bounce rate is because the costable acquiring a customer is very high and it's
49:10 Speaker 1.: actually much better to keep the customers you have than
49:13 Speaker 2.: trying to acquire new ones. And so if you have a very high turn, it shows that customers who are quite happy
49:20 Speaker 1.: to
49:23 Speaker 2.: buy products from us before have changed their minds so we must be doing something

49:28 Speaker 1.: not
49:32 Speaker 2.: as well as we used to or competitors are doing something that is attracting our customers. So this is actually a major area of concern and something that marketing are
49:46 Speaker 1.: monitoring.
49:49 Speaker 2.: very much all right so
49:51 Speaker 1.: last
49:54 Speaker 2.: week we talked so previous the previous slide we talked about some descriptives for marketing last week we actually talked about
50:02 Speaker 1.: diagnostics
50:05 Speaker 2.: and so we there were two things we look in diagnostics right either Deviation from the norm, so outliers. Deviation from expectations, so deviations is the first one, right? So something that is different from what we expected or relationships between Factors, right? So the first one with deviations, so identifying anomalies and outliers. Some examples in the product mix.
50:41 Speaker 1.: So the
50:43 Speaker 2.: expectation. This is a real good one, actually, because I see it all the time. So the expectation is that if something has been selling well, it will continue to sell well. So, you know, every year we expect it, say we had growth for the last three years, so we're expecting to have growth this year again. And so all of a sudden you have product sales are dropping instead of growing, and so this is considered to be...
51:17 Speaker 2.: because it contradicts the expectation. So, frankly, I know they've given you here one possible explanation, and it's true, maybe something happened to disrupt this growth, and so competitors are offering a product for the same price that is a better product, yes, this is a really
51:44 Speaker 2.: possible hypothesis which is external and the other one also external recession reduces customers discretionary income right so something we can't really control but And hopefully, next term, when we talk about, well, not me, but you talked this with Casey, about predictions, we should also think about how we use data to form our expectations, right? Because is it realistic that we expect for growth to go forever, right?
52:19 Speaker 1.: Is that

52:19 Speaker 2.: how the time series have shown us of other products, and should
52:25 Speaker 1.: we look at
52:28 Speaker 2.: the financial situation, discretionary funds before it surprises us. So this could have been actually identified before it happened. So you could look at how discretionary income would change and kind
52:49 Speaker 1.: of flag that we should have
52:51 Speaker 2.: lower sales. The same thing, markets will grow over time for a new product. So this is quite reasonable expectation when you have growing markets in general. For example, when smartphones first were introduced, there was just not enough supply for the demand. Everybody wanted smartphones. It was a hugely growing market. But after a bit of time, it starts plateauing. So, you know,
53:32 Speaker 2.: Market growth is low or almost non-existent. So when you hit that, of course, you have to think about is it the price being too high or we have more competition. Sales of a product will be the same at all stores. So if that is an expectation and one store sells less product than the other stores, that's anomaly. I actually find with these examples from the book, a lot of these anomalies, we can think about could we have thought about our expectations, you know, because maybe we could have looked at the demographics in the areas of those stores and think about
54:16 Speaker 1.: whether that product is
54:20 Speaker 2.: suitable for that demographic. Maybe in terms of their price sensitivity, or maybe in terms of their age, and they're not interested in those products, so
54:32 Speaker 1.: some of these things could
54:35 Speaker 2.: be diagnostics, but also very good candidates for forecasting. Now in terms of statistical
54:44 Speaker 1.: techniques.
54:46 Speaker 2.: So finding relationships between
54:49 Speaker 1.: measures
54:52 Speaker 2.: or variables. We talked about regression and correlation last week. So we have for correlation, to what extent promotional emails relate to sales revenue.

55:08 Speaker 1.: if we have
55:09 Speaker 2.: a greater number of destruction coming up people, if we have a greater number of promotional emails, are we going to have increase in sales revenue? And so again, all of the measures are around the four P's.
55:31 Speaker 1.: So product, place, promotion.
55:34 Speaker 2.: and price, and you try to think, are they related to each other, are they correlated, and are we targeting one, like the churn or,
55:44 Speaker 1.: say,
55:49 Speaker 2.: revenue, and see what contributes to it through regression.
55:54 Speaker 1.: And
55:55 Speaker 2.: then we have hypothesis testing, which we also discussed, testing whether There are some variations and causes for contributions, but we also have talked about pivot tables, where the way you summarize and group the information might give you an idea about the possible causes. And this is what you were asking me earlier. All right, since I have absolutely killed the time plan, let's have a break, and I will recalibrate what
56:33 Speaker 1.: I'm going, because today is the
56:35 Speaker 2.: last one.
56:37 Speaker 1.: So I
56:39 Speaker 2.: will think about my priorities, shall we say. And also you could enjoy the kids without being distracted. You could just put all of your attention on them. This is a perfect positive correlation, right?
57:10 Speaker 1.: Kids coming through,
57:12 Speaker 2.: distraction,
57:13 Speaker 1.: top.
57:15 Speaker 2.: Nobody listens to the lecture.
57:45 Speaker 2.: You can get up, you can come talk to me, move, talk, wake up.
57:57 Speaker 1.: Yes,
57:59 Speaker 2.: exactly. Get some blood flowing.
58:27 Speaker 2.: Hold on, let me take this out.
59:00 Speaker 2.: Okay, so I can relax the control so
59:05 Speaker 1.: that you can submit. So I'll do that.
59:13 Speaker 2.: I can do that.
59:17 Speaker 1.: I actually accomplished on your behalf. I don't know who you

are when I allotate them. For some reason I put a lot in there. At last I realized it so much. You submitted it. You submitted it, right? Try to
59:40 Speaker 2.: be nicer to yourself. Yeah,
59:43 Speaker 1.: exactly. This is what I was saying earlier when I came.
59:51 Speaker 2.: I could feel like people are getting more anxious and more stressed and,
59:56 Speaker 1.: you know, it's week five syndrome. But try to kind of remember that you've managed all of this before.
01:00:04 Speaker 2.: You could manage the game.
01:00:06 Speaker 1.: All right, so. I'll just. Thank you.
01:00:51 Speaker 1.: Thank you. foreign foreign That way we can still see. You
01:01:14 Speaker 2.: see, my issue is if I change the
01:01:17 Speaker 1.: due date, then people say, oh, I changed the due date. So I changed the due date. So right now definitely should work.
01:01:42 Speaker 1.: So
01:01:42 Speaker 2.: the rules
01:01:44 Speaker 1.: are being used again.
01:01:50 Speaker 2.: But because it's a test, I'm not that fast. But you just be aware that the next test is the second term.
01:02:00 Speaker 1.: Well, yeah, we actually are going to pull it.
01:02:26 Speaker 1.: So I don't know how to change.
01:02:53 Speaker 2.: All right, 30 seconds.
01:03:33 Speaker 2.: All right, let's start. I've looked, and although it looks like there are lots of slides, actually not as much
01:03:42 Speaker 1.: thinking
01:03:44 Speaker 2.: power required for this one. So we have a little bit on accounting that needs a bit of thinking power, and the rest of it is kind of
01:03:52 Speaker 1.: descriptive.
01:03:55 Speaker 2.: we can show the slides very quickly. All right, so last week and today, we talked about marketing and what are some of the unique measures and specific KPIs that are important to marketing, and just so that you have some other context to relate to. For example, last week we talked about loans, so we also need to think about financial, other financial measures. And so accounting,

01:04:29 Speaker 1.: how
01:04:32 Speaker 2.: many people have done accounting? Right, and we talked about, when we talked about how marketing relates to accounting, right? So what is accounting about? It's about collecting... preparing and summarizing and reporting financial information, right? And so the, as with every data and information, it is related to decisions. So we always have to think about whether it's marketing, whether it's accounting, whether it's operations management, it's always about some decision. That's why we had the information value chain early on. So, so. There's a definition, and that's exactly what I was saying. So measuring is about collecting. So where is the data recording?
01:05:30 Speaker 1.: or
01:05:31 Speaker 2.: preparing information and then reporting or communicating to decision makers. Now, depending on who the decision makers are and what the decision is about, there are four kind of broad areas of accounting that we need to think about. And again, I don't think you need to put this in your notes because I'm not going to ask you that. You do need to think about how the data would be different depending on which one you're dealing with. Because financial accounting focuses on reporting performance and financial information to external stakeholders, right? So you need to think about
01:06:24 Speaker 2.: who are the external stakeholders to an organization that need or require financial information.
01:06:33 Speaker 1.: So those could be
01:06:35 Speaker 2.: government agencies, you know,
01:06:38 Speaker 1.: the tax office, but also
01:06:41 Speaker 2.: if it's a listed company, it would be the exactly people who have shared stakeholders, so shareholders, so they need to know
01:06:51 Speaker 1.: what is the overall performance, right? They
01:06:54 Speaker 2.: don't need to know how we get there. And unless we want to explain a little bit why sales have gone up or down, right? But we are looking at performance measures that are relevant to those investors and external parties. And their decisions are not going to be decisions. that are about how the organization operates, there are going to be decisions whether they will continue investing, whether they will be willing to take a lower dividend or

things like that, right? So the information therefore tends to be targeted for those
01:07:44 Speaker 1.: decisions.
01:07:44 Speaker 2.: But then we also have management accounting, and management accounting is inward. looking.
01:07:52 Speaker 1.: So
01:07:52 Speaker 2.: actually very, very closely related to operations management and to information systems. Until I came to work in this department, I never really realized what management accountants do. So they focus on
01:08:13 Speaker 2.: collecting data and information about internal performances, so how processes are working, how controls are working, so for example, if you have management control that tries to reduce fraud or to try to reduce misuse of information. You see, I have always thought about that as an information system kind of security measure, but the management accountants consider this to be an internal control as well. So they look at any internal processes, how they're performing,
01:08:57 Speaker 1.: are
01:08:59 Speaker 2.: we efficient? mostly about efficiency but also effectiveness and all of these costs per product line and revenue per product line and what would be the budgets and how do we set it up would be something that management accounting would care. And they report to their internal superiors. So this is about how we manage the business internally. So zooming in and looking at
01:09:35 Speaker 2.: how different processes, different functions add value. And so the performance measures will be different for management accounting than for financial accounting. And then we have auditing, which frankly is just pure IS, but a bit more boring,
01:09:54 Speaker 1.: because what
01:09:57 Speaker 2.: that looks at is really monitoring and investigating whether practices have been complied with,
01:10:10 Speaker 1.: and
01:10:12 Speaker 2.: whether information is correct, and if they're conforming to standards, because in accounting you have standards, you have regulations, it's

a very highly regulated discipline, so just checking forensically, if you like. always looking at the current and past data to see whether it complies
01:10:38 Speaker 1.: with how
01:10:39 Speaker 2.: things should have been presented, whether there is any missing information, whether there are any outliers that could indicate issues. So really it's all about data.
01:10:51 Speaker 1.: And tax
01:10:53 Speaker 2.: accounting,
01:10:54 Speaker 1.: well,
01:10:57 Speaker 2.: It focuses on all transactions and data related to
01:11:05 Speaker 1.: tax
01:11:07 Speaker 2.: collection and reporting. And again, conforming with regulations. tax regulations. So one of the things I wanted to discuss in terms of financial or accounting, we had a presentation by food staffs last year from their decision support manager and he's an accountant by background. And he was explaining to us from an accounting point of view,
01:11:42 Speaker 1.: how they use the
01:11:44 Speaker 2.: point of sale data. So I thought, okay, actually a lot of what he said, we could have generated ourselves with our
01:11:55 Speaker 1.: brain power. And so
01:11:59 Speaker 2.: I want you to think about, first of all, what data we collect. at point of sale by information systems and how that can be useful to
01:12:16 Speaker 1.: decision
01:12:18 Speaker 2.: makers, either for external decision makers or internal. But let's start with internal because that's easier for us to think about.
01:12:31 Speaker 1.: So first
01:12:33 Speaker 2.: of all, what are point of sale systems? What do we normally call them?
01:12:47 Speaker 1.: Yes,
01:12:48 Speaker 2.: or FPOS, right? When you go to, but FPOS is only one part of it, which is related to the
01:12:55 Speaker 1.: payment,
01:12:57 Speaker 2.: right? But point of sale
01:12:58 Speaker 1.: is...

01:13:00 Speaker 2.: At the point where you are purchasing, a lot of information is collected, right? So I want you to think about, I want you to think about, don't look at that one, because these are some things that I generated quickly. Think about when you, and this was foodstuff, so let's think about a supermarket. What is the information,
01:13:26 Speaker 1.: data?
01:13:28 Speaker 2.: Actually data, what data is collected, recorded at the point of sale? So when you go and buy something, what data is recorded? And then we'll think about how that relates to... accounting and how that relates to marketing because your practice test is kind of similar
01:13:59 Speaker 1.: so I
01:14:01 Speaker 2.: want to practice this before you have your practice test so alright how long do you need because we're very very behind so
01:14:10 Speaker 1.: should we say
01:14:12 Speaker 2.: Three minutes. Three minutes to generate as much as you like. So it took me about two minutes to generate something. So without CHGPD, should I say. So you can do it.
01:14:24 Speaker 1.: So think about you are
01:14:27 Speaker 2.: paying for something. It is all electronically linked to lots of different systems, which we have already discussed before. So try to think about all the data that is collected with that purchase. So apparently there are about,
01:14:48 Speaker 1.: for memory, like 300
01:14:52 Speaker 2.: data points you get. So it's a lot. And we just need a few. I'm pleased to see people are practicing handwriting, because that's what you'll be doing next week.
01:17:20 Speaker 2.: All right. Should be enough time. So this example, in a way, kind of wraps up lots of things we've been doing already. And you might find it easier also if you think about, you know, the marketing product price.
01:17:41 Speaker 2.: promotion in place, or you might like to think about accounting and how they would be looking at some internal performance and think what performance.
01:17:50 Speaker 1.: But really think
01:17:53 Speaker 2.: about the data
01:17:54 Speaker 1.: that

01:17:54 Speaker 2.: the point of sale system needs and that it generates. And you could organize it around different categories. So what is one category
01:18:04 Speaker 1.: that will
01:18:08 Speaker 2.: get data about. So you are checking out something. Price of what? Of the item, right? So you might say, okay, we will have to have
01:18:23 Speaker 1.: product
01:18:25 Speaker 2.: data, right? Including the price, the product code.
01:18:32 Speaker 1.: What else?
01:18:34 Speaker 2.: Description. Description. Supplier, weight,
01:18:41 Speaker 1.: particular,
01:18:45 Speaker 2.: yeah, the product idea I think we mentioned already.
01:18:49 Speaker 1.: What else do we have?
01:18:51 Speaker 2.: So this is around product, right? What else do we have?
01:19:13 Speaker 2.: cancel.
01:19:14 Speaker 1.: Okay, so yes, we could
01:19:16 Speaker 2.: actually, at the point of say, we are also recording canceled transactions. So I would call this more of a transaction data, right, for the actual transaction, because the transaction would be around what was your
01:19:31 Speaker 1.: overall total.
01:19:35 Speaker 2.: For example, what were the different items or products? So we have the product, and then we have the actual transaction. So you could say,
01:19:44 Speaker 1.: for
01:19:46 Speaker 2.: products, a canceled product. Or you could say, for transactions, what were some cancellations? It doesn't matter really, because your idea is a good idea. items from a transaction that have been canceled. And on the same sort of note, you could also, you know how we were talking about the Commerce Commission and how they charge the wrong price? Well, we also could have, not could, we have the data on how many of the prices had to be manually changed by a staff member. Because they were incorrect. This is all going to be there.
01:20:35 Speaker 1.: And how
01:20:37 Speaker 2.: much was paid, the time and date of transaction, the place of the transaction, online, in a shop, the customer details, if they used loyalty cards,

01:20:54 Speaker 1.: So then,
01:20:56 Speaker 2.: in terms of product, we talked about all the things for inventory as a whole. So the inventory, we'll have the stock levels, so we will have the stock levels before and after the transaction. We'll have, you know, how it changes over time from the past data,
01:21:18 Speaker 1.: because as
01:21:19 Speaker 2.: soon as you record the sale, the inventory changes, we'll have the unique identifiers of the products you already mentioned, we'll have how often
01:21:33 Speaker 1.: and
01:21:35 Speaker 2.: at what times the thresholds for low stock are ahead. Anything else you
01:21:47 Speaker 1.: had on
01:21:50 Speaker 2.: We also know the
01:21:51 Speaker 1.: customer's purchasing pattern, what things he purchases.
01:21:56 Speaker 2.: So we also have the customer data. So
01:21:59 Speaker 1.: we have
01:22:00 Speaker 2.: the information about the customer, their purchase history, is that what you're talking about? Any loyalty program data, like what do they like as rewards? What do they respond well when they give it to you? For example, Woolworths are trying to do this. I mean, definitely not working for me, but they're trying to do something where if you buy this product, they give you extra. So clearly don't know enough about me to offer me the right thing. But they have that.
01:22:43 Speaker 2.: Yeah, so yes, from a financial point of view, we would have the amount that was paid, the amount that
01:22:55 Speaker 1.: it actually cost
01:22:59 Speaker 2.: us, the amount that... there is a tax on the GST, so all of that, all the financial information would be there. And then one thing that, did anyone think about the staff data?
01:23:19 Speaker 1.: So the
01:23:20 Speaker 2.: checkout operators always have to log in when they work and so you know who is doing it and you could actually see how long they take. And if they're using any discounts for themselves, all of that information is also

there, how much they get paid. So
01:23:47 Speaker 1.: trying to show
01:23:49 Speaker 2.: you that in POS data, we actually have data that is relevant to different functions within the organization. So now I want you to. based on this data, what is one
01:24:06 Speaker 1.: question that
01:24:09 Speaker 2.: can be addressed by this data that will be relevant to marketing, one question that will be relevant to
01:24:18 Speaker 1.: accounting,
01:24:19 Speaker 2.: and one question that would be relevant to kind of internal performance, whether management, accounting, or operations management.
01:24:36 Speaker 1.: Is my question clear?
01:24:38 Speaker 2.: So this is the specified question that is addressed by data. So we now
01:24:44 Speaker 1.: see what kind of data we have. So
01:24:46 Speaker 2.: if this was an exam question, it will be described. You'll be getting the information about what happens and the pause. So that gives you ideas about the data. is generated and now you need to think, what are some questions that we can formulate that will be helpful for marketing decisions and what decision for
01:25:14 Speaker 1.: accounting
01:25:15 Speaker 2.: decisions, what decision.
01:25:18 Speaker 1.: Sales revenue information can be important for the financial accounting.
01:25:25 Speaker 2.: What would be the question? So remember, in all of these tables, I was giving you lots and lots of questions, right?
01:25:34 Speaker 1.: So what would be a
01:25:36 Speaker 2.: question that we can immediately answer with this huge amount of data?
01:25:43 Speaker 1.: So
01:25:44 Speaker 2.: you were saying financial accounting, something that we want to... show the shareholders who are, say, in foodstuffs, they are cooperative, so they are members, so they care about financial performance,
01:25:59 Speaker 1.: what would be

01:26:00 Speaker 2.: something that shows the financial performance?
01:26:06 Speaker 1.: Usually
01:26:07 Speaker 2.: return investment, right? So that's all the costs and then all the
01:26:12 Speaker 1.: revenue and profit,
01:26:17 Speaker 2.: right? So these will be things that as a whole, not per product, as a whole. So that would be more external. They're all aggregated. So what would be then the question?
01:26:33 Speaker 1.: What is the annual sales?
01:26:35 Speaker 2.: Yeah, you could say, what is the annual profit? What is the year-over-year growth? That is also financial performance of the business as a whole, rather than individual product lines. External people don't care about our individual management of products. They want to know, as a whole, how are we doing?
01:27:04 Speaker 2.: What about marketing? Since we talked a lot about that, what are some questions related to marketing that we can
01:27:10 Speaker 1.: answer?
01:27:21 Speaker 2.: So you could have a count and say
01:27:23 Speaker 1.: when a
01:27:30 Speaker 2.: product is discounted, what is the,
01:27:34 Speaker 1.: well, how
01:27:36 Speaker 2.: many items are sold. And based on that, you could say, because you're kind of going with impact, but remember, you have to actually ask it in a way that the data can answer it. So you can say, how many, what is the difference, you know, what's the increase in terms of percentage or in terms of numbers to the previous period when it was not discounted? And what would that tell us?
01:28:10 Speaker 1.: Did it work?
01:28:11 Speaker 2.: You know, how many more we actually sold? And then from a financial point of view, it could also tell us whether it actually was worthwhile doing, right? Because there is, what you are gaining is
01:28:28 Speaker 1.: more
01:28:29 Speaker 2.: sales and
01:28:31 Speaker 1.: more

01:28:32 Speaker 2.: revenue, right? But we are reducing the price, so that means we have to get more economies of scale. What other questions, what did you have?
01:28:52 Speaker 1.: Okay, so how
01:28:57 Speaker 2.: many times, because remember, use words,
01:29:00 Speaker 1.: there are counts,
01:29:02 Speaker 2.: percentages, or some kind of measure, right? So how often comes as how many times
01:29:10 Speaker 1.: has a
01:29:12 Speaker 2.: customer visited A store, right? So this then is
01:29:18 Speaker 1.: focusing on a customer, and
01:29:21 Speaker 2.: why would we care for that particular customer how many times they have visited a store? Because you remember the information value chain? We always have to link the question to some additional understanding that would then lead to a
01:29:43 Speaker 1.: decision. So if
01:29:45 Speaker 2.: I know that you visit the store every day, how would that help? What is the decision? So please do think about these things when you generate the questions. Because I can come up with some ideas. But the point is for you to come up with ideas now, because we've done this before.
01:30:19 Speaker 1.: to this customer.
01:30:21 Speaker 2.: So don't offer this person a lot because they come anyway. Okay, so then you actually have to look at what they're buying, right? To see what they actually buy, whether they buy convenience foods.
01:30:46 Speaker 1.: So
01:30:47 Speaker 2.: this would be a decision specifically related to that customer and what promotion we are offering based on the time of the day and the product. So just saying how often they come by itself. It just shows us that this is a regular customer, but maybe you just buy a bottle of water every day. So you might not be that profitable. You know, maybe other customers come once a week, but they buy a lot more.
01:31:22 Speaker 2.: So you need to think, why do we care about a specific? So if it is related to a customer, yes, I agree. The decisions would be, because it's on the individual customer level, what we are offering that customer or which

category they fall in. So you'll be doing it for all the customers, right? And you'll see,
01:31:46 Speaker 1.: well,
01:31:49 Speaker 2.: for this demographic. So usually you'll be aggregating and say
01:31:53 Speaker 1.: something
01:31:56 Speaker 2.: like
01:31:56 Speaker 1.: what
01:32:00 Speaker 2.: describes our
01:32:03 Speaker 1.: most
01:32:04 Speaker 2.: frequent
01:32:05 Speaker 1.: customers,
01:32:07 Speaker 2.: right? The people who come a lot. What kind of products they buy, what times they come, right? And you need to think about
01:32:16 Speaker 1.: If they're
01:32:17 Speaker 2.: the ones we need to be targeting to buy more, maybe they don't buy a lot. So you actually try to understand the customers. But in terms of visiting the store, I would probably be interested also in how many, because remember we have counts, how many customers visit
01:32:37 Speaker 1.: at
01:32:39 Speaker 2.: a particular time and day and what they buy. Because then it's related to stocking decisions and staffing decisions. So you always have to think about, there are always questions we can ask, but you need to link them to decisions. And this is where if you start answering your question, you start describing a lot the question, then you don't have the decision that it's related to.
01:33:05 Speaker 2.: What I've seen in the past is people cross it out like, oh, too hard. Let's think of another question. So think about it first and then, or think really, okay, when I have this information about a particular customer and their pattern of behavior, what would we be doing with that? Because there is actually what we can do.
01:33:26 Speaker 2.: We can say,
01:33:27 Speaker 1.: well, clearly
01:33:29 Speaker 2.: they come a lot, they're interested enough to come, but

they don't buy a lot, right? So based on their purchasing behavior, we could offer promotions for products they're not buying. Next term, you talk about association analysis, but you can say people who buy this product also often buy this other product,
01:33:54 Speaker 1.: and so you can kind of give
01:33:56 Speaker 2.: discounts for the other product.
01:33:59 Speaker 1.: So what else can we get? What about the staff?
01:34:02 Speaker 2.: What did we talk about in terms of staff? Because this is more operations management, human resource. What are some decisions there? Do you remember the example where we talked about the workloads? I know, we've always done stuff before. So, you know, where they were overloaded and they wanted more people. And what were some of the questions we were asking there? Are they really overloaded or are some people taking much longer than others? So they need some training. So we can say when we get the information on how long people spend. take for an average transaction for a certain number of items? If there are people who take a lot longer, do they need extra training or
01:35:02 Speaker 1.: is
01:35:08 Speaker 2.: there something else going on? So decisions on performance management, for example, or on training. What else can we get out of this time spent processing transactions? We could compare it to the time and use of people at self-checkouts to
01:35:35 Speaker 1.: see whether one
01:35:38 Speaker 2.: is actually better. So maybe we would be better off having more self-checkouts with more people monitoring them. than putting the people, because this is about how do we manage our people resource, right? So you might think, okay, actually at self-checkouts, we are, for the same period of time, we have processed a lot more items with just one person. So if we actually put two
01:36:14 Speaker 1.: people, because...
01:36:15 Speaker 2.: We can see also how often people wait, right? Because when you have to have an approval for
01:36:22 Speaker 1.: a
01:36:23 Speaker 2.: product, you have to wait for the person monitoring the

self-checkouts to come and approve.
01:36:29 Speaker 1.: So if
01:36:32 Speaker 2.: you put more self-checkouts, you could actually see how much longer people would have to wait for approval. and then say, okay, but if we put two staff members, this is the tutorial this week, right? Sensitivity analysis. If we put two staff members, then that is still the same amount of transactions, but actually more transactions than having two staff members at
01:37:02 Speaker 1.: checkouts. So, you know, this is both.
01:37:10 Speaker 2.: human resource management and
01:37:12 Speaker 1.: productivity.
01:37:16 Speaker 2.: So lots and lots of questions. So try to think about that because the practice question would be kind of similar. So try to think about different decisions that
01:37:27 Speaker 1.: could
01:37:29 Speaker 2.: be made based on data that the business will have from the past data, because this is actually what a lot of businesses are dealing with. They're collecting a lot of data, but they don't know what to do with it, and that's the point of analytics, right? We generate questions, and we generate what could be something they can do better,
01:37:52 Speaker 1.: and
01:37:54 Speaker 2.: back it up with data. Now, for the next 13 minutes, We're going to talk about the last step. So we've already talked about specifying the question, where are we getting the data,
01:38:14 Speaker 1.: different types
01:38:16 Speaker 2.: of data analytics in terms of descriptive and diagnostic. And today
01:38:21 Speaker 1.: we will also kind
01:38:24 Speaker 2.: of recap a little bit because you've been doing this in tutorials and I've been showing you in techniques how we report the results so that
01:38:35 Speaker 1.: they give
01:38:38 Speaker 2.: the right message. So you always need to think about what is the message you are trying to convey with your data, and then you present it in a way that is clear. So the first point I wanted to make

01:38:55 Speaker 1.: is that
01:38:58 Speaker 2.: Reporting the data is part of data analytics, but it's also a really big part of making decisions. Because decisions, if they're good decisions, are made based on data. And decision makers are not analysts. So they don't want to see huge spreadsheets or huge databases, right? They want to see... what you've come up with as a conclusion and what's the data to back it up. So the way it works is obviously all the business functions as we were just talking about inventory.
01:39:35 Speaker 2.: HR, they generate some data when they perform processes,
01:39:39 Speaker 1.: and then that is usually in
01:39:42 Speaker 2.: huge data repositories, and then we analyze it, report it, and then the decision maker has to
01:39:49 Speaker 1.: approve
01:39:51 Speaker 2.: some action. And this is why I keep asking you, what's the decision, what are the actions? Very important to think about visualization and reporting
01:40:02 Speaker 1.: of data in that context.
01:40:04 Speaker 2.: because how you present the data would make a difference on decisions that are being made. So effectively you're also decision makers because
01:40:14 Speaker 1.: you are
01:40:16 Speaker 2.: Influencing the decision makers. So some basics of data visualizations.
01:40:22 Speaker 1.: It
01:40:23 Speaker 2.: presents the results of your analysis and you want to put the emphasis on your conclusions and your insights. This is why you're called business analytics and not just data analysts, because you're also looking at the implications for the business.
01:40:40 Speaker 1.: And.
01:40:41 Speaker 2.: People retain 95% of visual messages compared to 10% of text messages. This is actually a very significant difference, and this is why we sometimes try to have pictures, because we remember the pictures. So data visualization, just as a sort of funny anecdote, it's not new. Time series charts have been there forever. So they were just not done by computers and

spreadsheets, but they have been, you see this is the first one I could find, 1786.
01:41:24 Speaker 2.: And they were looking at the, this is quite interesting in the current times, we're constantly talking about trade.
01:41:33 Speaker 1.: deficits and
01:41:34 Speaker 2.: surpluses. So the same thing. They were looking at the line of imports and the line of exports between
01:41:43 Speaker 1.: Denmark and
01:41:47 Speaker 2.: Norway.
01:41:48 Speaker 1.: And so how the line
01:41:52 Speaker 2.: of exports was shifting
01:41:56 Speaker 1.: and
01:41:57 Speaker 2.: how the line of imports was shifting. And so that gave them the trade balance. So this is a typical time series line. Imports over time and exports over time.
01:42:13 Speaker 1.: So very quickly, you have five
01:42:17 Speaker 2.: seconds to tell me how many five, how many digits five
01:42:25 Speaker 1.: are
01:42:26 Speaker 2.: in this one.
01:42:27 Speaker 1.: Is it zero? I don't know.
01:42:32 Speaker 2.: You give me a number. It's quite hard, isn't it? How about now? This is the point I really want to make about visualization.
01:42:46 Speaker 1.: And this
01:42:48 Speaker 2.: is a point I want you to constantly, this is your critical thinking. When somebody presents information to you, this is why when you read the news, you need to think, this is what they've done for you. They have decided to highlight a particular item from the data, and you want to think, Is this the question I want to ask? Have they provided me with the data for this question? So this is when you are the consumer of the data. when you are the decision makers. But when you are the analyst providing the information, also think which of the information I want to make so easy to see that they don't have to go through tables and tables to find it. I want to summarize it. I want to give them the total at the top. I want to show what is the most profitable product because this is the point I'm trying to make. So effectively visualization is about selecting the elements of your results that are most pertinent to the

question.
01:43:55 Speaker 2.: Because otherwise, we are here, and we have no idea what
01:44:00 Speaker 1.: is
01:44:00 Speaker 2.: important. Even if we're looking at it, we don't know what the answer is. But when you do that, you didn't need five seconds for that. It's very clear. This, of course, is assuming that I have highlighted all the fives.
01:44:17 Speaker 1.: So lots
01:44:18 Speaker 2.: of different visualization tools that are being used. This time you're using Power BI, and next term we'll be using Tableau, which is part of Salesforce, and it's one of the most used products by businesses. But whichever tool you're using, because people always focus on tools, the ways to visualize data are always the same. They're just better at doing it.
01:44:47 Speaker 2.: the actual ways you can show it, they're always the same. So you could
01:44:52 Speaker 1.: offer the
01:44:54 Speaker 2.: large data set in a table and have this situation where people have to look through the table. So think about who you're offering that to, usually to somebody who might want to actually manipulate the data as well. Or you could actually start Merging data sets and grouping them in a way that people can see easier the categories that they are going to be interested in. Or we could have visuals in terms of charts and graphs and dashboards that actually tell the story.
01:45:33 Speaker 2.: So the chart components. I want you to distinguish between charts in terms of the data type. So what was the data type that we know about? What were the two main types? Yeah, we have structured and unstructured data, but in terms of... What's in the data? What does unstructured data mostly consist of? Text or numbers and within structured? What data types do we have within structured?
01:46:22 Speaker 2.: to revise guys categorical and numerical do you remember that one right so depending on that one you actually will be presenting it in in different ways because when you have categories right
01:46:35 Speaker 1.: most
01:46:38 Speaker 2.: of the time so here's categorical versus numerical I thought we were just recapping things we know with

01:46:45 Speaker 1.: these slides
01:46:48 Speaker 2.: So, for example, you might use a pie chart for categorical, right? It has all these different categories in different colors. And what is great is that you can immediately see the proportion, right? Because they're all out of 100. And so if
01:47:07 Speaker 1.: you wanted to
01:47:08 Speaker 2.: put forward a message saying, hey, look, this one dominates. This has a really... significant percentage, or these are the top two players, top two products in this case, then that's easy. When would be a bad idea to use a pie chart? I kind of gave you the answer just
01:47:34 Speaker 1.: now.
01:47:38 Speaker 2.: When you have too many categories and when you don't have, because remember, what is the story we are trying to put forward, right? If we want to show that one is significantly greater, but if you have lots of categories and they're all about the same, then pie charts actually can't really give you that message. So that's why you always have to think about what am I trying to say with my data. And that's why we kind of make you do these different charts. so that you can see which one actually puts forward your message best. So for example,
01:48:13 Speaker 1.: here,
01:48:15 Speaker 2.: if you, if you want to show here, it's, it's very obvious, right? These are percentages of the population, and you could definitely see how that population is dominated by one type, and this is a very clear view on it.
01:48:33 Speaker 1.: But if you have, and
01:48:36 Speaker 2.: then of course we have
01:48:37 Speaker 1.: others,
01:48:39 Speaker 2.: this is a kind of slightly funky one, you could actually put the numbers, but when you have lots of categories, it becomes a little bit overwhelming.
01:48:50 Speaker 1.: We start, you know,
01:48:52 Speaker 2.: really having to struggle to
01:48:54 Speaker 1.: see how
01:48:54 Speaker 2.: it's working. While this one is a lot easier because we have one really obvious bigger slice, and then this is the second one, so not too

many categories and significant variation.
01:49:11 Speaker 1.: But,
01:49:12 Speaker 2.: so I put the advantages. Let's look at this one. When we have When we have elements, they're quite similar. So, for example, here, we have three. They're very similar. It's really hard to figure out which one is the biggest. Which one do you think is the biggest?
01:49:39 Speaker 1.: Supplier B.
01:49:40 Speaker 2.: Supplier B. So this is another way that pie charts, when they're too similar, can skew. So be careful with
01:49:56 Speaker 1.: that. So when you have more categories,
01:49:58 Speaker 2.: it's much better to have a
01:50:01 Speaker 1.: bar chart,
01:50:02 Speaker 2.: and then you could actually easier, it's much easier to compare the percentages. And you've been also doing stacked bar charts, I think, or it's coming up, I'm not sure. But definitely also the clustered ones.
01:50:23 Speaker 1.: Been doing with different
01:50:25 Speaker 2.: categories and comparing it within a period of time for different categories. So again, it depends what you want to show. If you want to show, look, over the years, this is how we've had these three different products grow, then you do the clustered one, because it's over the years for different categories.
01:50:52 Speaker 2.: But if you want to show for one period of time how different products are popular, then the simple bar chart is enough.
01:51:04 Speaker 1.: So,
01:51:08 Speaker 2.: One I also wanted to show you is the charts for categorical data that try to do things with colors or heat maps. So this is either a tree map or a heat map where
01:51:21 Speaker 1.: the,
01:51:23 Speaker 2.: instead of a pie chart, you have a rectangular and then you have the proportion covered by Again, similar issues as with pie charts, that if you have some that are about equal, it's a bit harder to see
01:51:48 Speaker 1.: them.
01:51:48 Speaker 2.: But actually, if
01:51:52 Speaker 1.: you don't have too many

01:51:53 Speaker 2.: categories, it's quite a clear pattern. So in the same way with spatial data, very often you would have the map and then you have on the map different dots that show the magnitude. So of course when you have series like here for Louisiana versus
01:52:17 Speaker 1.: Arizona,
01:52:20 Speaker 2.: well, oh no, Arkansas, right, then it's clear that it's different. But these two are almost the same. Right, you say it. So it is great if you want to see it across a geographical area, but if you want to see the proportions, it's very difficult. But this one, for example, is really good because what we are trying to see is across different employees over different days. what we were talking about with the post data, for example, right? And you can see which ones are
01:53:06 Speaker 1.: going
01:53:07 Speaker 2.: through more sales and which days of the week we seem to have more sales. So we are starting to see a pattern. This is, remember, descriptive. It gives us ideas for a hypothesis immediately, visually, rather than searching for it within a table.
01:53:27 Speaker 2.: So it's not about the exact proportion of numbers, it's about ways of you representing the data that you could easily see the patterns and get some ideas about hypothesis and what's going on.
01:53:45 Speaker 1.: So this is a heat
01:53:48 Speaker 2.: map of a web page which
01:53:51 Speaker 1.: actually shows
01:53:53 Speaker 2.: Well, first of all, where do people most visit, right? So this one is the start here,
01:53:59 Speaker 1.: so that's great.
01:54:01 Speaker 2.: Obviously, people are going to the next page. Also looking at this particular topic that they like. This one must be them just sitting here with the,
01:54:13 Speaker 1.: there's
01:54:15 Speaker 2.: no text there, so maybe they're just. Resting their eyes so
01:54:20 Speaker 1.: you could you could see What
01:54:23 Speaker 2.: is the pattern is mostly on
01:54:24 Speaker 1.: this side not much happening here?

01:54:27 Speaker 2.: So what we have is half of the page is
01:54:31 Speaker 1.: wasted from
01:54:33 Speaker 2.: a web design point of view Nobody's looking at this. Everybody's going here So we probably need to think about rebalancing And if this is the one that people are looking most at, why is it right at the bottom? I mean, why are we not making it more central, more visible? All right, now what about numerical data?

View original transcript at Tactiq.